NeedleFinder 2026: Automatic Catheter Reconstruction with Sub-Millimeter Shaft Accuracy on MRI for Interstitial Gynecologic Brachytherapy in Seconds

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ABSTRACT

Objective. In MRI-guided interstitial gynecologic brachytherapy, a clinical team inserts 15 to 30 thin plastic catheters into a pelvic tumor and acquires a 3T T2-weighted MRI scan while the patient is under anesthesia. Before radiation can be planned, a physicist must trace the 3D path of every catheter on the scan. Doing this by hand takes about 45 to 50 minutes. Existing automatic methods based on voxel segmentation struggle on T2-weighted MRI because plastic catheters appear only as tiny dark dots that look much like blood vessels, and brief signal dropouts break single catheters into disconnected pieces. Here we present NeedleFinder 2026, an algorithm that reconstructs the catheter bundle automatically, to sub-millimeter shaft accuracy, in a few seconds.

Approach. NeedleFinder 2026 combines imaging physics with three stages of machine learning. First, an analytical 3D filter (Stage 0) locates the entry plane of the perineal template automatically and measures circular dark-core symmetry on each slice—highlighting round catheter cross-sections while ignoring one-sided tissue borders. Second, a native-resolution 3D residual U-Net (Stage 1) predicts the sub-pixel center and local 3D tilt of each catheter on every slice. To prevent this network from mistaking pelvic veins for catheters, we pre-train it on 51 external pelvic MRI scans, 50 of them needle-free, injected with synthetic catheter voids and fine-tune it on consensus traces from two expert teams. Third, a graph-based tracking pipeline (Stage 2) links detections across slices, stitches fragments broken by signal dropouts, evaluates the entire 15-to-30-catheter bundle together using a graph transformer, and pinpoints each catheter's deepest tip using a 1D axial neural network equipped with calibrated uncertainty intervals. We evaluated the pipeline across 54 clinical 3T MRI scans (1,063 catheters) using strictly nested 5 × 5 patient-level cross-validation.

Main results. NeedleFinder 2026 detected 91.4% of all 1,063 catheters with 86.9% precision and an F1 score of 89.1% (87.4% and 89.4% after adding three refinements borrowed from the literature; we quote both configurations as a range below) (and 90.0%–90.1% F1 on the 759 full-depth treatment shafts), outperforming our re-implementations of five peer-reviewed MRI/curvilinear methods and one recent tracking preprint on the identical nested splits by 6.0 to 50.7 F1 points ($p \leq 1.1 \times 10^{-5}$). Along the catheter shaft, the median in-plane error was 0.47 mm (mean 0.58–0.59 mm; 0.44 mm median on full-depth shafts). At the deepest tip, the median 3D error was 2.26–2.27 mm (mean 6.23 mm), with 98.8% of

the squared tip error lying longitudinally along the catheter track. Across 40 scans traced independently by two human expert teams, one-sided non-inferiority tests, with Team 1 as reference, showed that the algorithm is statistically non-inferior to human inter-observer agreement on both shafts ($p = 1.3 \times 10^{-8}$) and tips ($p = 2.6 \times 10^{-7}$). Post-implant CT scans from 38 paired patients confirmed that 52.0% of the algorithm's apparent "false alarms" on MRI were physical catheters visible on the later CT that the primary human team had not traced on MRI. In simulated high-dose-rate ^{192}Ir treatment planning on standardized target envelopes, moving only the two tips per patient that the algorithm flagged as most uncertain to their reference depth (an oracle correction, modeled at 18 seconds of review) lowered the median tumor dose error ($|\Delta D_{90}|$) from 4.2% to 2.8% (mean from 8.2% to 5.9%), while differences in dose to proxy bladder and rectum shells remained 13% to 14% median after review. Without retraining, the gynecologic pipeline also found 13 of 14 needles with no false alarms on a single 3T prostate brachytherapy scan, reached 92.9% F1 on balanced steady-state MRI (2 scans, 47 catheters) but only 56.1% F1 on post-contrast T1 MRI (12 scans, 217 catheters), and 82.8% F1 on 23 serial needle-insertion scans from 4 patients, and recovered 68.9% of catheters on post-implant CT. End-to-end reconstruction takes 4.15 s per patient on a workstation CPU (1.26 s on the 20 standard-resolution scans, 5.85 s on the 34 high-resolution scans) and 2.67 s on a GPU.

Significance. Combining a simple physical symmetry prior with multi-stage graph learning turns a manual tracing bottleneck of 45 to 50 minutes into a 4-second automatic reconstruction (production code path, timed headless) followed by a brief, modeled uncertainty-guided check in 3D Slicer.

RESEARCH HIGHLIGHTS

- 3D dark-core prior + graph networks reconstruct MRI catheter bundles in 4.15 s.
- Beats 6 methods on 54 patients (1,063 tubes): 89.4% F1, 0.47 mm median shaft.
- Non-inferior to human experts on 40 dual-rater scans; 52% of FPs confirmed on CT.
- Split-conformal 3D tip spheres guide an 18 s check cutting dose error to 2.8%.
- Transfers zero-shot to prostate MRI, wbSSFP, 4D insertion scans, and 3D Slicer.

Keywords: Interstitial brachytherapy; Catheter reconstruction; Susceptibility MRI; Geometric deep learning; Graph transformer; Conformal prediction; 3D Slicer.

1 INTRODUCTION

1.1 *The Clinical Bottleneck in MRI-Guided Brachytherapy*

For locally advanced cervical and vaginal cancer, high-dose-rate (HDR) brachytherapy—delivering radiation directly inside the tumor—is essential for cure (Pötter *et al.*, 2018, 2021). The international EMBRACE-I study demonstrated 92% five-year local tumor control by delivering a median dose of 90 Gy

(EQD2) to the high-risk tumor volume (Pötter *et al.*, 2021), and the EMBRACE-II protocol prescribes 90 to 95 Gy to the target while strictly limiting dose to the adjacent bladder, rectum, and sigmoid colon (Pötter *et al.*, 2018). When a tumor is large, asymmetric, or extends into surrounding tissues, a central uterine tandem alone cannot cover it safely. Clinicians therefore perform *interstitial* brachytherapy: alongside the central applicator, they insert 15 to 30 thin plastic catheters through a perineal template or freehand directly into the tumor (Kapur *et al.*, 2012; Mastmeyer *et al.*, 2017b; Zaffino *et al.*, 2019; Chino *et al.*, 2025).

T2-weighted (T2W) MRI is the gold standard for visualizing gynecologic tumors and surrounding soft-tissue organs, and international guidelines (GEC-ESTRO and AAPM Task Group 303) recommend MRI-based treatment planning (JC *et al.*, 2012; Jacobsen *et al.*, 2022; Prisciandaro *et al.*, 2022; Richardson *et al.*, 2025). Ideally, the entire workflow—imaging, tumor contouring, catheter reconstruction, and radiation dose optimization—takes place on the T2W planning MRI alone (Malajovich *et al.*, 2020; Shaaer *et al.*, 2020; Lee *et al.*, 2023). In practice, however, many clinics still acquire an additional CT scan and fuse it to the MRI simply because plastic catheters are so difficult to see on MRI (Hellebust *et al.*, 2010; Ladbury *et al.*, 2023), or rely on specialized positive-contrast lumen markers (Zhang *et al.*, 2019; Ning *et al.*, 2020; Shaaer *et al.*, 2020) or zero-echo-time sequences (Kaza *et al.*, 2022; Lee *et al.*, 2023) that require extra scan time and image registration.

Why is reconstructing catheters on a standard T2W MRI so challenging, and why does its accuracy matter so much?

1. **Radiation dose changes steeply over millimeters.** The ^{192}Ir source steps through each catheter in 2.5 to 5.0 mm increments, and its dose falls off roughly with the inverse square of distance. A 1 mm shift along the catheter path or a 1.5 mm shift toward an organ at risk can alter bladder or rectal dose metrics (D_{2cc}) by 5% to 10% (Tanderup *et al.*, 2008; Schindel *et al.*, 2013; C *et al.*, 2014; Singh *et al.*, 2026), and swapping or mis-tracing catheters can shift target and urethral doses by 5% to 20% (Rylander *et al.*, 2017; Nunez *et al.*, 2021). Even experienced clinicians make errors when digitizing catheters under time pressure (S *et al.*, 2009; Androulakis *et al.*, 2025).
2. **Catheters look like tiny dark voids surrounded by dark anatomy.** A 2 mm (6F) plastic catheter contains no mobile water protons. On an axial T2W MRI slice, its plastic wall and lumen produce a small proton-density signal void, 2 to 5 pixels across, accompanied by a mild magnetic susceptibility ($\Delta\chi$) transition at the catheter–tissue interface (Richart *et al.*, 2018). Unfortunately, the female pelvis is full of other dark structures: flow-void blood vessels, air pockets in the rectum or vagina, calcifications, and the dark connective-tissue planes (fascia) between muscles.
3. **Manual tracing takes tens of minutes while the patient waits with the implant in place.** Because each scan contains 28 to 160 axial slices (1.2 to 6.0 mm thick) and 15 to 30 catheters that bend, cross, or touch one another, tracing every catheter slice by slice takes 46 to 51 minutes on average in reported MRI-only gy-

necologic workflows (Shaaer *et al.*, 2020, 2022)—nearly half of the 100-minute operating-room procedure (Roumeliotis *et al.*, 2026).

Our goal is to replace this manual bottleneck with an automatic reconstruction algorithm that operates directly on standard clinical 3T T2W MRI, matches human expert precision along both the shaft and the tip, flags its own uncertain tips for rapid human verification, and finishes in a few seconds.

1.2 *A Quick Guide for Readers Outside Medical Physics or Machine Learning*

Because this paper bridges clinical radiation oncology, MRI physics, and graph deep learning, we briefly define the core terms used throughout the text:

- **Catheter (or needle), shaft, and tip.** We use *catheter* and *needle* interchangeably for the hollow 2 mm plastic tubes inserted into the tumor. The **shaft** is the main body of the tube passing through the tissue; the **tip** (or *distal/cranial end*) is the deepest point the tube reaches inside the body, where the radioactive source stops first.
- **Obturator and tandem.** Larger central plastic guides placed in the vagina (*obturator*) and uterus (*tandem*) around which the thinner interstitial catheters are arranged.
- **Voxel, in-plane resolution, and slice thickness.** A 3D MRI scan is a stack of 2D horizontal cross-sections (*axial slices*). Within each 2D slice (*in-plane*), the image is sharp (pixel size 0.31 to 0.88 mm). Between slices (along the vertical body axis, z), the spacing is much coarser (slice thickness 1.2 to 6.0 mm). This anisotropy explains why finding the sideways (*in-plane*) center of a catheter is easier than judging the exact height (z) where its tip ends between two slices.
- **Recall, precision, and F1 score.** *Recall* is the percentage of real catheters the algorithm successfully finds. *Precision* is the percentage of algorithm-reported tracks that correspond to a real catheter (rather than a false alarm on a blood vessel). The *F1 score* is the harmonic mean of recall and precision; it is high only when the algorithm finds almost all real catheters while inventing very few false ones.
- **Shaft error (E_{shaft}) and tip error (E_{tip}).** For every detected catheter that matches an expert trace (within 3.5 mm), *shaft error* measures the sideways distance (in mm) between the automatic curve and the expert curve along their shared length. *Tip error* measures the 3D distance (in mm) between the predicted deepest tip and the expert's deepest tip.
- **Strictly nested cross-validation (out-of-fold evaluation).** To test our models on all 54 patients without ever testing on data used for training or tuning, we split the 54 patients into 5 outer groups (folds). In each round, 43 or 44 patients are used for training and tuning (via an internal 80/20 split or 5-fold inner cross-fitting), and the remaining 10 or 11 patients are held in a locked vault and evaluated *once*. Every patient is thus evaluated strictly "out-of-fold" (as an unseen patient).

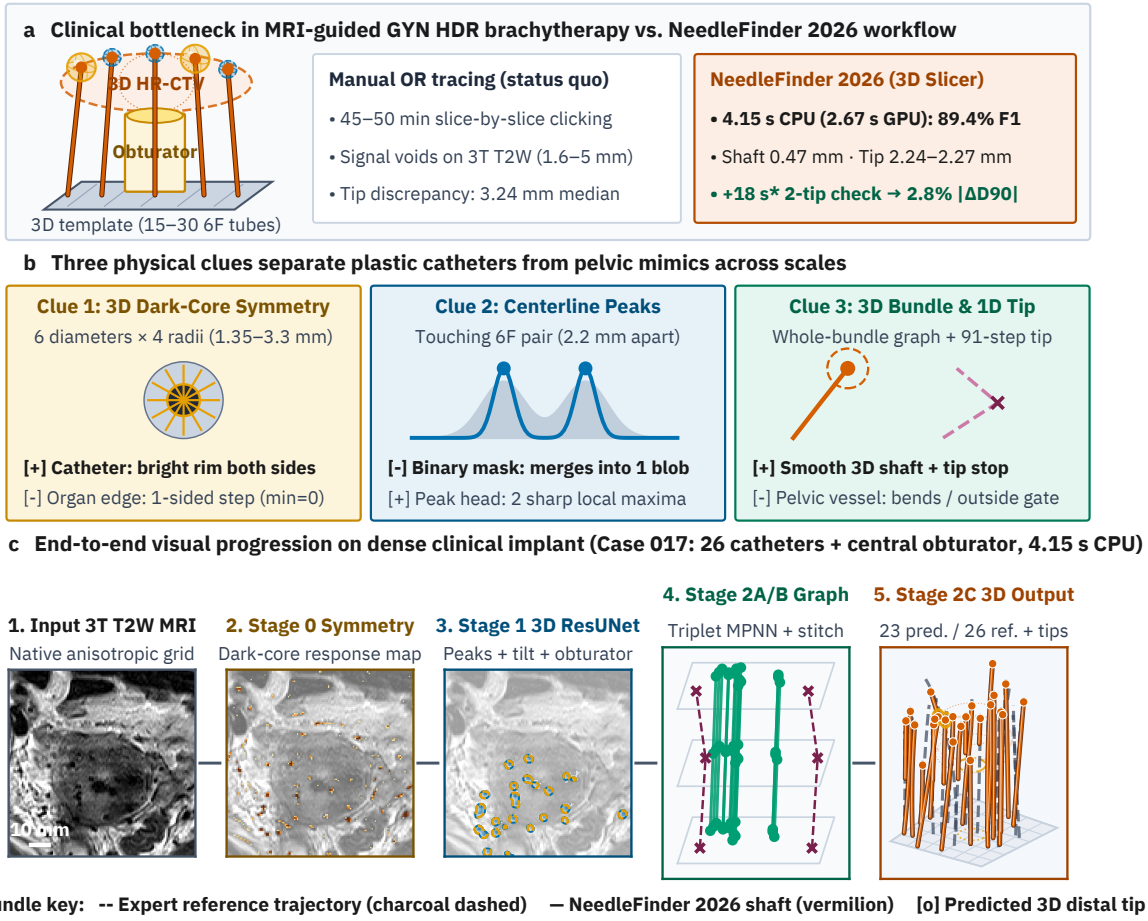


FIGURE 1 **Combining a physical 3D dark-core symmetry prior with multi-scale graph learning turns a 45–50 minute manual tracing bottleneck into a 4.15 s automatic reconstruction in 3D Slicer. (a) Clinical bottleneck vs. NeedleFinder 2026 workflow:** In MRI-guided interstitial gynecologic brachytherapy, 15 to 30 thin 6F (2.0 mm) plastic catheters are inserted through a perineal template and vaginal obturator into the high-risk tumor target (HR-CTV). Because plastic tubes contain no mobile protons, they appear only as tiny signal voids on 3T T2-weighted MRI (Δz = 1.6–5.0 mm), requiring 45 to 50 minutes of manual slice-by-slice clicking while the patient waits under anesthesia. **(b) Physical clues across three scales:** (Clue 1, Stage 0) A catheter cross-section is a circular dark core surrounded by brighter tissue across all six diameters (`[+]`), whereas an organ border is a one-sided step edge (`[-]`); (Clue 2, Stage 1) when two catheters touch (`2.2 mm` apart), binary segmentation masks merge into a single blob (`[-]`), whereas regressing 2D Gaussian centerline peaks keeps the two centers distinct (`[+]`); (Clue 3, Stage 2) evaluating the full 15-to-30-catheter bundle together and profiling the cranial termination in 0.5 mm steps separates smooth catheters (`[+]`) from bending pelvic vessels (`[-]`). **(c) End-to-end visual progression on an unseen test patient (Case 017, 26 expert reference trajectories):** From left to right: native 3T T2W MRI (`10 mm` scale bar); Stage 0 dark-core symmetry response S_0 (`0.27 s` Rust CPU); Stage 1 sub-pixel centerline peaks, shaft mask, and obturator contour (`2.58 s` CPU / `1.33 s` GPU); Stage 2A/2B triplet message-passing graph and collinear fragment stitching (`1.25 s` CPU / `1.05 s` GPU); and Stage 2C 3D reconstructed bundle (`vermilion solid`, 23 predicted) overlaid on expert traces (`charcoal dashed`, 26 reference) with predicted 3D distal tips (`0.06 s` CPU / `0.03 s` GPU; `4.15 s` total CPU headless, `2.67 s` GPU).

– **Dosimetric metrics (D_{90}, D_{2cc}, and 3D γ).** In brachytherapy planning, D_{90} is the minimum radiation dose covering 90% of the high-risk tumor volume (which clinicians want to keep high), while D_{2cc} is the dose to the most-irradiated 2 cm^3 of a nearby healthy organ like the bladder or rectum (which clinicians want to keep low). The 3D *gamma (γ) pass rate* (2%/2 mm) measures the percentage of points in the 3D volume where the dose planned on automatic catheters agrees with the reference plan within 2% dose or 2 mm distance.

1.3 *Why Existing Approaches Fall Short on T2-Weighted MRI*

Over the past decade, automatic catheter reconstruction has evolved through three generations (reviewed in (Sargazi *et al.*, 2026; Vijayan *et al.*, 2026)):

Generation 1: Geometric and mechanical tracking from manual clicks (2012–2019). Early work in our Brigham and Women's Hospital (BWH) program introduced image-guided catheter tracking in the Advanced Multimodality Image Guided Operating (AMIGO) suite (Kapur *et al.*, 2012; Chen and Egger, 2014). Starting from a manual click at each catheter tip, cone-search trackers (Pernelle *et al.*, 2013) and mechanical rod-and-spring splines (Mastmeyer *et al.*, 2017a; Mastmeyer *et al.*, 2017b) achieved ~1.4–1.5 mm accuracy on T2W MRI. Similar geometry-guided or template-registered methods work well on CT, ultrasound, or rigid applicators (W *et al.*, 2013; Dise *et al.*, 2015; J *et al.*, 2015; Hrinivich *et al.*, 2019; Deufel *et al.*, 2020; Zheng *et al.*, 2021). However, they still require a human to click every catheter tip first, and they often lose track when catheters bend sharply or cross dark vessels.

Generation 2: 3D voxel segmentation with clustering (2019–2026). To eliminate manual clicks, Zaffino *et al.* (2019) trained a 3D U-Net on 50 BWH gynecologic MRI scans to classify every voxel as "catheter" or "background," then grouped the segmented voxels into tubes using density clustering (DBSCAN). Subsequent studies applied 3D U-Nets and nnU-Net (Isensee *et al.*, 2021) across CT, ultrasound, and prostate or marker-assisted MRI (Jung *et al.*, 2019b; Jung *et al.*, 2019a; Andersén *et al.*, 2020; Dai *et al.*, 2020; Li *et al.*, 2020; Wang *et al.*, 2020; Zhang *et al.*, 2020a; Zhang *et al.*, 2020b, 2020c; Zhang *et al.*, 2020d; Hu *et al.*, 2021; Liu *et al.*, 2022; Shaaer *et al.*, 2022; Weishaupt *et al.*, 2022; Xie *et al.*, 2023; Aleong *et al.*, 2024; Wang *et al.*, 2024; Zhou *et al.*, 2024; Goulet *et al.*, 2025; Kostoulas *et al.*, 2025; Moradi *et al.*, 2025; Quetin *et al.*, 2025; Kim *et al.*, 2026). Yet even when trained with topology-preserving losses such as centerline-Dice (clDice) (Hu *et al.*, 2019; Shit *et al.*, 2021; Kirchhoff *et al.*, 2024), pure voxel segmentation suffers from two fundamental failure modes on T2W MRI:

1. *Over-fragmentation:* If a catheter's dark void fades on even a single slice due to partial-volume averaging, the segmented mask breaks into 2 to 4 separate pieces.
2. *Touching catheters and vascular false alarms:* When two catheters touch (*catheter adhesion*), their voxel masks merge into a single blob; meanwhile, dark pelvic veins and tissue edges are segmented as dozens of short false catheters.

Generation 3: Structure-aware tracking, gap-bridging, and uncertainty-weighted fitting (2020–2026). To mitigate segmentation breaks on MRI, Dai

et al. (2020) introduced a deeply supervised 3D attention-gated U-Net with axial total-variation (TV) continuity regularization, Kostoulas *et al.* (2025) combined a 3D residual U-Net with iterative collinear gap-bridging and false-part pruning, and Aleong *et al.* (2024) and Kim *et al.* (2026) leveraged 5-model epistemic uncertainty ensembles and inverse-variance-weighted polynomial trajectory fitting. On pelvic CT, Xiao *et al.* (2026a, 2026b) addressed catheter adhesion by combining an implant-region gating module with 3D Hough initialization and synchronous slice-to-slice tracking. In computer vision, message-passing graph networks (Brasó and Leal-Taixé, 2020; Ben-Haim and Riklin Raviv, 2022; Shit *et al.*, 2022), keypoint regression (Zhou *et al.*, 2026), and synthetic-to-real transfer (Arapi *et al.*, 2023) have similarly improved curvilinear tracking. However, a mask-driven tracker applied to T2W MRI without a physical circular-symmetry prior or whole-bundle reasoning still trusts the underlying segmentation mask. Dark pelvic veins inside the implant region are tracked as catheters. The tracker has no mechanism to ask whether the resulting collection of 25 trajectories makes sense *together* as a physical implant, nor to resolve sub-slice tip terminations on thick MRI slices. (Hardware-based electromagnetic or active MR tracking avoids image ambiguity (Mehrtash *et al.*, 2014; Wang *et al.*, 2015; Arcos *et al.*, 2017; Dürrbeck *et al.*, 2022; Cantin *et al.*, 2024; Deufel *et al.*, 2024; Beaulieu *et al.*, 2025; Roumeliotis *et al.*, 2026), but requires specialized catheters and operating-room hardware.)

1.4 *How NeedleFinder 2026 Solves These Problems*

NeedleFinder 2026 bridges all three generations on the complete 54-patient, 1,063-catheter BWH gynecologic MRI cohort—to our knowledge the largest annotated clinical MRI dataset for interstitial brachytherapy to date (Figure 1). Each stage of our architecture is designed to solve a specific failure mode of prior work:

1. **Physics before learning (Stage 0):** Before any neural network runs, an analytical 3D circular dark-core symmetry filter separates round dark cross-sections (catheters and vessels) from one-sided dark boundaries (fascial planes and organ walls), while an automatic template-plane locator finds the base of the implant without needing manual bounding boxes.
2. **Centerline peaks and self-supervised pelvic pre-training (Stage 1):** Instead of relying on binary masks alone, our native-resolution 3D U-Net regresses sharp sub-pixel *centerline peaks* and *3D tangent directions*—so two touching catheters still produce two distinct peaks. To prevent the network from confusing pelvic veins with catheters, we pre-train it on 51 external pelvic MRI volumes (from 50 subjects) by injecting realistic synthetic catheter voids while explicitly penalizing real pelvic vessels and fascia as hard negatives, and then fine-tune it using consensus traces from two independent expert teams.
3. **Graph linking, fragment stitching, bundle-level reasoning, and 1D tip localization (Stage 2):** A triplet message-passing graph network links slice detections while penalizing sharp kinks, and a geometric stitcher reconnects catheter fragments separated by 1-to-3-slice signal dropouts. Next, a bundle

graph transformer evaluates all candidate trajectories simultaneously in the context of the whole implant. Finally, a dedicated 1D axial neural network samples a fine 0.5 mm multi-scale profile along the end of each trajectory to pinpoint the exact distal tip and output a calibrated split-conformal uncertainty interval, optionally fused with sub-voxel inverse-variance-weighted polynomial refinement (Aleong *et al.*, 2024; Kim *et al.*, 2026), axial TV continuity (Dai *et al.*, 2020), and iterative multi-segment gap-bridging (Kostoulas *et al.*, 2025).

Beyond geometric benchmarking against five peer-reviewed MRI/curvilinear methods and one recent tracking preprint reproduced on the identical nested splits, we evaluate NeedleFinder 2026 across four clinical translation dimensions: (i) three-cornered-hat observer adjudication against two human expert teams and paired post-implant CT scans; (ii) AAPM TG-43U1 ^{192}Ir radiation dosimetry; (iii) uncertainty-guided clinician tip triage; and (iv) zero-shot generalization to needle-free pelvic MRI, 3T prostate brachytherapy MRI, secondary MRI pulse sequences, serial intra-procedural insertion scans, and pelvic CT. Directly extending our group's multi-planar mechanical rod tracking in *Medical Image Analysis* (Mastmeyer *et al.*, 2017b) and 3D U-Net voxel clustering in *Physics in Medicine & Biology* (Zaffino *et al.*, 2019) to the full 54-patient AMIGO cohort, this study simultaneously addresses open challenges across three communities: in **brachytherapy**, enabling marker-free T2W MRI-only catheter digitization and dosimetrically calibrated tip verification; in **computer vision and geometric deep learning**, resolving touching curvilinear tubes ($\leq$ 3.0 mm) under extreme axial anisotropy ($\Delta z/\Delta x$ up to 15×) via hybrid 3D CNN peak regression, triplet message-passing, and whole-bundle graph transformers; and in **image-guided therapy**, delivering 4.15 s GPU-free reconstruction in 3D Slicer alongside 1.31 s causal 4D intra-procedural tracking and in-vivo tissue-push biomechanics.

2 PATIENT COHORTS AND CLINICAL DATASETS

Table 1 summarizes all clinical benchmark, inter-observer, external pre-training, and cross-domain validation cohorts analyzed in this study:

1. **Primary gynecologic MRI cohort (54 patients, 1,063 catheters).** We analyzed 54 intraoperative 3T T2W turbo spin-echo MRI volumes acquired during MRI-guided interstitial gynecologic brachytherapy in the AMIGO suite at Brigham and Women's Hospital (Harvard Medical School, Boston, MA, USA) under Institutional Review Board approval (Pernelle *et al.*, 2013; Mastmeyer *et al.*, 2017b; Zaffino *et al.*, 2019). This cohort consolidates the 48 recoverable cases from our 2013 and 2019 studies with 6 newly curated implants. Patients were implanted using a Syed–Neblett perineal template surrounding a central vaginal obturator ($R_{\mathrm{obt}} \approx$ 10–15 mm) and scanned with 6F polymeric interstitial catheters (with plastic or non-ferromagnetic obturators in situ, creating a proton-density signal void and mild interface susceptibility contrast on T2W MRI). Expert medical

TABLE 1 **Clinical imaging cohorts and evaluation protocols.** Primary evaluation uses strictly nested 5×5 patient-level out-of-fold (OOF) cross-validation with 100 % reference-free axial cropping (`k_lim_auto`) on 54 clinical 3T T2W MRI datasets from Brigham and Women's Hospital (BWH AMIGO).

Cohort / Subset	Source	Pts	Cath.	XY (mm)	Δz (mm)	Evaluation role
BWH GYN full cohort[a]	BWH (3T T2W)	54	1063	0.31–0.88	1.2–6.0	Nested 5 × 5 OOF benchmark
Long shafts (primary)	BWH (3T T2W)	54	759	0.31–0.88	1.2–6.0	Full-depth active dwell shafts
Short / obturator (secondary)	BWH (3T T2W)	54	304	0.31–0.88	1.2–6.0	Partial-FOV & template tracks
Adhesion subgroup[b]	BWH (3T T2W)	23	571	0.31–0.88	1.2–6.0	Touching / crossing shaft test
Dual-observer subset[c]	BWH (3T T2W)	40	723	0.31–0.88	1.2–6.0	Inter-observer floor (`CY+HT`, `LB/AMe`)
External SSL & specificity[d]	MSD + BWH	50 (51)	0	0.47–0.88	1.6–4.8	3D SSL (51) & 0-needle test (48)
Prostate 3T implant (`Case 100`)[e]	BWH (3T T2W)	1 (1)	14	0.47	1.6	Zero-shot cross-organ transfer
Secondary GYN MRI sequences[e]	BWH (wbSSFP/2D)	54 (243)	929	0.31–0.94	2.0–6.0	Zero-shot multi-sequence test
4D serial insertion (`Cases 68–71`)[e]	BWH (3T T2W)	4 (23)	108	0.31–0.75	1.6–3.0	Causal 4D tracking & biomechanics
Paired post-implant Pelvic CT[e]	BWH (Pelvic CT)	38 (40)	801	0.49–0.98	1.25–2.5	CT adjudication & cross-modality

[a] Consolidated superset of the cohorts of Pernelle *et al.* (2013, 10 patients) and Zaffino *et al.* (2019, 50 patients), extended with new implants; 9–42 reference trajectories per patient (20 standard-resolution 320 × 256 scans and 34 high-resolution 640 × 416 / 768 × 768 scans). In the Pts column, parentheses indicate total volumes when multiple scans per subject are analysed.

[b] Severe catheter-adhesion subgroup: the 23/54 patients in which ≥ 4 long or ≥ 6 total reference catheters pass within ≤ 3.0 mm of another catheter on shared axial slices. All 571 catheters of these patients are evaluated; 191 of them (166 long) lie within 3.0 mm of a neighbour (233 across the whole cohort).

[c] All 40/40 dual-traced clinical scenes independently contoured by two expert medical physics teams (`CY+HT`: 723 total trajectories = 505 `CY` long + 218 `HT` short/obturator; `LB/AMe`: 513 trajectories, with 465 consensus-fused with `CY`, 12 matched to `HT` within ≤ 5.0 mm [11 within ≤ 3.5 mm], and 36 unique `LB/AMe` trajectories forming the 759-trajectory union).

[d] Fifty-one external pelvic 3T T2W MRI volumes from 50 distinct subjects: 48 public CC-BY-SA 4.0 volumes from MSD `Task05_Prostate` (0.60–0.75 mm in-plane, 3.0–4.0 mm slices; zero implanted catheters, evaluated for false-positive specificity) plus 3 non-benchmark BWH pelvic T2W volumes from 2 distinct subjects (`Case 061` [2 series, 0.88 mm in-plane, 4.8 mm slices] and `Case 100 Prostate` [1 series, 0.47 mm in-plane, 1.6 mm slices], zero patient overlap with the 54 benchmark cases) used for self-supervised 3D U-Net pre-training.

[e] Cross-domain validation cohorts summarized in Table 4 and detailed in Appendix D (Table 8): zero-shot 3T prostate brachytherapy MRI (`Case 100`, 14 plastic needles); 243 secondary MRI volumes across the 54 sessions (including 2 `3D wbSSFP` volumes [47 catheters] and 49 thick-slice `2D AX TSE T2` volumes [882 physically visible catheters]); 23 sequential 4D intra-procedural insertion scans across 4 patients (`Cases 68–71`, 108 final catheters); and 40 post-implant Pelvic CT scans (38 paired with benchmark MRI patients, 801 reference catheters).

physicists traced 1,063 reference trajectories (9 to 42 per patient). These comprise **759 full-depth interstitial shafts** (which penetrate into the tumor to carry the active radiation dwell positions; archive code `CY` in Table 1) and **304 shorter auxiliary tracks** sitting inside the central vaginal obturator lumen (including any central tandem or internal obturator guide channels, which Stage 1 also masks at the outer obturator rim via P_{obt}) or only partially entering the field of view (archive code `HT`). Twenty scans were acquired at standard in-plane resolution (320 × 256 matrix, 0.75–0.88 mm pixels) and 34 at high resolution (640 × 416 to 768 × 768 matrix, 0.31–0.55 mm pixels), with slice thickness ranging from 1.2 to 6.0 mm.

2. **Severe catheter-adhesion subgroup (23 patients, 571 catheters).** Following Xiao *et al.* (2026a), we identified the 23 patients in whom catheters are tightly packed or touching—defined as having at least 4 long shafts (or at least 6 total catheters) that pass within ≤ 3.0 mm in-plane centerline distance of a neighbor-

ing catheter on shared slices. All 571 catheters in these 23 patients are evaluated together (191 of them touch a neighbor within ≤ 3.0 mm, out of 233 such touching catheters across all 54 patients).

3. **Dual-expert inter-observer cohort (40 patients).** Forty of the 54 scans were traced independently by two separate expert groups: **Expert Team 1** (723 trajectories: 505 long shafts and 218 short obturator tracks) and **Expert Team 2** (archive code `LB/AMe`: 513 trajectories, of which 460 are long shafts). Across these 40 scans, 424 long-shaft pairs (and 476 all-catheter pairs) match one-to-one within 3.5 mm between the two human teams, while their de-duplicated union contains 759 distinct trajectories (723 from Team 1 plus 36 additional catheters traced only by Team 2).
4. **External pelvic MRI cohort for self-supervised pre-training and false-alarm testing (51 volumes from 50 subjects).** To pre-train our 3D network on pelvic anatomy and test false-alarm rates on the 48 public scans containing *zero* catheters, we assembled 51 strictly external 3T T2W pelvic MRI volumes having zero patient overlap with the 54 benchmark cases: all **48 public prostate MRI scans** from the Medical Segmentation Decathlon (`MSD Task05_Prostate` (Antonelli *et al.*, 2022); 0.60–0.75 mm pixels, 3.0–4.0 mm slices) plus **3 non-benchmark BWH pelvic T2W volumes** from two separate patients (`Case 061` [2 series] and `Case 100 Prostate` [1 series]).
5. **Cross-organ, multi-sequence, 4D insertion, and paired CT validation cohorts.** To test how well the gynecologic-trained pipeline generalizes without retraining (Table 1, bottom four rows; Table 4, Part D; Appendix D, Table 8), we evaluated: (i) a clinical 3T T2W **prostate brachytherapy implant** (`Case 100`, 14 plastic needles); (ii) **243 secondary MRI volumes** acquired during the 54 gynecologic procedures, including 3D balanced steady-state free precession (`3D wbSSFP / True-FISP`, 2 volumes, 47 catheters), 12 post-contrast axial T1 VIBE volumes (217 catheters), and 49 thick-slice 2D axial T2W (`2D AX TSE T2`, 2.4–6.0 mm slices, 882 physically visible catheters) scans; (iii) **23 serial 4D intra-procedural MRI scans** acquired across 4 patients (`Cases 68–71`, 5 to 7 sequential scans per patient as needles were progressively inserted, 108 final catheters); and (iv) **40 post-implant pelvic CT volumes** (38 of them paired directly with benchmark MRI patients, 801 reference catheters).

3 ARCHITECTURE OF NEEDLEFINDER 2026

A human physicist tracing catheters on MRI uses three kinds of clues in sequence: first, looking for small round dark spots on each 2D slice; second, checking whether those dark spots line up across adjacent slices; and third, verifying that the full collection of 15 to 30 curves forms a plausible implant bundle and stopping each curve where the dark void disappears at the tip. NeedleFinder 2026 mirrors this exact hierarchy in three stages (Figures 1 and 2).

3.1 *Stage 0: Physical Dark-Core Symmetry and Automatic Template Cropping*

The physical intuition. Why do standard neural networks confuse muscle borders and organ walls with catheters? Because both are dark on T2W MRI. However, their *geometry* is different: a muscle border is a dark edge with bright tissue on one side and dark tissue on the other, whereas a catheter cross-section is a tiny dark core surrounded by brighter tissue in *every* direction.

Stage 0 turns this geometric observation into a fast 3D filter $S_0(x, y, z)$ with no learned parameters, implemented in multi-threaded native Rust. At every voxel $\mathbf{p}$ on axial slice z, we sample six diameters passing through $\mathbf{p}$ at angles $\theta_d = d\pi/6$ ($d \in \{0, \dots, 5\}$) across four physical radii $r \in \mathcal{R} = \{1.35, 1.85, 2.5, 3.3\}$ mm (converted to pixels using the scan's in-plane spacing). Along each diameter d and radius r, we take the *darker* of the two opposite ring points:

$$m_{d,r} = \min\big(I(\mathbf{p} + r\mathbf{u}_d),\ I(\mathbf{p} - r\mathbf{u}_d)\big).$$

If $\mathbf{p}$ sits on a one-sided edge, at least one end of the diameter remains dark, keeping $m_{d,r}$ small. We then sort the six directional values ($m_{(1),r} \le \dots \le m_{(6),r}$) and compute a robust ring brightness that requires the surrounding tissue to be bright in almost all directions while tolerating a single dark neighbor (such as a touching twin catheter):

$$S_{\text{ring}}(\mathbf{p}) = \max_{r \in \mathcal{R}} \left[0.45\, m_{(2),r} + 0.55\, m_{(3),r}\right].$$

Combining this positive ring-to-core contrast with the relative darkness of the 3×3 smoothed center intensity $\bar{I}_{3\times3}(\mathbf{p}) \in [0, 1]$ yields the raw dark-core symmetry score,

$$S_0(\mathbf{p}) \propto \max\big(0,\ S_{\text{ring}}(\mathbf{p}) - \bar{I}_{3\times3}(\mathbf{p})\big) \cdot \left[1 - \bar{I}_{3\times3}(\mathbf{p})\right],$$

which is normalized to $[0, 1]$ per volume. Because S_0 takes the maximum over radii up to 3.3 mm and uses the second- and third-darkest of the six directional values, it is, by construction, tolerant of realistic oblique catheter tilt ($\alpha \le 25^\circ$–30° relative to the slice normal, where the in-plane elliptical cross-section elongates by only $\sec \alpha \le 1.15$; see Appendix A for the formal tilt-invariance derivation). On very thick slices ($\Delta z = 4$–6 mm) where steep tilt causes stronger partial-volume elongation, we first upsample the volume along z to 1.6 mm spacing and rely on Stage 2A fragment stitching to bridge dimmed slices.

Automatic reference-free cropping of the template plane. Below the patient's perineum, the inferior slices of the MRI scan contain outside air and the plastic needle-guiding template. Previously, pipelines relied on manually drawn bounding boxes to crop out these bottom slices, which is unavailable in a fully automatic workflow and occasionally clipped tilted templates too high—cutting off the bases of 42 catheters across our cohort. We replaced this with a reference-free axial template locator. After standardizing the volume to caudo-cranial slice ordering via its DICOM/NRRD direction cosines, the locator scans upward from the inferior boundary and detects the step-change in 90th-percentile soft-tissue intensity alongside the emergence of clustered dark-core candidates (S_0) within the central field of view. Across all 54 patients, it places the crop boundary a median of 2.0 slices deeper

than the manually drawn boxes (mean 1.91 slices deeper, IQR 1.0; 64.1 vs. 62.2 active slices on average), eliminating all 42 in-field-of-view base truncations and preserving **100.0% of in-field-of-view catheter length** (99.65% of total unclipped catheter length) without ever consulting manual annotations. Stage 0 and automatic cropping run in **0.27 s** per volume on a 24-core CPU (0.07 s on standard scans, 0.38 s on high-resolution scans).

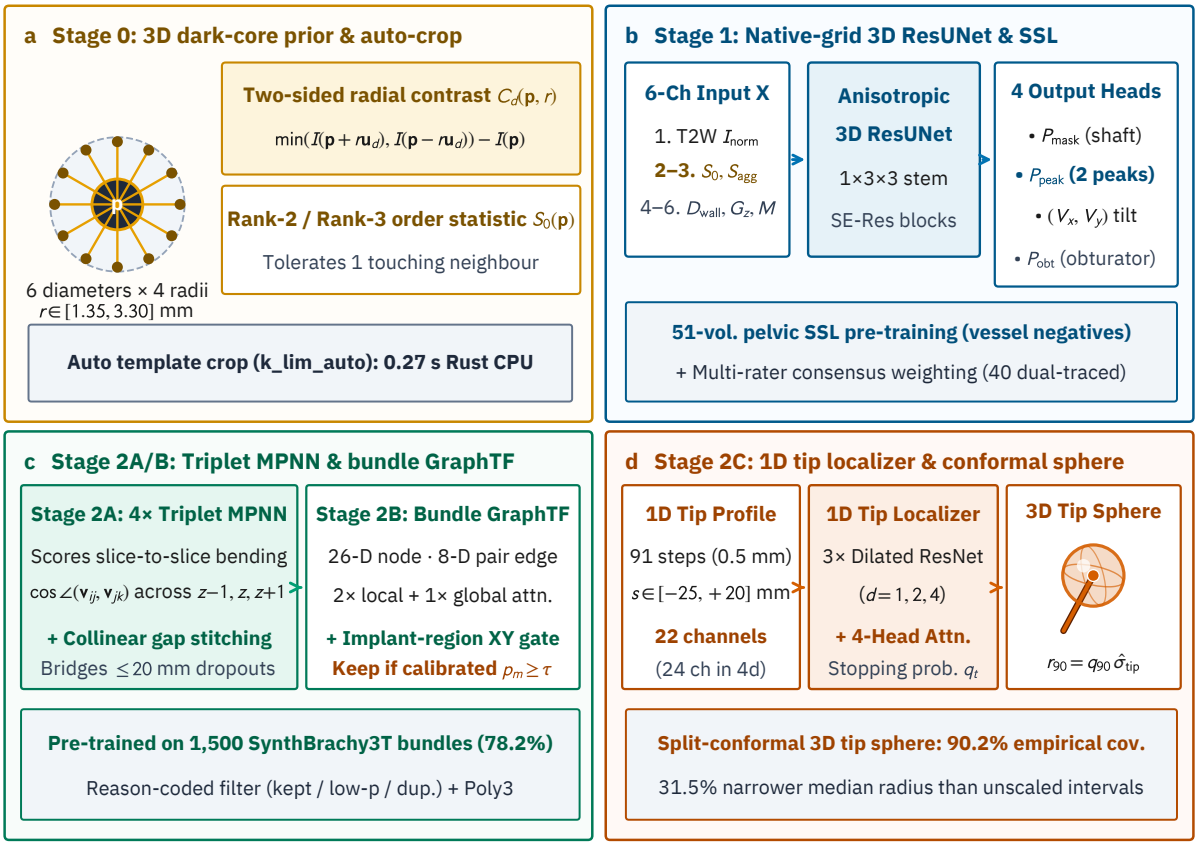


FIGURE 2 **Four-stage physical and neural mechanism of NeedleFinder 2026. (a) Stage 0 analytical dark-core prior and template auto-crop:** At each voxel $\mathbf{p}$, six diameters ($\theta_d = d\pi/6$) across four physical radii ($r \in [1.35, 3.30]$ mm) evaluate two-sided radial contrast $\min(I(\mathbf{p}+r\mathbf{u}_d), I(\mathbf{p}-r\mathbf{u}_d)) - I(\mathbf{p})$; the rank-2/rank-3 order statistic $S_o(\mathbf{p})$ suppresses one-sided organ borders while tolerating a touching neighbor, and `k_lim_auto` locates the caudal template entry slice automatically (`0.27 s` Rust CPU). **(b) Stage 1 native-resolution anisotropic 3D ResUNet (`NeedleUNet3D`):** Operating directly on the native voxel grid (1×3×3 in-plane stem and SE-Res blocks), the network is pre-trained on 51 external pelvic MRI volumes (penalizing real vessels and fascia as hard negatives) and fine-tuned on dual-expert consensus traces to predict four heads: tubular mask (P_{mask}), sub-pixel centerline peaks (P_{peak}), local 3D slope (V_x, V_y), and obturator mask (P_{obt}). **(c) Stage 2A & 2B graph tracking, collinear gap stitching, and bundle transformer:** A 4-layer triplet message-passing network scores slice-to-slice bending angles $\cos\angle(\mathbf{v}_{ij}, \mathbf{v}_{jk})$ across $z-1, z, z+1$; a collinear stitcher bridges ≤ 20 mm signal dropouts; and a 26-D node / 8-D edge bundle graph transformer (pre-trained on `SynthBrachy3T`) outputs calibrated confidence $p_m \geq \tau$ with reason codes. **(d) Stage 2C dedicated 1D axial tip localizer and split-conformal sphere:** Sampling 91 steps (0.5 mm over [−25, +20] mm) across 22 channels (24 in 4d) along the cranial tangent $\hat{\mathbf{t}}$, a 3-block dilated 1D ResNet ($d = 1, 2, 4$) + 4-head self-attention predicts the stopping distribution q_t, sub-slice tip offset Δs (`2.27 mm` median 3D error), and split-conformal 3D tip radius $\hat{r}_{90} = \hat{q}_{90}\hat{\sigma}_{tip}$ (`90.2%` empirical coverage at 90% nominal).

3.2 *Stage 1: Native-Resolution 3D U-Net with Self-Supervised and Multi-Rater Learning*

Why native resolution and multiple output heads matter. Many 3D medical segmentation pipelines first resample every scan to coarser, near-isotropic voxels such as $1 \times 1 \times 1$ mm^3. For a 2 mm catheter imaged at 0.35 mm pixel resolution, resampling blurs the sharp dark core into the surrounding tissue. Conversely, applying a standard $3 \times 3 \times 3$ convolution directly on the native grid mixes features across up to 18 mm vertically (z) but only 1 mm horizontally (x, y).

Our Stage 1 network (Figure 2a) therefore operates directly on the native voxel grid using an anisotropic 3D residual U-Net with squeeze-and-excitation channel attention at four scales (24, 48, 96, and 144 channels). Crucially, it pools spatially *only in-plane* (x, y) at the first two scales so that thick axial slices are never compressed early. The network takes six input channels—normalized T2W intensity, Stage 0 dark-core symmetry S_0, ring contrast, smoothed intensity, normalized axial height z, and radial distance from the scan center—and predicts four complementary outputs:

1. **Tubular mask (P_{mask}):** The probability that a voxel lies within a catheter cross-section (trained with focal binary cross-entropy, up-weighted 19-fold on catheter voxels and on false Stage 0 vessel peaks, plus soft Dice loss).
2. **Sub-pixel centerline peak (P_{peak}):** A narrow 2D Gaussian heatmap ($\sigma = 0.70$ mm) centered on the exact intersection of each catheter with the slice (trained with peak-weighted mean-squared error). **Why this matters:** When two catheters touch, their binary masks (P_{mask}) merge into one blob, but their two centerline peaks (P_{peak}) remain distinct.
3. **Local 3D direction field ($V_x, V_y = \Delta x/\Delta z, \Delta y/\Delta z$):** The local tilt of the catheter as it passes from slice to slice (trained with smooth L_1 loss and an axial consistency penalty), telling the tracker where to look on the next slice.
4. **Obturator segmentation (P_{obt}):** A dedicated mask for the large central vaginal obturator ($R_{\text{obt}} \approx 10$–$15$ mm) and its outer cylindrical plastic wall, preventing its curved dark rim from generating false peripheral needle detections while allowing central tandem or internal obturator guide channels (`HT`) to be identified and distinguished from full-depth interstitial shafts (`CY`).

How we teach Stage 1 to ignore blood vessels and noisy single-rater labels. We introduce two training innovations for Stage 1:

– **Self-supervised pre-training on 51 external pelvic MRI scans with anatomical hard negatives.** Before training on any benchmark brachytherapy patient, we pre-train the 3D U-Net for 45 epochs across the 51 external needle-free pelvic MRI volumes (48 public MSD prostate scans + 3 non-benchmark BWH pelvic scans). In each training step, we digitally inject 8 to 20 realistic synthetic 3D catheter trajectories—curved dark cores with partial-volume bright rims and random signal dropouts—while simultaneously masking random 3D tissue patches for reconstruction. Because 50 of these 51 external scans contain *zero* real catheters, **every real pelvic blood vessel, bowel loop, or fascial border that**

triggers a high Stage 0 symmetry score ($S_0 > 0.25$) outside the synthetic tubes serves as an anatomical hard negative (the one prostate implant scan contributes its 14 unlabeled needles as background, which we accept as label noise). The network is thus pushed to separate a plastic catheter void from a pelvic vein before it ever sees the benchmark cohort.

- **Multi-rater consensus supervision.** On the 40 training scans traced by both human expert teams, training on a single team's annotations penalizes the network whenever it finds a real catheter that Team 1 overlooked but Team 2 traced, and passes along single-observer clicking jitter. Within each outer training fold (never touching test patients), we pair matching trajectories between Team 1 and Team 2 (≤ 3.5 mm, $n = 465$ pairs) and supervise Stage 1 along their precision-weighted consensus centerline (weight 1.15). Catheters traced only by Team 1 receive weight 1.00, and additional catheters traced only by Team 2 ($n = 35$ after excluding 465 primary-matched and 13 obturator-overlapping traces; this training-time match pairs each Team 2 trace, clipped to the automatic axial crop, with a Team 1 long shaft only, so its counts differ slightly from the evaluation-time pairing of all unclipped traces of both teams in the cohort description above, 476 pairs) receive soft positive weight 0.55 instead of being treated as background false alarms.
- **Inner 5-fold ensembling and 4-view test-time augmentation.** Within each outer training fold, we train 5 inner-fold Stage 1 models with rich 3D MRI augmentations (random 3D bias fields, slice-thickness blurring, slice dropouts, and flips) and average their predictions across four flip symmetries (I, X, Y, XY) at inference.

3.3 *Stage 2: Graph Tracking, Fragment Stitching, Bundle Scoring, and 1D Tip Localization*

Once Stage 1 has highlighted candidate catheter centers and directions on every slice, Stage 2 solves three sequential geometric problems (Figure 2c, d):

3.3.1 *Step 2A: Connecting slice detections and stitching split fragments.*

On each slice, local peaks of P_{peak} and S_0 become graph nodes (each described by 16 local image and direction features). Directed edges connect nodes across 1 to 4 slices (up to 24 mm vertically) whenever their in-plane displacement agrees with the Stage 1 direction field.

Why use a graph network alongside Kalman tracking instead of greedy slice-to-slice linking alone? Because two consecutive links can each look reasonable on their own yet form a sharp, unphysical kink when joined together. First, a fast native C++17/Rust bidirectional Kalman tracker traces candidate 3D paths from the Stage 0 and Stage 1 seeds in both directions and merges near-duplicate paths within 1.65–2.05 mm while preserving genuinely touching twin catheters. Second, a **4-layer triplet message-passing network** (hidden width 64) scores every node and edge in the slice graph using attention-weighted messages from both its incoming and outgoing neighboring edges—explicitly rewarding smooth, collinear trajectories, refining sub-pixel coordinates, and penalizing kinks.

Finally, our failure analysis of earlier pipelines revealed that mid-shaft signal dropouts (where a catheter crosses a dark vessel or air pocket over 1 to 3 slices) frequently split one physical catheter into a lower fragment and an upper fragment—causing both a false-positive extra fragment and a severe tip undershoot on the lower track. We therefore added a **collinear split-fragment stitcher**: whenever a lower track fragment ends and an upper fragment begins within an axial gap of $\leq$ 20 mm, and their linearly extrapolated positions at the midpoint agree within $\leq$ 2.2 mm with matching 3D slopes (Δslope $\leq$ 0.14), the two fragments are seamlessly stitched into a single full-length candidate (creating 102 stitched full-length trajectories across the 54 patients).

3.3.2 *Step 2B: Judging the whole implant bundle together.* After Step 2A, the stitched candidate pool recovers **97.4%** of all real catheters (98.8% of long shafts), and the greedily linked Stage 2A tracks alone (Appendix B, Table 5) still contain an average of 6.6 unmatched tracks per patient. How can the algorithm tell a real catheter apart from a dark vein that happens to look tubular over a few slices? By looking at the **whole bundle at once**: real catheters enter through a common perineal template corridor, run roughly parallel to the central obturator, maintain a consistent spacing, and terminate along a coherent surface covering the tumor.

We summarize each candidate trajectory by a 26-feature token (length, straightness, Stage 0/1/2A confidence percentiles, entry position and tilt relative to the bundle median, and tip contrast gradient) and describe every pair of trajectories by 8 geometric interaction features (minimum 3D clearance, shared-slice separation, angular divergence, and overlap). A **bundle graph transformer** applies two layers of edge-biased local attention between neighboring tracks followed by global self-attention across the entire candidate bundle, outputting a calibrated probability $p_m \in [0, 1]$ that each candidate is a real catheter. Candidates outside a self-calibrating implant radius (98th percentile of high-confidence tracks plus 16 mm) require a higher confidence threshold to pass. To enable zero-shot and few-patient learning, Stage 2B can also be pre-trained on **SynthBrachy3T**—a physics-based generator of 300 simulated candidate bundles containing realistic touching twins, short obturator channels, split fragments, and vascular false alarms (detailed in Appendix A)—before fine-tuning on the clinical training fold.

3.3.3 *Step 2C: Pinpointing the deepest tip, quantifying its uncertainty, and fusing literature priors.* Even when a catheter's shaft is traced to within half a millimeter, finding its exact deepest (cranial) tip on thick-slice T2W MRI is difficult: the dark void tapers off gradually across 2 to 6 mm thick slices, or collides with dark bowel gas just above the uterus.

To resolve the tip with sub-slice precision, we built a **dedicated 1D axial tip localizer** (Figure 2d). Starting at the top of each candidate trajectory and continuing along its 3D tangent direction, the network samples a **91-step 1D profile** from -25.0 mm below the initial tip to $+20.0$ mm above it, in fine 0.5 **mm sub-slice steps**. At every 0.5 mm step, it extracts 22 multi-scale measurements: center and ring T2W intensities, local annular contrast at two radii, first and second axial deriva-

tives (∂_z, ∂_{zz}), Stage 0 symmetry S_0, Stage 1 mask and peak probabilities, obturator probability, and depth relative to neighboring catheter tips. A **3-block dilated 1D ResNet** (kernel size 5, dilations 1, 2, 4) followed by **4-head 1D self-attention** reads this 91-step sequence and predicts both a 91-bin probability distribution over the exact stopping point and a continuous sub-millimeter offset Δs; this predicted offset shifts the trajectory's cranial termination point along its 3D tangent vector before the full 3D curve is smoothed with a cubic polynomial in z.

Crucially, the sharpness (entropy) of this 91-bin tip distribution, combined with the local image contrast drop and slice thickness Δz, provides an immediate out-of-fold **tip uncertainty score** $\hat{\sigma}_{\text{tip}}$. To turn this raw score into a physical error bar in millimeters without assuming Gaussian errors, we use **split-conformal calibration**: on the held-out inner validation patients of each fold, we compute the empirical $(1-\alpha)$ quantile $\hat{q}_{1-\alpha}$ of the normalized non-conformity ratios $r_i = E_{\text{tip},i}/\hat{\sigma}_{\text{tip},i}$. Multiplying any unseen test tip's uncertainty score by $\hat{q}_{1-\alpha}$ yields an adaptive 3D prediction radius $\hat{r}_{1-\alpha} = \hat{q}_{1-\alpha}\,\hat{\sigma}_{\text{tip}}$ that covers the true tip at the target nominal rate (e.g., 90% or 95%) on average, without any Gaussian assumption, provided calibration and test tips are drawn alike. Because tips cluster within patients, we verify this coverage empirically on the outer test folds (Results). The same score ranks which few tips in an implant warrant a quick visual check.

Finally, because recent peer-reviewed MRI brachytherapy studies have proposed complementary post-processing priors—specifically **axial total-variation (TV) continuity** (Dai *et al.*, 2020), **iterative multi-segment gap-bridging** (Kostoulas *et al.*, 2025), and **5-model epistemic uncertainty gating with inverse-variance-weighted polynomial fitting** (Aleong *et al.*, 2024; Kim *et al.*, 2026)—we also tested whether integrating these three ideas directly inside Stage 2 (`4d: NeedleFinder 2026 + Literature Fusion` in Table 2) improves our pipeline even further:

1. **Sub-voxel dark-core and inverse-variance-weighted cubic-polynomial shaft fitting (Aleong *et al.*, 2024; Kim *et al.*, 2026):** On each axial slice z_k along a kept trajectory, we refine the in-plane coordinate toward the precision-weighted sub-voxel centroid of the Stage 1 peak and Stage 0 dark-core response, and weight the cubic polynomial fit by $w_k = (P_{\text{mask}}(z_k) + 0.5\,S_0(z_k))/(0.04 + \sigma^2_{\text{mask}}(z_k))$, where $\sigma_{\text{mask}}(z_k)$ is the voxel-wise epistemic standard deviation across the 5 inner-fold Stage 1 models.
2. **Directional axial TV continuity and epistemic uncertainty features (Dai *et al.*, 2020; Aleong *et al.*, 2024):** We append σ_{mask} and its axial derivative $\partial_z \sigma_{\text{mask}}$ as channels 23 and 24 of the 91-step 1D tip localizer and penalize candidate tracks exhibiting high axial TV jitter or high epistemic uncertainty in Stage 2B.
3. **Iterative multi-segment gap-bridging and polynomial residual split suppression (Kostoulas *et al.*, 2025):** We extend our two-fragment stitcher to iteratively bridge 3-segment trajectories across multi-slice dropouts whenever a single cubic polynomial fits the combined path within a 1.35 mm RMS residual, and suppress lower-confidence collinear sub-segments that lie within 2.6 mm of a longer kept catheter's polynomial fit.

4 EVALUATION PROTOCOL, BASELINES, AND CLINICAL TRANSLATION

4.1 *Strictly Nested* 5 × 5 *Cross-Validation and Baselines*

In small clinical datasets, subtle leaks—such as picking the best training epoch on the test fold, using manual crop boxes, or training a second-stage network on over-fitted first-stage predictions—can inflate reported accuracy. We eliminated all three sources of optimism using a **strictly nested 5 × 5 patient-level cross-validation protocol** across the 54 patients:

1. **Locked outer test folds and disjoint inner validation.** The 54 patients are partitioned into 5 outer folds (43–44 outer-train patients and 10–11 outer-test patients per fold). Inside each outer fold, the outer-train patients are split 80/20 into an inner-training set (34–35 patients) and an inner-validation set (9 patients). Every hyperparameter, early-stopping checkpoint, temperature calibration factor, and conformal quantile is selected *exclusively* on the 9 inner-validation patients. Each outer-test patient is evaluated once.
2. **100% reference-free inference.** Every test volume is cropped solely by the automatic Stage 0 template-plane locator, with zero access to clinical scene files or manual annotations.
3. **Eliminating two-stage stacking bias via inner out-of-fold cross-fitting (4c).** In any two-stage machine learning pipeline, Stage 1 makes slightly cleaner predictions on the patients it was trained on than on brand-new patients. If Stage 2 is trained directly on those slightly optimistic Stage 1 maps (4a and 4b), it might misjudge how noisy Stage 1 will be at test time. To measure how much stacking bias our pipeline carries, we also trained 5 × 5 = 25 inner-fold Stage 1 models so that every training patient receives strictly out-of-fold Stage 1 maps before Stage 2 is trained (reported as configuration 4c in Table 2).

To ensure a thorough, apples-to-apples comparison on the exact same 54-patient nested folds and automatic crop bounds, we re-implemented **five peer-reviewed MRI/curvilinear methods** and **one recent structure-aware tracking preprint** from their published descriptions and trained each under our nested protocol (all six were written in PyTorch from the published descriptions without any original authors' code; the 2019 pipeline is our own earlier method in vectorized form; the numbers below are therefore for our re-implementations):

- **Zaffino *et al.* 2019 (*Physics in Medicine & Biology*, 3D U-Net + DBSCAN) (Zaffino *et al.*, 2019):** Isotropic 1 mm^3 resampling, 3D U-Net voxel segmentation, principal-axis clustering with DBSCAN, and spline fitting.
- **Shit *et al.* 2021 (*CVPR*, ResUNet + clDice + Skeleton) (Isensee *et al.*, 2021; Shit *et al.*, 2021):** A native-resolution 3D residual U-Net trained with focal cross-entropy, soft Dice, and topology-preserving centerline-Dice (clDice) loss, followed by 3D skeletonization and polynomial fitting.
- **Dai *et al.* 2020 (*Medical Physics*, 3D Attention U-Net + Axial TV + cubic fitting) (Dai *et al.*, 2020):** A deeply supervised 3D attention-gated U-Net trained on T2W MRI with multi-scale Dice/cross-entropy and axial total-

variation continuity regularization ($\mathscr{L}_{\text{TV-}z}$), followed by slice-wise centroid linking and cubic polynomial smoothing.

- **Kostoulas *et al.* 2025 (*Proc. SPIE Medical Imaging*, 3D ResUNet + Iterative Gap-Bridging) (Kostoulas *et al.*, 2025):** A native-resolution 3D residual U-Net paired with multi-peak blob splitting, iterative caudo-cranial collinear gap-bridging across segmentation dropouts (≤ 18 mm), false-part pruning, and polynomial fitting.
- **Aleong *et al.* 2024 / Kim *et al.* 2026 (*Medical Physics* / *Brachytherapy*, 5-Model Uncertainty Ensemble + Weighted cubic fitting) (Aleong *et al.*, 2024; Kim *et al.*, 2026):** Within each outer fold, a 5-model inner 3D U-Net ensemble estimates both mean catheter probability $\bar{p}$ and voxel-wise epistemic uncertainty σ_p; candidate tracks are extracted on the uncertainty-penalized map, filtered by ensemble disagreement ($\bar{\sigma}_p/\bar{p}$), and smoothed via inverse-variance-weighted cubic polynomial regression.
- **Xiao *et al.* 2026 (Structure-Aware Sync-Track, preprint; see also peer-reviewed JBHI counterpart (Xiao *et al.*, 2026b)) (Xiao *et al.*, 2026a):** Implant-region gating, direction-constrained 3D Hough transform initialization, synchronous inward slice tracking, and cubic polynomial smoothing, evaluated both on out-of-fold MRI segmentation masks and on ground-truth masks (as an oracle upper bound).

4.2 *Geometric, Observer, Dosimetric, and Generalization Metrics*

Geometric matching and error metrics. For each patient, predicted and reference trajectories are matched one-to-one using the Hungarian algorithm, forbidding pairs whose mean in-plane distance across shared slices exceeds 3.5 mm. On matched pairs, we measure **in-plane shaft error** (E_{shaft}, across 40 equidistant shared-slice stations), **3D symmetric curve distance** (D_{avg}), **95th-percentile Hausdorff distance** (HD_{95}), and **3D distal tip error** (E_{tip}, Euclidean distance between predicted and reference cranial endpoints). Our algorithm reconstructs *all* visible plastic tubes in the scan—both the 759 full-depth treatment shafts and the 304 shorter guide tubes inside the vaginal obturator. When we evaluate recall on the 759 full-depth shafts alone, an algorithm prediction that accurately traces one of the 304 real obturator tubes is not a false alarm on a blood vessel; we therefore report **obturator-adjusted precision (P_{adj}) and $F1_{\text{adj}}$**, which avoid penalizing valid detections of those secondary tubes. While Table 2 summarizes pooled error distributions across all matched catheters, all paired two-sided Wilcoxon signed-rank tests are conducted at the **patient level** ($N = 54$ paired patient means, or $N = 23$ in the adhesion subgroup) to respect within-patient clustering (assigning fixed worst-case penalties of 3.5 mm shaft, 10 mm curve, and 20 mm tip/HD_{95} if a baseline yields zero matches on a scan), complemented by 1,000-resample patient-level bootstrap 95% confidence intervals. We report all p-values uncorrected for the roughly forty comparisons in this paper. The main benchmark differences ($p \leq 1.1 \times 10^{-5}$) survive any standard correction; the smaller effects at $p \approx 0.01$–0.05 (literature fusion versus the base pipeline,

Human Team 2 under Protocol A, and the flipped-reference tip test) would not, and we present them as suggestive.

Three-cornered-hat observer adjudication and non-inferiority testing. If an algorithm disagrees with Human Team 1 by 0.5 mm, how much of that discrepancy is the algorithm's error and how much is Team 1's own hand-clicking jitter? With only two observers, it is impossible to tell. However, when *three* sources trace the same catheter—Team 1, Team 2, and out-of-fold NeedleFinder 2026 ($n = 432$ three-way matched catheters)—we observe three pairwise difference variances (V_{12}, V_{13}, V_{23}) for three unknown internal error variances ($\sigma_1^2, \sigma_2^2, \sigma_3^2$). **Grubbs' (1948) three-cornered-hat estimator** solves this 3 × 3 linear system ($\hat{\sigma}_1^2 = \frac{1}{2}(V_{12} + V_{13} - V_{23})$) to isolate each source's intrinsic, reference-free error standard deviation, which we compute via classic maximum likelihood (RMS), outlier-resistant median absolute deviation (MAD), and 15% Winsorized estimators. The estimator assumes that the three sources err independently. This holds only approximately here: NeedleFinder is trained, in other folds, on traces from both teams, so its errors may share their habits. A violated assumption can push one estimated variance below zero, which we then report as 0.00 mm.

We also test whether the algorithm's disagreement with Team 1 is statistically non-inferior to Team 2's disagreement with Team 1 ($H_0 : E_{\mathrm{NF,Team\,1}} - E_{\mathrm{Team\,2,Team\,1}} \geq \Delta$ at one-sided significance level $\alpha = 0.05$). Pre-specified clinical non-inferiority margins were set to $\Delta = 0.25$ mm on shafts (roughly half the finest pixel spacing) and $\Delta = 1.00$ mm (median) / 1.50 mm (mean) on 3D tips (corresponding to half a slice thickness and the 1–1.5 mm TG-43U1 dosimetric sensitivity threshold (Tanderup *et al.*, 2008; Schindel *et al.*, 2013)). We evaluate non-inferiority at both the patient level ($N = 40$) and the catheter level ($n = 432$) using one-sided paired Wilcoxon tests and two-sided 90% ($1 - 2\alpha$) patient-clustered bootstrap confidence intervals (10,000 resamples). Finally, whenever the two human teams disagree on a catheter tip by ≥ 5.0 mm on MRI ($n = 70$ catheters across 38 patients with paired CT), we use the patient's post-implant pelvic CT scan as an independent physical tie-breaker. One caveat applies to every comparison against Team 1: NeedleFinder learned from Team 1's traces on other patients, whereas Team 2 did not, so part of the algorithm's closer agreement with Team 1 may reflect a shared tracing convention rather than higher accuracy. The CT adjudication is the one check that depends on neither team.

AAPM TG-43U1 HDR ^{192}Ir dosimetry and simulated 1-click tip triage. To translate millimeter geometry errors into clinical radiation dose differences across all 54 patients without confounding from inter-physician contouring variability (and because historical DICOM RTSTRUCT/RTPLAN files were not archived), we simulated HDR ^{192}Ir delivery using the AAPM TG-43U1 line-source formalism on standardized, implant-conformal anatomical envelopes. Each catheter was loaded with 5.0 mm dwell steps along its cranial 45 mm (starting 3.5 mm below the physical tip, active source length $L = 3.5$ mm) on a 1.5 mm isotropic 3D dose grid. A single-fraction prescription of $D_{90} = 7.0$ Gy was optimized via non-negative least-squares onto the implant-conformal high-risk target volume ($\mathrm{HR\text{-}CTV}_{\mathrm{implant}}$, defined as the 8.0 mm radial envelope around the active reference implant) while con-

straining 2 cm^3 proxy shells for the anterior bladder, posterior rectum, and cranial sigmoid/fundal organ-at-risk (OAR) zones. We evaluated two clinical protocols:

- **Protocol A (Fixed-Plan Transfer):** Reference dwell times optimized on the expert catheters are transferred onto the one-to-one matched predicted catheters (with missed reference catheters contributing zero dwell delivery) to isolate pure geometric-to-dosimetric sensitivity.
- **Protocol B (Counterfactual Plan Re-Optimization):** Modeling the end-to-end clinical planning workflow, dwell times are optimized from scratch on *all* algorithm-predicted catheters, including any false-positive tracks that a physicist would normally delete before planning, and the resulting dose is evaluated on the reference target envelope and OAR proxy shells.
- **Simulated 1-click clinician tip review:** Because 98.8% of tip error is longitudinal along the already-reconstructed 3D shaft, we also model an interactive verification step in which the physicist inspects the top $k \in \{1, \dots, 10\}$ tips per patient ranked by out-of-fold uncertainty $\hat{\sigma}_{\text{tip}}$ and clicks the true distal termination along the shaft (modeled by the retrospective 1-click oracle $E_{\text{tip}} \to \sqrt{\Delta r_{\perp}^2 + \min(|\Delta s_{\parallel}|, 0.85\ \text{mm})^2}$ whenever $E_{\text{tip}} > 2.0$ mm, leaving E_{shaft} unchanged, at an assumed, untimed cost of 8 s per flagged tip in 3D Slicer plus 2 s to open the review list). Only the $k = 2$ configuration was carried through the dose engine, where each flagged tip is moved along its shaft to the reference tip depth; larger k were evaluated on tip geometry alone.

Computing environment. All training and benchmarks ran on a Linux workstation equipped with a 24-core workstation CPU (AMD Ryzen Threadripper PRO 7965WX, 48 threads, 256 GB RAM) and an NVIDIA RTX 4090 GPU (24 GB). For clinical deployment, the complete pipeline is packaged as a one-click module in **3D Slicer 5.10** (Fedorov *et al.*, 2012) using native Rust (Stage 0) and ONNX Runtime (Stages 1 and 2), requiring no GPU.

5 RESULTS

5.1 *Overall Detection and Geometric Accuracy Against Published Methods*

Table 2 compares NeedleFinder 2026 against all five peer-reviewed MRI/curvilinear baselines and the Xiao *et al.* 2026 tracking preprint across all 54 patients (1,063 catheters), on the 759 full-depth active-dwell shafts, and on the 23-patient severe catheter-adhesion subgroup (with the full 12-stage architectural ablation ladder reported in Appendix B, Table 5). Figure 3 displays the complete per-catheter error distributions.

Four clear lessons emerge from the benchmark:

1. Pure voxel segmentation fragments catheters and hallucinates vessels. The native-resolution `ResUNet + clDice` baseline (Shit *et al.*, 2021) detects nearly every catheter location (92.4% recall) and traces matched segments accurately (0.43 mm median shaft error). However, because single-slice signal dropouts break catheter

TABLE 2 **Catheter reconstruction accuracy on 54 BWH patients (1 063 catheters), strictly nested 5 × 5 patient-level out-of-fold.** Compares **NeedleFinder 2026** against five peer-reviewed journal/conference methods and one recent preprint reproduced on the identical nested splits across the whole cohort, the 759 long interstitial shafts, and the severe catheter-adhesion subgroup (see Appendix B, Table 5, for the complete 11-row stage-by-stage ablation ladder). Bold: best; underline: second best within each block. Shaded: **NeedleFinder 2026** variants (pre-specified $\tau = 50\,\%$).

	Detection, 1-to-1 (%)			E_{shaft} (mm)			E_{tip} (mm)		Time (s)
Method	Rec.	Prec.	F1	Mean	Median	P95	Mean	Median	CPU
Whole cohort (54 patients, 1 063 catheters)									
PMB 2019 3D U-Net (Zaffino *et al.*)[a]	88.8	78.0	83.1	**0.52**	**0.42**	**1.11**	10.75	3.39	14.80
ResUNet + clDice + Skel. (Shit *et al.* 2021)[b]	92.4	24.5	38.7	0.55	0.43	1.36	10.08	3.10	3.85
Dai *et al.* 2020 (3D Att. U-Net + Axial TV)[c]	**93.6**	42.2	58.2	0.57	0.43	1.45	10.61	3.77	4.05
Kostoulas *et al.* 2025 (ResUNet + Gap-Bridge)[d]	89.5	53.0	66.5	0.57	0.43	1.45	10.55	3.47	3.92
Aleong 2024 / Kim 2026 (5-Model Unc. Ens.)[e]	89.2	56.9	69.5	0.58	0.44	1.42	10.23	4.32	11.40
Xiao *et al.* 2026 Sync-Track (*preprint*)[f]	87.7	39.2	54.2	0.66	0.47	1.82	15.65	6.04	**3.42**
NeedleFinder 2026 (no SynthBrachy3T, 4a)[g]	92.5	85.3	88.7	0.59	0.47	1.35	6.39	2.26	4.15
NeedleFinder 2026 + Synth + 1DTip (4b)[h]	91.4	86.9	89.1	0.59	0.47	1.37	**6.23**	2.27	4.15
NeedleFinder 2026 + Inner-OOF Cross-Fit (4c)[h]	92.3	86.1	89.1	0.59	0.47	1.37	6.25	**2.24**	4.15
NeedleFinder 2026 + Lit. Fusion (4d)[i]	91.4	**87.4**	**89.4**	0.58	0.47	1.36	**6.23**	2.26	4.18
Long interstitial shafts (54 patients, 759 catheters, obturator-adjusted precision)[j]									
PMB 2019 3D U-Net (Zaffino *et al.*)	86.7	87.6	87.2	**0.52**	**0.40**	**1.23**	14.93	4.04	–
ResUNet + clDice + Skel. (Shit *et al.* 2021)	91.3	51.5	65.8	0.57	0.43	1.40	14.67	3.53	–
Dai *et al.* 2020 (3D Att. U-Net + Axial TV)	84.6	83.0	83.8	0.58	0.41	1.74	12.53	4.38	–
Kostoulas *et al.* 2025 (ResUNet + Gap-Bridge)	86.7	79.1	82.7	0.56	**0.40**	1.50	11.14	3.77	–
Aleong 2024 / Kim 2026 (5-Model Unc. Ens.)	85.5	80.5	82.9	0.58	0.41	1.70	11.73	4.50	–
Xiao *et al.* 2026 Sync-Track (*preprint*)	84.2	53.7	65.6	0.64	0.46	1.84	16.02	6.90	–
NeedleFinder 2026 (no SynthBrachy3T, 4a)	**93.4**	87.1	90.2	0.56	0.45	1.24	7.47	2.49	–
NeedleFinder 2026 + Synth + 1DTip (4b)	91.3	**88.7**	90.0	0.57	0.45	1.27	7.41	**2.41**	–
NeedleFinder 2026 + Inner-OOF Cross-Fit (4c)	92.5	88.6	**90.5**	0.56	0.45	1.27	**7.17**	2.44	–
NeedleFinder 2026 + Lit. Fusion (4d)	91.3	**88.7**	90.0	0.56	0.44	1.25	7.40	2.43	–
Severe catheter-adhesion subgroup (23 patients, 571 catheters)[k]									
PMB 2019 3D U-Net (Zaffino *et al.*)	94.6	79.3	86.3	0.54	0.43	1.22	11.88	3.70	–
ResUNet + clDice + Skel. (Shit *et al.* 2021)	96.0	32.3	48.4	**0.52**	0.42	**1.18**	11.58	3.37	–
Dai *et al.* 2020 (3D Att. U-Net + Axial TV)	**96.2**	52.0	67.5	0.53	0.42	1.23	11.88	4.20	–
Kostoulas *et al.* 2025 (ResUNet + Gap-Bridge)	94.4	67.4	78.6	0.54	**0.41**	1.37	11.26	3.61	–
Aleong 2024 / Kim 2026 (5-Model Unc. Ens.)	93.9	72.2	81.7	0.55	0.43	1.27	10.97	4.37	–
Xiao *et al.* 2026 Sync-Track (*preprint*)	91.6	49.0	63.9	0.64	0.47	1.83	17.69	7.52	–
NeedleFinder 2026 (no SynthBrachy3T, 4a)	94.6	87.2	90.8	0.58	0.45	1.36	**7.03**	2.48	–
NeedleFinder 2026 + Synth + 1DTip (4b)	93.2	**89.6**	91.3	0.58	0.46	1.41	7.08	2.51	–
NeedleFinder 2026 + Inner-OOF Cross-Fit (4c)	94.6	89.1	**91.8**	0.57	0.46	1.34	7.30	2.47	–
NeedleFinder 2026 + Lit. Fusion (4d)	93.2	**89.6**	91.3	0.57	0.46	1.40	7.08	**2.44**	–

Rec./Prec./F1: strict one-to-one Hungarian matching (≤ 3.5 mm shaft threshold); E_{shaft}: in-plane centerline distance on matched slices; E_{tip}: 3D distal-tip distance (P95 on whole cohort: 27.27 mm for **NeedleFinder 2026** 4b and 27.42 mm for 4d vs. 43.23 mm for Aleong/Kim, 45.76 mm for clDice, 46.68 mm for Dai 2020, 50.23 mm for Kostoulas 2025, 52.70 mm for PMB 2019, and 62.69 mm for Xiao 2026).

[a] Peer-reviewed MRI baseline (Zaffino *et al.* 2019, *Phys. Med. Biol.*): $1 \times 1 \times 1\,mm^3$ isotropic 3D U-Net, DBSCAN clustering, and polynomial fitting.

[b] Peer-reviewed tubular topology baseline (Shit *et al.* 2021, *CVPR*): native-resolution 3D ResUNet trained with Dice + focal + soft-skeleton clDice loss, followed by ridge extraction and chaining.

[c] Peer-reviewed MRI baseline (Dai *et al.* 2020, *Med. Phys.*): deeply supervised 3D attention-gated U-Net trained with multi-scale Dice/BCE and weighted axial total-variation (TV) continuity regularization, followed by slice-to-slice centroid linking and cubic polynomial smoothing.

[d] Peer-reviewed MRI baseline (Kostoulas *et al.* 2025, *Proc. SPIE Med. Imaging*): native 3D ResUNet with multi-peak blob splitting, iterative collinear gap-bridging across discontinuous MRI slices (≤ 18 mm gap), false-fragment pruning, and 3rd-order polynomial fitting.

[e] Peer-reviewed MRI ensemble baseline (Aleong *et al.* 2024, *Med. Phys.* / Kim *et al.* 2026, *Brachytherapy*): 5-model inner-

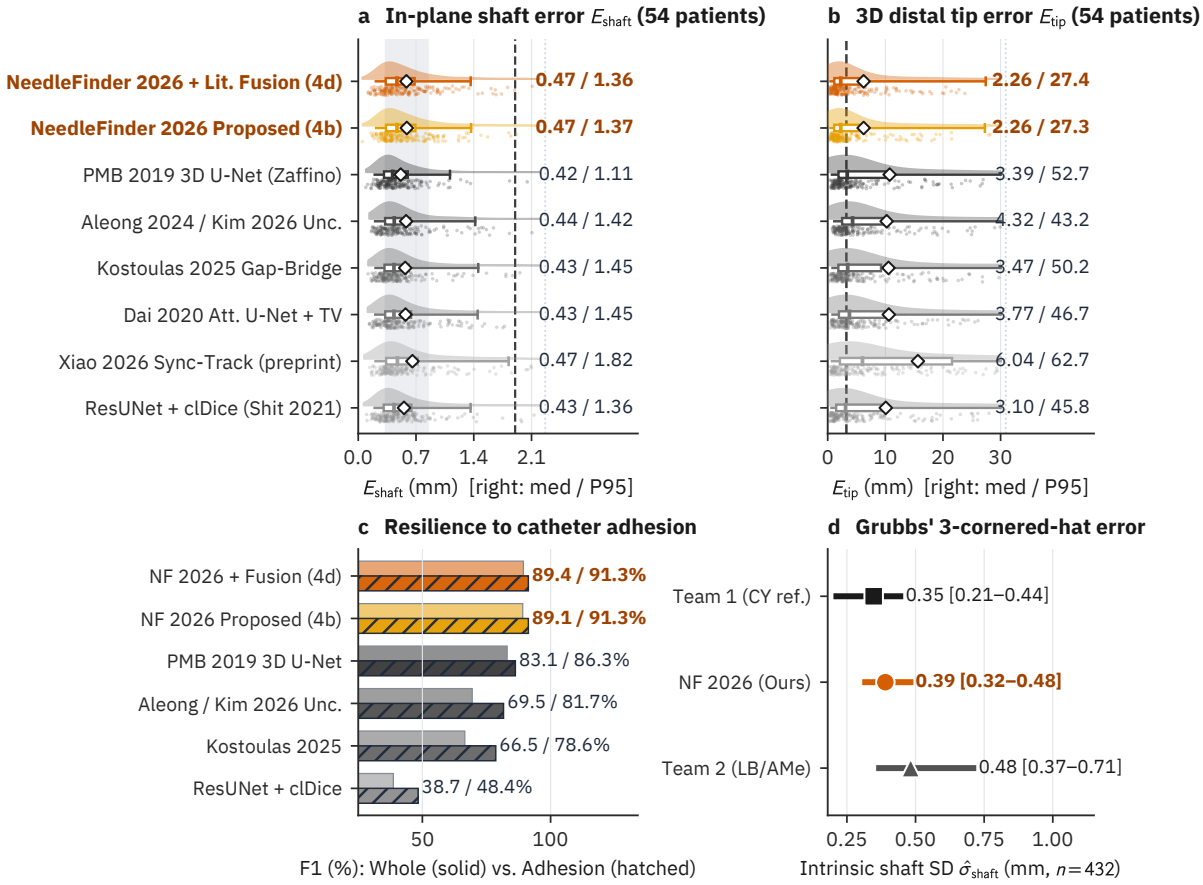


FIGURE 3 **Per-catheter shaft and tip error distributions, adhesion resilience, and Grubbs' intrinsic observer error. (a)** In-plane shaft error (E_{shaft}) and **(b)** 3D distal tip error (E_{tip}) across all 54 patients for each method's matched catheters. Each raincloud shows the full-distribution probability density (top), median and interquartile range (white box), 5th-to-95th percentile span (whiskers), mean (white diamond), and individual catheters (dots); the dedicated right-hand tabular column reports `median / 95th percentile` in mm across the full unclipped distribution. Shaded grey bands and dashed vertical lines indicate the human inter-observer agreement between Expert Team 1 and Expert Team 2 on the 40 dual-traced scans (IQR 0.34–0.85 mm and P95 1.90 mm on shafts; median 3.24 mm on tips). **(c)** Whole-cohort F1 (solid bars, 1,063 catheters) versus 23-patient severe catheter-adhesion subgroup F1 (hatched bars, 571 catheters). **(d)** Reference-free Grubbs' (1948) three-cornered-hat intrinsic shaft error standard deviation $\hat{\sigma}_{shaft}$ (with 10,000-resample bootstrap 95% confidence intervals) across $n = 432$ three-way matched catheters.

masks into pieces and dark pelvic veins are segmented as tubes, it outputs 4,015 fragments across the 54 scans—yielding a precision of only 24.5% and an F1 score of 38.7%.

2. Domain-specific MRI post-processing (axial TV, gap-bridging, and uncertainty ensembles) cuts false tracks by more than half, but cannot replace physical symmetry or bundle reasoning. The three recent peer-reviewed MRI brachytherapy methods substantially improve upon raw clDice skeletonization on native-resolution T2W volumes:

– **Dai *et al.* 2020** (3D Attention U-Net + axial TV regularization (Dai *et al.*, 2020)) achieves the highest raw baseline recall (**93.6%**, 995/1,063; 0.43 mm me-

dian shaft error, 3.77 mm median / 10.61 mm mean tip error) and cuts output fragments from 4, 015 to 2, 358, raising F1 to **58.2%** (83.8% $F1_{adj}$ on long shafts).

- **Kostoulas *et al.* 2025** (3D ResUNet + iterative collinear gap-bridging and false-part pruning (Kostoulas *et al.*, 2025)) reconnects broken shaft segments across slice dropouts and prunes isolated fragments, lifting whole-cohort precision to 53.0% and F1 to **66.5%** (89.5% recall, 0.43 mm median shaft error, 3.47 mm median / 10.55 mm mean tip error).
- **Aleong *et al.* 2024 / Kim *et al.* 2026** (5-model epistemic uncertainty ensemble + disagreement filtering and inverse-variance-weighted cubic fitting (Aleong *et al.*, 2024; Kim *et al.*, 2026)) further suppresses high-uncertainty vascular false alarms, reaching **69.5% whole-cohort F1** (89.2% recall, 56.9% precision; **81.7% F1** in the severe-adhesion subgroup; 0.44 mm median shaft error, 4.32 mm median / 10.23 mm mean tip error).
- Meanwhile, when given ground-truth oracle masks (Appendix B, Table 5, bottom row), the **Xiao *et al.* 2026 Sync-Track** algorithm (Xiao *et al.*, 2026a) achieves 97.7% recall, 90.0% F1, and 0.11 mm median shaft error—indicating that our re-implementation of synchronous tracking separates touching masks when segmentation is perfect—but drops to **54.2% F1** (39.2% precision) on real out-of-fold MRI masks. Thus, without Stage 0 dark-core symmetry and Stage 2B whole-bundle context, the three recent native-resolution post-processing pipelines and mask-based tracking remain below 70% whole-cohort F1 because dark pelvic veins inside the implant region look locally tubular, while the isotropic 2019 U-Net + DBSCAN pipeline reaches 83.1% (Table 2). The gap is one of precision, not recall: the three newer re-implementations find as many catheters as the 2019 pipeline or more (89.2% to 93.6% vs. 88.8% recall) but emit 1.4 to 1.9 times as many tracks (1,667 to 2,358 vs. 1,210 predictions) from out-of-fold voxel masks of only 0.39 to 0.43 Dice, so their precision falls to 42% to 57% against 78%. Two caveats apply: the 2019 pipeline is our own earlier method, whose isotropic resampling and DBSCAN minimum-cluster rule were developed on this cohort, and the newer methods were re-implemented from their papers without their authors' tuning.

3. NeedleFinder 2026 achieves the highest F1 score and lowest tip error of any method run on real out-of-fold MRI, in every cohort; adding three literature priors gives small further gains. At the default operating threshold (τ = 50%), NeedleFinder 2026 (base configuration, Table 2) detects **91.4% of all 1,063 catheters** (972/1, 063) with **86.9% precision** (972/1, 118) and **89.1% F1** (lowering the threshold to τ = 35% raises recall to 93.6% but lowers precision to 83.4% and F1 to 88.2%). Furthermore, incorporating the three complementary ideas from Dai *et al.* (2020), Kostoulas *et al.* (2025), and Aleong *et al.* / Kim *et al.* (2024; 2026) directly into Stage 2 (`4d: NeedleFinder 2026 + Literature Fusion`) eliminates 6 residual split-fragment false positives with zero lost true positives, lifting precision to **87.4%** (972/1, 112) and whole-cohort F1 to **89.4%** (a 0.3-point gain, uncorrected p = 0.028 vs. base `4b`). Across all 54 patients, NeedleFinder 2026 (`4b`–`4d`) outperforms all six external baselines by **+6.0 to +50.7 F1 points**: +6.0–6.3 points over Zaffino *et al.*

2019 (83.1% F1, $p = 1.1 \times 10^{-5}$), +19.6–19.9 points over Aleong *et al.* / Kim *et al.* (69.5% F1, $p = 5.0 \times 10^{-10}$), +22.6–22.9 points over Kostoulas *et al.* 2025 (66.5% F1, $p = 1.8 \times 10^{-10}$), +30.9–31.2 points over Dai *et al.* 2020 (58.2% F1, $p = 3.8 \times 10^{-9}$), +34.9–35.2 points over Xiao *et al.* 2026 (54.2% F1, $p = 3.0 \times 10^{-10}$), and +50.4–50.7 points over ResUNet + clDice (38.7% F1, $p = 5.2 \times 10^{-10}$).

Along the catheter shaft, NeedleFinder 2026 reaches a **median in-plane error of** 0.47 **mm** (mean 0.59 mm in 4b; inverse-variance-weighted sub-voxel fitting in 4d lowers the mean by less than 0.01 mm to 0.58 mm and the 95th percentile to 1.36 mm, a consistent but practically negligible change, $p = 1.8 \times 10^{-8}$). At the deepest tip, our 1D axial tip localizer and fragment stitcher cut **median 3D tip error to** 2.26–2.27 **mm** (1.94 mm median axial tip error, mean 6.23 mm), reducing mean tip error by 39%–42% compared with Zaffino *et al.* 2019 (10.75 mm mean / 3.39 mm median, $p = 1.7 \times 10^{-8}$), Dai *et al.* 2020 (10.61 mm mean / 3.77 mm median, $p = 2.7 \times 10^{-7}$), Kostoulas *et al.* 2025 (10.55 mm mean / 3.47 mm median, $p = 6.9 \times 10^{-7}$), and Aleong *et al.* / Kim *et al.* (10.23 mm mean / 4.32 mm median, $p = 3.3 \times 10^{-7}$), and by 60% compared with Xiao *et al.* 2026 (15.65 mm mean / 6.04 mm median, $p = 6.2 \times 10^{-10}$). Nearly three-fifths (**58.7%–59.0%**) of matched catheters have a 3D tip error within ≤ 3.0 mm (roughly one slice thickness). On the **759 full-depth active-dwell shafts**, NeedleFinder 2026 reaches **91.3% recall**, **88.7%–89.0% obturator-adjusted precision**, and **90.0%–90.1%** $F1_{adj}$, with a median shaft error of 0.44–0.45 **mm** (mean 0.56–0.57 mm) and a median tip error of 2.39–2.41 **mm**.

4. The advantage is largest when catheters touch, and inner cross-fitting shows no measurable stacking bias. On the **23-patient severe catheter-adhesion subgroup** (571 catheters), where touching masks cause DBSCAN clustering and greedy trackers to merge or swap paths, NeedleFinder 2026 reaches **91.3% F1** (93.2% recall, 89.6% precision, 0.46 mm median shaft error, 2.50–2.51 mm median tip error), outperforming Zaffino *et al.* 2019 (86.3% F1, $p = 0.0012$), Aleong *et al.* / Kim *et al.* (81.7% F1, $p = 2.9 \times 10^{-5}$), Kostoulas *et al.* 2025 (78.6% F1, $p = 1.9 \times 10^{-6}$), Dai *et al.* 2020 (67.5% F1, $p = 5.2 \times 10^{-5}$), Xiao *et al.* 2026 (63.9% F1, $p = 4.8 \times 10^{-7}$), and ResUNet + clDice (48.4% F1, $p = 4.8 \times 10^{-7}$). Moreover, training Stage 2 on strictly out-of-fold inner Stage 1 maps (4c in Table 2) yields the exact same **89.1% whole-cohort F1** (92.3% recall) and even slightly higher accuracy on long shafts (**90.5%** $F1_{adj}$) and touching catheters (**91.8% F1**), showing that two-stage stacking optimism does not inflate our headline results.

5.2 *How Each Architectural Stage Contributes, and Learning from Few Patients*

Why every stage is needed (Appendix B, Table 5; Figure 4a). Tracing the sequential stage progression (detailed row-by-row in Appendix B, Table 5) shows how each mechanism resolves a specific physical failure mode of the stage before it: the parameter-free Stage 0 dark-core filter alone reaches **75.3% F1** (82.8% recall, 0.47 mm median shaft error, 9.91 mm mean tip error); adding the self-supervised, multi-rater Stage 1 3D U-Net lifts recall to **94.2%** (97.0% on long shafts) while emitting 11.6 vascular false alarms per scan (74.4% F1); Stage 2A triplet message passing suppresses 43% of false tracks via collinearity (**82.5% F1**); Stage 2B whole-bundle graph scor-

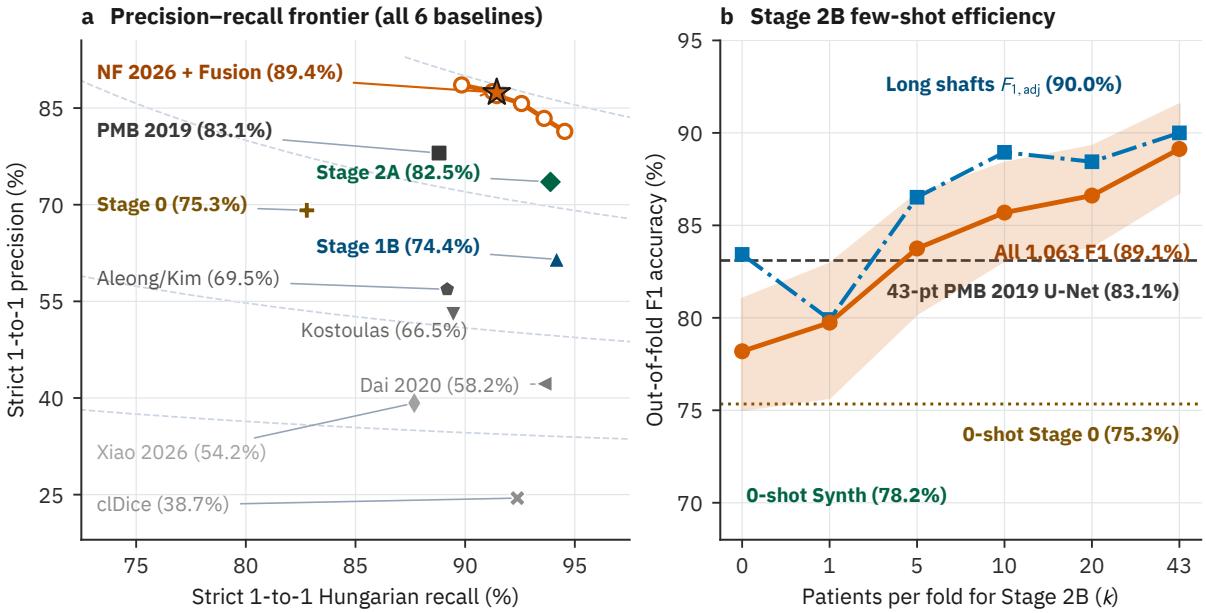


FIGURE 4 **(a) Precision–recall progression** across pipeline stages, published baselines, and the NeedleFinder 2026 confidence threshold sweep on all 54 patients. Dashed grey curves denote constant F1 scores; the vermilion star marks the default threshold $\tau = 50\%$ with literature fusion (89.4% F1). **(b) Leak-free clinical label efficiency:** F1 score (with 95% bootstrap confidence bands) as a function of the number of annotated clinical training patients ($k \in \{0, 1, 5, 10, 20, 43\}$) used to fine-tune the Stage 2B bundle transformer.

ing and collinear gap stitching raise precision to **86.9%** and F1 to **89.1%**; Stage 2C 1D sub-slice tip profiling (4b) cuts mean 3D tip error from 6.92 mm to 6.23 **mm** (2.27 mm median); and Literature-Inspired Fusion (4d (Dai *et al.*, 2020; Aleong *et al.*, 2024; Kostoulas *et al.*, 2025; Kim *et al.*, 2026)) refines mean shaft error to 0.583 **mm** ($p = 1.84 \times 10^{-8}$), median tip error to 2.26 **mm**, and precision to **87.4%** (**89.4% F1**, $p = 0.028$).

How many annotated clinical patients are needed to train the bundle scorer? (Figure 4b). Because annotating 3D catheter bundles is labor-intensive for a new clinic, we varied the number of annotated training patients $k \in \{0, 1, 5, 10, 20, 43\}$ used to fine-tune Stage 2B inside each fold (with zero blending from any full-cohort model). With **zero clinical bundle labels** ($k = 0$), pre-training Stage 2B purely on our synthetic `SynthBrachy3T` bundles lifts whole-cohort F1 from 75.3% (Stage 0) to **78.2% F1**—and reaches **83.4%** $F1_{\mathrm{adj}}$ **on long shafts** (+6.3 points over Stage 0). Fine-tuning on just $k = 1$ **patient** reaches **79.7% F1**; $k = 5$ **patients reach 83.8% F1** (86.5% **on long shafts), already matching the 43-patient fully supervised PMB 2019 U-Net baseline** (83.1% **F1; a 0.7-point difference we do not claim as significant)**; $k = 10$ reaches **85.7% F1**; $k = 20$ reaches **86.6% F1**; and all $k = 43$ training patients reach **89.1% F1** (88.7% without `SynthBrachy3T`, showing that synthetic bundle pre-training provides its largest boost in the zero-to-few-patient regime while still improving dosimetric tip accuracy at full training size).

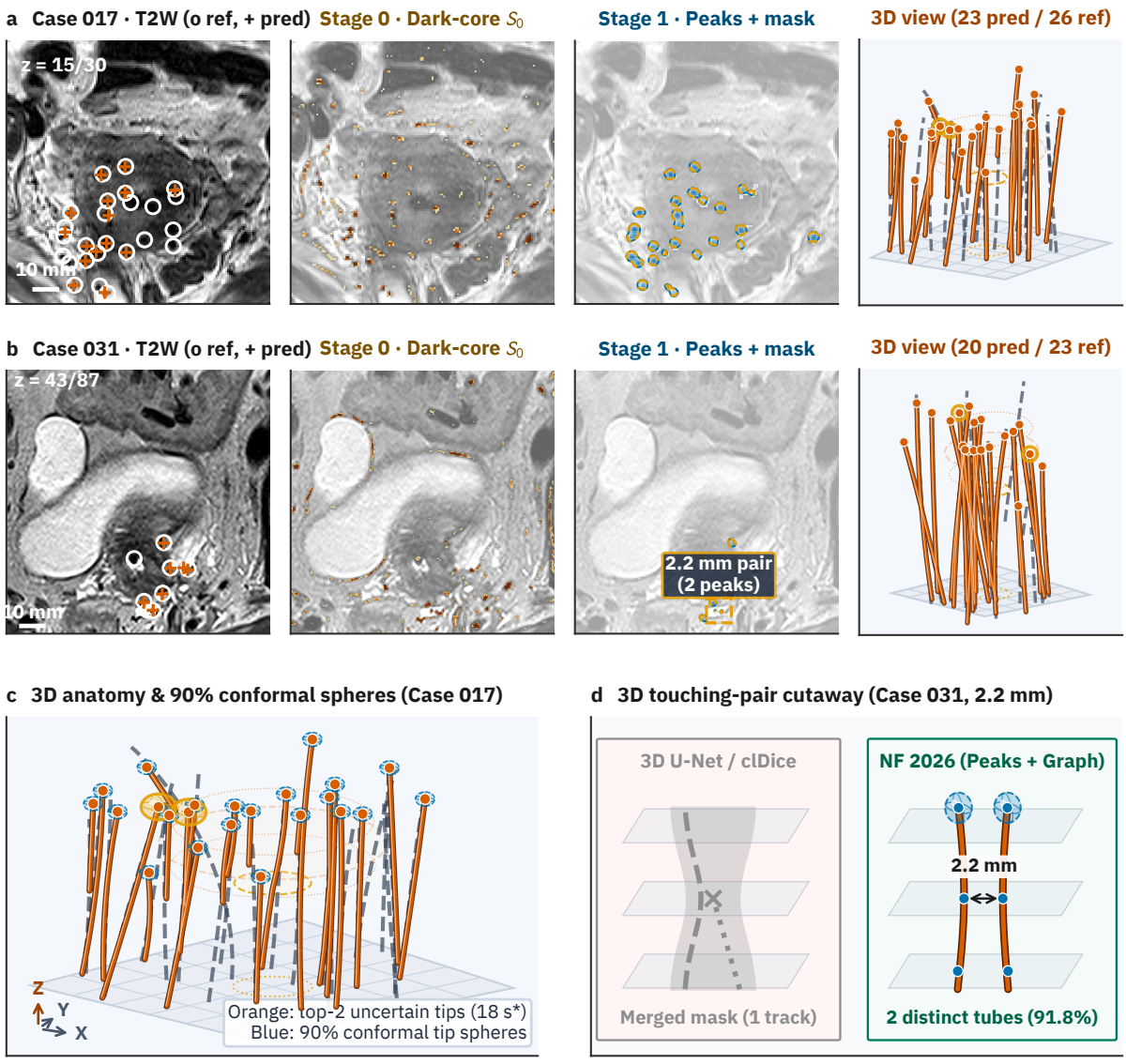


FIGURE 5 **Step-by-step 2D slice progression and true 3D perspective anatomical reconstruction on two unseen test patients. (a) Case 017:** A dense implant with 26 reference trajectories. **(b) Case 031:** A severe-adhesion implant with 23 catheters, 10 of which touch a neighbor within ≤ 3 mm (highlighted by the 2.2 mm touching-pair callout box resolved into 2 distinct centerline peaks). From left to right in **(a, b)**: axial T2W MRI slice showing expert intersections (white circles) and out-of-fold NeedleFinder 2026 detections (vermilion crosses); the Stage 0 circular symmetry map S_o; Stage 1 sub-pixel centerline peaks (blue) with mask contours (solid) and the central obturator gate (dashed orange); and the 3D perspective bundle (vermilion tubes) overlaid on the expert reference (charcoal dashed). **(c) True 3D perspective anatomy and 90% split-conformal tip spheres on Case 017:** Shaded 3D rendering of the implant-conformal HR-CTV target envelope (rose wireframe), central vaginal obturator cylinder, perineal template grid, 26 expert reference tracks vs. 23 predicted 3D tubes, and adaptive 3D split-conformal 90% tip spheres ($\hat{r}_{90} = \hat{q}_{90}\hat{\sigma}_{\text{tip}}$), highlighting the two widest-uncertainty review targets (#7 and #19, magenta spheres) flagged for the 18 s simulated clinician check. **(d) 3D perspective touching-pair cutaway on Case 031 (2.2 mm separation):** Whereas a 3D U-Net / clDice voxel mask merges across the shared axial slices into a single fused track (left, grey), NeedleFinder 2026 sub-pixel centerline peaks (P_{peak}) and triplet graph tracking resolve two distinct 3D tubes across the full insertion depth (right, vermilion + sky-blue; 91.8% F1 on the adhesion subgroup). *(An interactive 3D rotatable viewer for Case 017 and Case 031 is embedded in the HTML companion.)*

Figure 5 shows how these stages work visually in both 2D axial cross-sections and true 3D anatomical space on two challenging out-of-fold test cases: while Stage 0 lights up both catheters and surrounding pelvic vessels, Stage 1 isolates sharp individual peaks even where adjacent catheters touch (2.2 mm apart in `Case 031`, Figure 5d), and Stage 2 reconstructs clean, smooth 3D bundles equipped with calibrated 3D split-conformal tip spheres ($\hat{r}_{90}$) that visually guide rapid clinician verification (Figure 5c).

5.3 *Is the Algorithm as Accurate as a Second Human Expert? (Three-Cornered-Hat & CT Adjudication)*

Because a 2 mm catheter is only a few pixels wide and slices are up to 6 mm thick, two human experts tracing the same MRI scan never agree perfectly (S *et al.*, 2009; Kallis *et al.*, 2019; Karius *et al.*, 2024). A common benchmark for an automatic method is that its discrepancies from a human expert be no larger than the discrepancies *between* two independent human experts (Hrinivich *et al.*, 2019; Andersén *et al.*, 2020); meeting it is necessary for clinical use but does not by itself establish it. Table 3 tests this across **all 40 doubly traced clinical scans** using four complementary analyses:

1. Direct comparison against human inter-observer agreement (Table 3, top two blocks). When Expert Team 2 (460 long traces) is compared directly against Expert Team 1 (505 long traces) across the 40 scans, 424 long catheters match within 3.5 mm (84.0% recall, 92.2% precision, 87.9% F1). Between these two human teams, the **in-plane shaft discrepancy is** 0.49 **mm median** (0.70 ± 0.58 mm mean, 1.90 mm P95) and the **3D tip discrepancy is** 3.24 **mm median** (6.28 ± 7.29 mm mean, 21.88 mm P95).

Evaluated strictly out-of-fold on the exact same 40 scans against Team 1 (723 tracks), **NeedleFinder 2026 achieves a smaller shaft discrepancy (**0.47 **mm median,** 0.58 ± 0.39 **mm mean,** 1.33 **mm P95) and a smaller 3D tip discrepancy (**2.10 **mm median,** 5.26 ± 8.78 **mm mean,** 19.99 **mm P95) than the disagreement between the two human teams** (the human comparison is restricted to long shafts while the algorithm's includes obturator tracks; the paired comparison on identical catheters follows below), while matching 90.3% of Team 1's trajectories (87.6% F1). Furthermore, when evaluated against the **union of both human teams** (759 distinct trajectories), **16 NeedleFinder predictions that looked like "false alarms" against Team 1 alone turn out to be real catheters independently traced by Team 2**, lifting precision by +2.1 percentage points to 87.1%.

2. Reference-free Grubbs' three-cornered-hat intrinsic error estimation (Table 3, third block). To measure each source's true internal error without treating Team 1 as infallible, we applied Grubbs' three-cornered-hat estimator across the $n = 432$ catheters matched three-way among Team 1, Team 2, and NeedleFinder 2026 (formed by intersecting the 476 all-catheter human pairs with out-of-fold NeedleFinder predictions within 3.5 mm). The intrinsic outlier-resistant (MAD-scaled) in-plane shaft standard deviations are 0.35 **mm for Human Team 1**, 0.39 **mm for NeedleFinder 2026**, and 0.48 **mm for Human Team 2**. For axial tip stopping

TABLE 3 **Human inter-observer agreement versus NeedleFinder 2026 across all 40/40 dual-annotated clinical 3T MRI scenes.** Expert teams CY+HT (n = 723 total trajectories: 505 CY long + 218 HT short/obturator) and LB/AMe (n = 513, of which 460 are long) digitized the same 40 volumes independently; **NeedleFinder 2026** rows are strictly nested 5 × 5 out-of-fold predictions on the same 40 scenes.

	1-to-1 Detection (≤ 3.5 mm)				E_{shaft} (mm)				E_{tip} (mm)	
Comparison pair (eval. vs. reference)	n_{match}	Rec.	Prec.	F1	Mean	SD	Median	P95	Mean	Median
Human inter-observer agreement										
LB/AMe vs. CY long (460/505)[a]	424	84.0	**92.2**	**87.9**	0.70	0.58	0.49	1.90	6.28	3.24
LB/AMe vs. CY+HT all (513/723)	476	65.8	92.8	77.0	0.73	0.61	0.50	2.09	6.30	3.21
Automated out-of-fold predictions vs. observer teams across all 40 scenes										
NeedleFinder 2026 (τ=50 %) vs. CY+HT (768/723)	653	90.3	85.0	87.6	**0.58**	**0.39**	**0.47**	1.33	**5.26**	**2.10**
NeedleFinder 2026 (τ=35 %) vs. CY+HT (825/723)	669	**92.5**	81.1	86.4	0.59	0.40	**0.47**	**1.32**	5.54	2.20
NeedleFinder 2026 (τ=50 %) vs. CY+HT+LB (768/759)[b]	669	88.1	87.1	87.6	0.59	0.40	**0.47**	1.36	5.44	2.17
NeedleFinder 2026 (τ=50 %) vs. LB/AMe (768/513)[c]	465	90.6	60.5	72.6	0.77	0.62	0.51	2.11	7.87	3.97

Grubbs' three-cornered-hat intrinsic error SD (n = 432) & Pelvic CT adjudication (n = 70)[d]

Source (3-way matched)	n	3W Sh.	3W Tip	Role	σ_{RMS}	σ_{MAD}	$\sigma_{z,MAD}$	$\sigma_{z,W}$	CT	Win
Human Rater 1 (CY primary)	432	Ref.	Ref.	Ref.	**0.39**	**0.35**	**0.00**[d]	**3.77**	4.31	–
NeedleFinder 2026 (τ=50 %, OOF)	432	**0.43**	**2.33**	Eval.	0.56	0.39	2.98	4.94	**4.20**	**78.6 %**
Human Rater 2 (LB/AMe)	432	0.49	3.18	R2	0.86	0.48	3.55	5.09	8.00	–

One-sided non-inferiority tests vs. human inter-observer discrepancy (N = 40, n = 432)[e]

Endpoint (mm)	Δ	NF	LB	Diff.	CI lo	CI hi	Pass	p_{boot}	p_{pt}	p_{cath}
In-plane shaft mean (E_{shaft})	+0.25	**0.56**	0.72	−0.16	−0.31	−0.02	Yes	< 10^{-4}	1.3e−8	1.6e−41
In-plane shaft median (E_{shaft})	+0.25	**0.43**	0.49	−0.06	−0.19	+0.01	Yes	< 10^{-4}	8.2e−10	1.6e−41
3D distal tip median (E_{tip})	+1.00	**2.33**	3.18	−0.85	−1.46	−0.52	Yes	< 10^{-4}	2.6e−7	2.2e−14
3D distal tip mean (E_{tip})	+1.50	**5.51**	6.10	−0.60	−1.80	+0.57	Yes	0.0014	1.1e−4	6.8e−19

[a] Strict long-shaft inter-observer comparison (424 matched pairs out of 460 LB/AMe long vs. 505 CY long: 83.96 % recall, 92.17 % strict precision [93.6 % obturator-adjusted], 87.88 % strict F1 [88.5 % obturator-adjusted], 0.70 ± 0.58 mm shaft error [median 0.49 mm, IQR 0.50 mm, P95 1.90 mm] and 6.28 ± 7.29 mm 3D tip error [median 3.24 mm, IQR 6.02 mm, P95 21.88 mm]).

[b] Evaluated against the de-duplicated union of both expert teams (723 CY+HT + 36 unique LB/AMe trajectories = 759 union catheters), showing that 16 apparent false positives against CY+HT alone are genuine catheters traced by LB/AMe (precision rises from 85.0 % to 87.1 %).

[c] Because LB/AMe primarily traced long interstitial shafts (513 traces vs. 723 for CY+HT), unadjusted precision against LB/AMe alone penalizes valid short obturator predictions; recall is 90.6 %.

[d] Reference-free Grubbs' (1948) three-cornered-hat intrinsic source error SD across the n = 432 three-way matched catheters (formed by intersecting the 476 all-catheter human pairs with OOF **NeedleFinder 2026** predictions within ≤ 3.5 mm). Columns: 3W Sh. and 3W Tip give median pairwise error vs. CY (mm); σ_{RMS} is classic MLE shaft SD (mm); σ_{MAD} is robust MAD-scaled shaft SD (mm); $\sigma_{z,MAD}$ is robust MAD axial tip SD (mm, where Rater 1 reaches the 0.001 mm lower-bound clamp [95 % CI 0.001–2.02 mm] because median signed Δz_{R1-NF} = 0.00 mm); $\sigma_{z,W}$ is 15 % Winsorized axial tip SD (mm); CT and Win give median 3D tip error (mm) against independent post-implant Pelvic CT across the n = 70 catheters with $|\Delta z_{CY-LB}| \geq 5.0$ mm on MRI and the percentage of disputed catheters (78.6 %, 55/70) where **NeedleFinder 2026** lies closer to CT than single-rater MRI.

[e] One-sided non-inferiority tests ($H_0 : E_{NF,CY} - E_{LB,CY} \geq \Delta$) at α = 0.05 across the N = 40 dual-traced patients (n = 432 three-way matched catheters), reporting two-sided 90 % ($1 - 2\alpha$) patient-clustered bootstrap CIs (10,000 resamples), patient-level paired Wilcoxon p_{pt} (N = 40), and catheter-level paired Wilcoxon p_{cath} (n = 432; bootstrap p_{boot} = 0.0014 for tip mean, < 10^{-4} for all other endpoints).

height (σ_z), NeedleFinder 2026 achieves a lower intrinsic error than Human Team 2 under both the robust MAD estimator (2.98 **mm vs.** 3.55 **mm**, where Team 1's estimate collapses to the 0.00 mm lower bound (bootstrap 95% CI 0.0 to 2.0 mm) because its axial disagreements with NeedleFinder 2026 and with Team 2 together barely exceed the disagreement between Team 2 and NeedleFinder 2026, so the subtraction returns a non-positive variance; this is the signature of correlated errors between two sources, expected here because NeedleFinder 2026 was fine-tuned toward Team 1's tip conventions, so we read these axial-tip values as a ranking rather than as exact error bars) and the 15% Winsorized estimator, which yields strictly positive intrinsic standard deviations of 3.77 **mm for Team 1**, 4.94 **mm for NeedleFinder 2026**, and 5.09 **mm for Team 2**.

3. Formal one-sided statistical non-inferiority (Table 3, fourth block). Across all $N = 40$ dual-traced patients ($n = 432$ three-way matched catheters), NeedleFinder 2026 passes formal one-sided non-inferiority testing relative to human inter-observer disagreement on every pre-specified endpoint at both the patient level and the catheter level. Specifically, the algorithm's error against Team 1 is smaller than Team 2's error against Team 1 on **shaft mean** (0.56 vs. 0.72 mm, $p_{\text{pt}} = 1.3 \times 10^{-8}$), **shaft median** (0.43 vs. 0.49 mm, $p_{\text{pt}} = 8.2 \times 10^{-10}$), **3D tip median** (2.33 vs. 3.18 mm, $p_{\text{pt}} = 2.6 \times 10^{-7}$), and **3D tip mean** (5.51 vs. 6.10 mm, $p_{\text{pt}} = 1.1 \times 10^{-4}$, bootstrap $p_{\text{boot}} = 0.0014$; full 90% confidence intervals and catheter-level p-values in Table 3). When the reference role is flipped so that Team 2 serves as the reference, the result is weaker. On 3D tip median the algorithm passes at the catheter level ($p_{\text{cath}} = 9.6 \times 10^{-5}$) and by patient-clustered bootstrap ($p_{\text{boot}} = 0.046$) but not by the patient-level Wilcoxon test ($p_{\text{pt}} = 0.073$), and on 3D tip mean it does not pass (7.53 vs. 6.10 mm, $p_{\text{pt}} = 0.12$). This asymmetry is expected, because Stage 1 was fine-tuned toward Team 1's tracing conventions, and it means the non-inferiority claim holds against the team whose conventions the model learned.

4. Independent adjudication on paired post-implant Pelvic CT scans. On CT, catheters and metallic stylets have unmistakable Hounsfield contrast. Using the 38 patients who had both intraoperative MRI and post-implant Pelvic CT, we adjudicated the $n = 70$ catheters where Human Team 1 and Human Team 2 disagreed on the MRI tip position by ≥ 5.0 mm. Measured against the independent CT tip position, **NeedleFinder 2026 achieved a median 3D tip error of** 4.20 **mm**, comparable to **Human Team 1 (**4.31 **mm; difference not tested)** and smaller than **Human Team 2 (**8.00 **mm)**, and lying closer to the CT tip position than the less accurate of the two human teams on **78.6% (**55/70**)** of disputed catheters. Across all 100 MRI tracks predicted by NeedleFinder 2026 on these 38 patients that did not match Team 1, **52.0% (**52/100**) corresponded to a physical catheter on the paired CT scan** (whether each was deliberately left untraced on MRI cannot be determined retrospectively).

5.4 *Interpretability, Confidence Calibration, and Physical Causes of Residual Errors*

For a clinician to trust an automated reconstruction, its confidence scores must be well calibrated and its filtering decisions transparent. Across all 1, 960 stitched can-

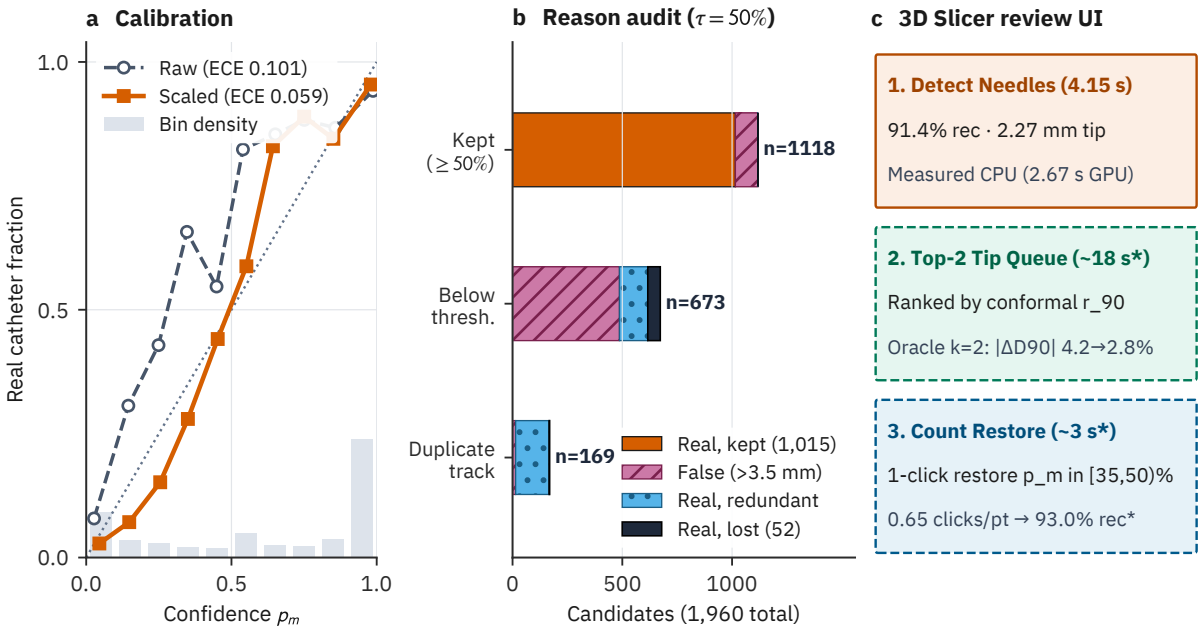


FIGURE 6 **(a) Confidence calibration** across all 1,960 stitched candidate tracks in the 54 patients, before (dashed) and after (solid vermilion) inner-fold temperature scaling. **(b) Reason-coded filtering decisions** at the default threshold $\tau = 50\%$ (colorblind-safe hatched palette), showing how many candidates are kept or pruned by each rule and whether pruned tracks were false alarms, redundant duplicates of an already-kept catheter, or lost real catheters. **(c) Proposed 3D Slicer clinical verification workflow** combining 4.15 s measured headless CPU detection with modeled uncertainty-guided top-2 tip review (⋆ = simulated timing/oracle).

didate tracks entering Stage 2B (36.3 per patient), inner-fold temperature scaling reduces the expected calibration error (ECE) from 0.101 to 0.059 (and to 0.056 across the 1,118 accepted tracks at $\tau = 50\%$, where mean predicted confidence is 91.9% and closest-reference precision is 90.8%; Figure 6a). Every discarded track is tagged with an explicit reason code (Figure 6b): at $\tau = 50\%$, 673 candidates are removed for falling below the confidence threshold (eliminating 80.6% of all false candidates) and 169 are removed as duplicates of an already-kept track (suppressing 153 redundant collinear or touching fragments at the cost of only 2 lost catheters).

Our catheter-level audit of all residual errors across the 54 patients (itemized in Appendix B) confirms three physical patterns: (i) among the 146 unmatched predictions (2.7/patient), 31.5% (46/146) are residual split segments, 20.5% (30/146) match Expert Team 2 within 5.0 mm (52.0% confirmed on paired CT), and only **45.2% (66/146, or 1.22/patient)** are genuine vascular/fascial false alarms (> 6.0 mm); (ii) among the 91 missed catheters (1.7/patient), **97.4% (1,035/1,063)** of all catheters (**98.8% [750/759]** of long shafts) exist in the Stage 2A candidate pool, with 69.2% (63/91) filtered at the $\tau = 50\%$ bundle threshold (recoverable to **93.6% recall** at $\tau = 35\%$, precision 83.4%); and (iii) **98.8% of total squared 3D tip error** (99.3% in the > 10 mm tail, vs. 0.72 mm median lateral error) is purely longitudinal along the catheter shaft, driven by mid-shaft signal dropouts (41.4%) or dark bowel gas above the tip (25.3%).

5.5 *Specificity on Needle-Free Scans and Few-Second Runtime*

Zero-catheter specificity test (Table 4, Part C; Appendix C, Table 6; Figure 7a). When deployed on the 48 public 3T T2W prostate MRI scans (`MSD Task05_Prostate`) that contain *zero* catheters, an ungated 3D U-Net or bundle transformer mistakes dark pelvic veins for needles, producing 4.10 to 6.38 false tracks per scan. Activating the analytical Stage 0 circular dark-core symmetry gate eliminates over 92% of these vascular false alarms: on a variant of the pipeline whose Stage 1 pre-training excluded the 48 public scans (so that they are strictly unseen), false alarms drop to **0.33 per scan** (16 across 48 scans, median 0, with **79.2% of scans having zero false alarms** and 91.7% having ≤ 1). When the 48 needle-free MSD volumes are also included as unlabeled background anatomy during self-supervised synthetic-tube pre-training (a Stage 1 pre-trained with those scans as background and gated by Stage 0, with zero clinical catheter labels), false alarms drop further to **0.125 per scan** (6 across 48 scans, **87.5% zero-false-alarm scans**, and **100.0% with at most 1 false track**; full 5-row specificity breakdown in Appendix C, Table 6).

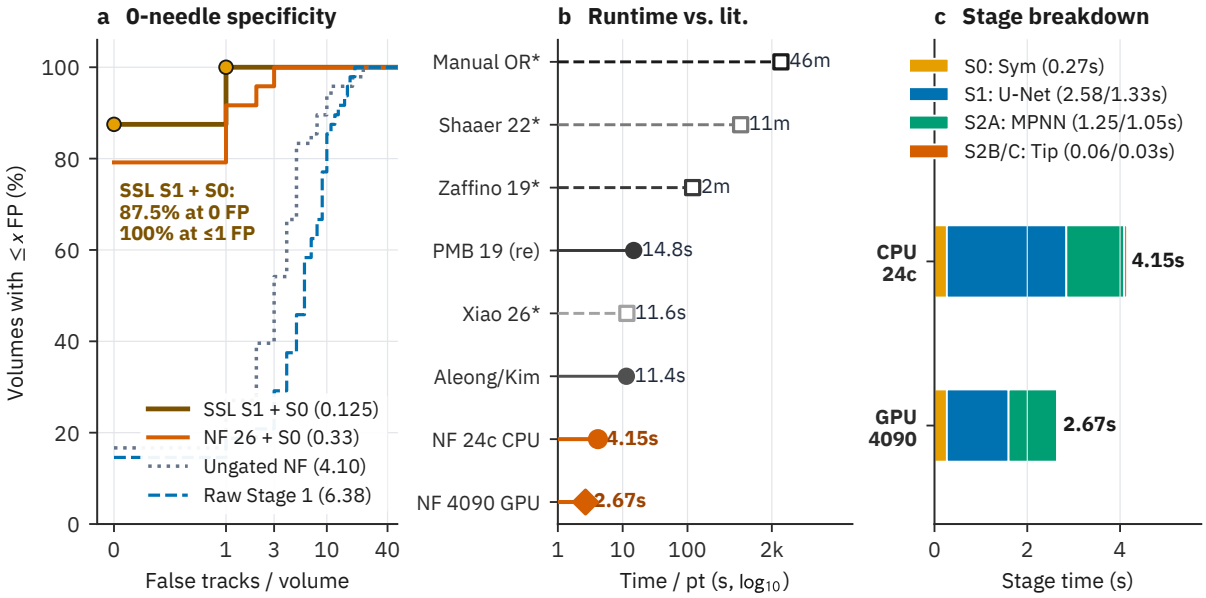


FIGURE 7 **(a) Specificity on 48 needle-free pelvic MRI scans** (`MSD Task05_Prostate`), showing the percentage of scans with at most x false tracks with and without Stage 0 dark-core symmetry gating. **(b) End-to-end runtime per patient** (log scale) compared with published MRI and CT methods. **(c) Stage-by-stage runtime breakdown** across all 54 patients on a 24-core workstation CPU (4.15 s mean; 1.26 s on standard-resolution scans, 5.85 s on high-resolution scans) and on one RTX 4090 GPU (2.67 s mean).

Runtime across standard and high-resolution clinical volumes (Figure 7b, c). Measured end-to-end from raw uncached volume loading to final 3D trajectories across all 54 patients, NeedleFinder 2026 takes **4.15 s on average on a 24-core workstation CPU** (1.26 s on the 20 standard-resolution 320 × 256 scans and 5.85 s on the 34 high-resolution 640 × 416 / 768 × 768 scans) and **2.67 s on a single RTX 4090 GPU** (0.67 s standard-resolution, 3.85 s high-resolution). On CPU, Stage 0

takes 0.27 s (6.5%), Stage 1 takes 2.58 s (62.2%; reduced to 1.33 s on GPU), Stage 2A takes 1.25 s (30.1%), and Stage 2B/2C bundle scoring plus 1D tip localization takes 0.06 s (1.5%). This represents a **29-to-130-fold speedup** over the 2 to 9 minutes reported for our 2019 U-Net + DBSCAN pipeline (Zaffino *et al.*, 2019) (3.6× faster than our modernized re-implementation at 14.80 s) and is 2.8–4.3× faster than the 11.6 s reported by Xiao *et al.* on their own hardware (Xiao *et al.*, 2026a).

5.6 *Clinical Translation: Radiation Dosimetry, Uncertainty-Guided Triage, and Cross-Domain Generalization*

Finally, we asked four practical questions that determine whether an algorithm is genuinely useful in clinical radiation oncology (summarized in Table 4 and Figure 8, with complete protocol and sequence breakdowns in Appendix D, Tables 9 and 8):

5.6.1 *Does geometric accuracy translate into accurate radiation dose delivery? (Table 4, Part A; Appendix D, Table 9; Figure 8a).*

Not all catheter errors matter equally to the radiation dose: a tip error on a peripheral catheter carrying almost zero dwell time has

TABLE 4 **Summary of clinical translation: AAPM TG-43U1 ^{192}Ir dosimetry, split-conformal tip triage, zero-needle specificity, and zero-shot cross-domain generalization.** Consolidates the headline metrics across our four clinical validation studies (complete per-protocol, per-k, and per-sequence breakdowns are provided in Appendices C and D, Tables 6–8).

Dimension	Configuration / Cohort	N	Primary metric	Secondary	E_{shaft}/E_{tip}	Takeaway
1. AAPM TG-43U1 HDR ^{192}Ir dosimetry (7.0 Gy prescription, 5.0 mm dwell steps; Appendix D, Table 9)						
Prot. A (fixed)	PMB 2019 3D U-Net	54	$\lvert\Delta D_{90}\rvert$: 27.6/27.2 %	$\gamma_{2/2}$: 64.0 %	0.42/3.39	Tip dose shift
Prot. A (fixed)	**NeedleFinder 2026** (4b, k=0)	54	$\lvert\Delta D_{90}\rvert$: 15.2/11.5 %	$\gamma_{2/2}$: 76.6 %	0.47/2.27	p=4.9×10^{-6}
Prot. B (re-opt)	Human Team 2 (`LB/AMe`)	40	$\lvert\Delta D_{90}\rvert$: 20.9/4.0 %	$\gamma_{2/2}$: 59.3 %	0.49/3.24	Human floor
Prot. B (re-opt)	**NeedleFinder 2026** (4b, k=0)	54	$\lvert\Delta D_{90}\rvert$: 8.2/4.2 %	$\gamma_{2/2}$: 66.0 %	0.47/2.27	3.2 % (N=40)
Prot. B + k=2*	**NeedleFinder 2026** + k=2 (18 s*)	54	**$\lvert\Delta D_{90}\rvert$: 5.9/2.8 %**	**$\gamma_{2/2}$: 74.3 %**	0.47/1.95	p=2.4×10^{-5}
2. Split-conformal 3D tip intervals & uncertainty-guided clinician review (n = 972 matched catheters; Appendix D, Table 9)						
90 % sphere	Unreviewed (k=0, $\hat{r}_{90}$)	54	**90.2 % emp. cov.**	12.11 mm r	2.27/27.3	−31.5 % width
Review (k=5*)	Top-5 $\hat{\sigma}_{tip}$/pt (42 s*)	54	> 10 mm: 8.0 %	≤ 3 mm: 72.6 %	1.67/14.3	Halves > 10 mm
Review (k=8*)	Top-8 $\hat{\sigma}_{tip}$/pt (66 s*)	54	> 10 mm: 5.0 %	≤ 3 mm: 78.5 %	**1.51/9.90**	P95 < 10 mm
3. Out-of-distribution specificity & zero-shot cross-domain generalization (Appendices C & D, Tables 6 & 8)						
0-needle MRI	MSD `Task05_Prostate`	48	**0.33/0.125 FP/vol**	79.2/87.5 % 0-FP		– vs. 4.10–6.38
3T Prostate	`Case 100` (0-exp. auto)	1	**96.3 % F1 (13/14)**	100.0 % Prec.	0.33/0.99	100 % manual
3D wbSSFP	Steady-state MRI	2	**92.9 % F1 (46/47)**	97.9 % Rec.	0.72/3.40	vs. 10.62 mm
2D AX TSE T2	Δz=2.4–6.0 mm (n=882)	49	76.9 % F1	88.4 % Rec.	1.07/4.89	vs. 57.6 %
3D T1 VIBE	Bright-vessel shift	12	56.1 % F1	58.5 % Rec.	1.87/8.08	Stage 0: 61.5 %
4D insertion	Causal Kalman (`68–71`)	23	82.8 % F1	**88.4 % Prec.**	0.75/3.57	+12.4 % (1.31 s)
Paired CT	Dual-polarity (≤ 5 mm)	40	79.2 % Rec.	52.0 % conf.	2.01/2.78	78.6 % on 70

In Block 1, $\lvert\Delta D_{90}\rvert$ reports mean/median percentage target dose error and E_{shaft}/E_{tip} reports median in-plane shaft / median 3D tip error in mm; in Block 2, E_{shaft}/E_{tip} reports median / 95th-percentile 3D tip error (E_{tip}) in mm; in Block 3, it reports median shaft / median 3D tip error in mm. Asterisks (*) denote modeled 1-click oracle tip review (8 s/tip + 2 s queue overhead).

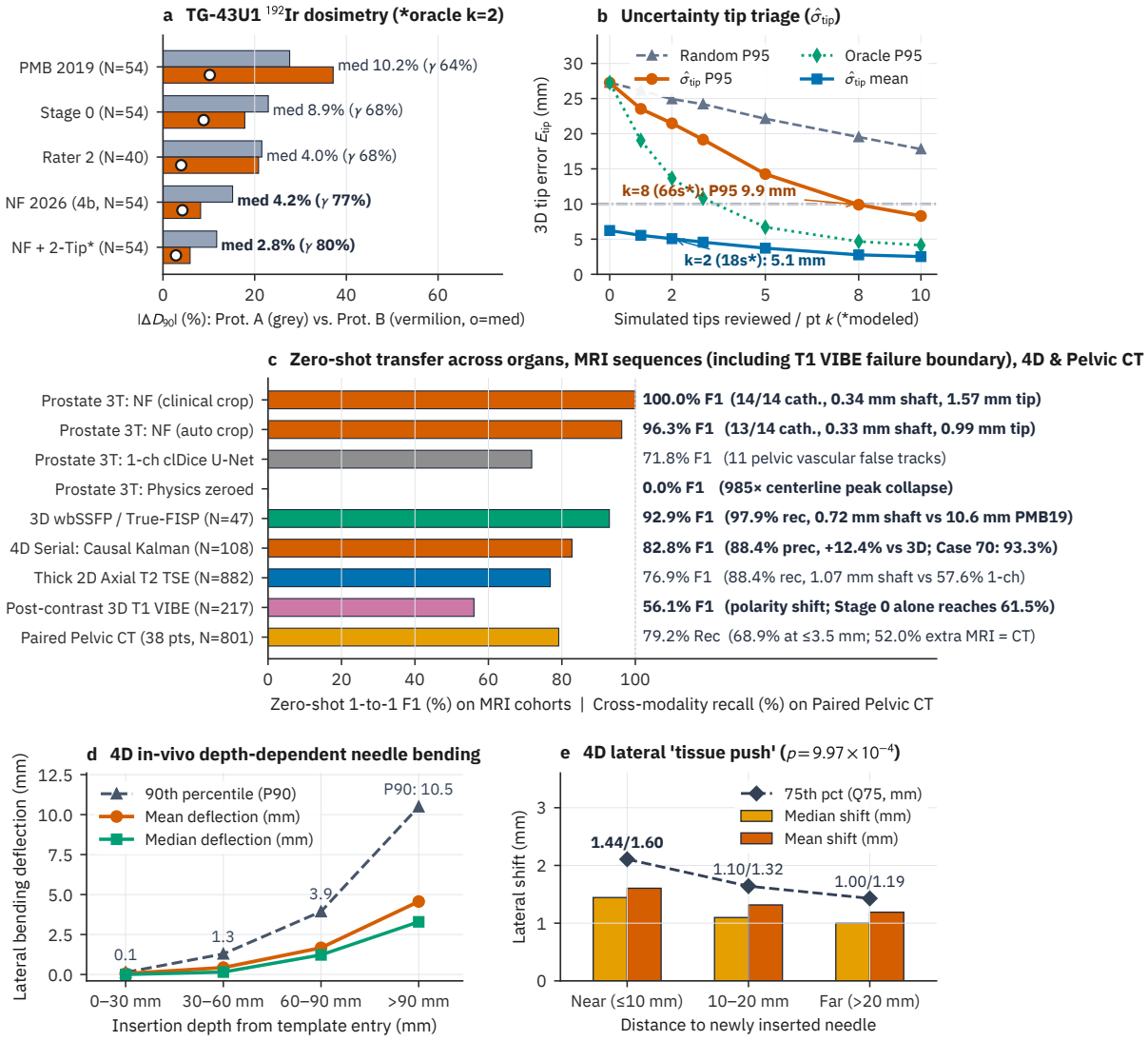


FIGURE 8 **Clinical translation and cross-domain generalization. (a) AAPM TG-43U1 ^{192}Ir dosimetric accuracy** across 54 patients under Protocol A (fixed reference dwell plan transferred to predicted catheters, grey bars) and Protocol B (dwell times re-optimized on predicted catheters and delivered to the true anatomy, vermilion bars with white circles marking the median), alongside 3D γ(2%/2 mm) pass rates ($\star$ = simulated 2-tip oracle). **(b) Simulated clinician tip-review curve:** 95th-percentile and mean 3D tip error as a physicist reviews the top $k \in \{0, \ldots, 10\}$ tips per patient ranked by predicted uncertainty $\hat{\sigma}_{\text{tip}}$ (1-click tip relocation oracle, vs. random order and largest-error oracle). **(c) Zero-shot generalization** to 3T prostate brachytherapy MRI (`Case 100`), 3D wbSSFP, thick-slice 2D AX TSE T2, post-contrast 3D T1 VIBE (showing the polarity-shift failure boundary where Stage 0 alone reaches 61.5% F1 vs. 56.1% for the T2W-trained full pipeline), 4D intra-procedural causal Kalman tracking (`Cases 68–71`), and 38 paired Pelvic CT cases (40 CT volumes). **(d) In-vivo 4D depth-dependent needle bending** across 23 serial scans (`Cases 68–71`), showing non-linear lateral shaft deflection versus insertion depth from the template entry. **(e) In-vivo 4D lateral "tissue push" displacement** of already-implanted catheters as a function of distance to a newly inserted needle ($p = 9.97 \times 10^{-4}$).

little effect, whereas a tip error on a heavily weighted catheter inside the tumor shifts the target dose substantially. Across our 54 patients, **dwell-time-weighted 3D tip error** ($E_{\text{tip}}^{\text{dwell-w}}$—the average 3D tip error across a patient's catheters weighted by each catheter's share of total radiation dwell time) correlates more strongly with target dose error $|\Delta D_{90}|$ than unweighted tip error does (Spearman $\rho = 0.863$ vs. 0.786 across the 54 patients; we did not test the difference between the two correlations).

NeedleFinder 2026 cuts dwell-weighted tip error nearly in half compared with PMB 2019 (5.72 mm vs. 10.51 mm mean; 4.66 mm vs. 8.36 mm median), and is lower than Human Team 2's on the 40 dual-traced scans (5.18 mm vs. 6.27 mm mean; difference not tested). Consequently:

- Under **Protocol A (transferring a fixed dwell plan)**, NeedleFinder 2026 cuts mean target dose error $|\Delta D_{90}|$ from 27.6% (PMB 2019) to **15.2%** (median **11.5%**, $p = 4.9 \times 10^{-6}$) and achieves a **76.6% global 3D γ(2%/2 mm) pass rate** (**84.3%** at γ(3%/3 mm))—and lower error than Human Team 2 on the 40 dual-traced scans (13.5% vs. 21.6% $|\Delta D_{90}|$, uncorrected $p = 0.028$; 79.0% vs. 68.1% $\gamma_{2\%/2\text{mm}}$, uncorrected $p = 0.011$; suggestive, see Methods).
- Under **Protocol B (clinical workflow: re-optimizing dwell weights on the predicted catheters)**, dwell optimization compensates for minor longitudinal shifts, cutting unreviewed target $|\Delta D_{90}|$ to **8.2% mean / 4.2% median** (vs. 37.1%/10.2% for Zaffino *et al.* 2019; on the 40 dual-traced scans, 7.0%/3.2% for NeedleFinder 2026 vs. 20.9%/4.0% for Human Team 2). When a simulated physicist corrects just the **top 2 uncertainty-flagged tips per patient** (each moved to its reference depth), target $|\Delta D_{90}|$ drops to **5.9% mean / 2.8% median** (paired Wilcoxon $p = 2.4 \times 10^{-5}$; improved in 37 of 54 patients, worse in 8, unchanged in 9), and the global $\gamma_{2\%/2\text{mm}}$ pass rate rises from 66.0% to 74.3% ($p = 9.5 \times 10^{-7}$; 78.2% on the 40 dual-traced scans). Gamma against the reference plan is lower under Protocol B than under Protocol A because a re-optimized plan is a different dose distribution, not a worse one.
- **Organ-at-risk proxy doses under Protocol B.** Because the re-optimized plan is a different plan, dose to the proxy organ shells moves more than the target dose does. Across the 54 patients, the unreviewed $|\Delta D_{2\text{cc}}|$ on the bladder, rectum, and sigmoid proxy shells is 17.5%, 18.3%, and 12.5% median (24.6%, 26.2%, and 29.5% mean); correcting the two most uncertain tips lowers these to 13.7%, 13.1%, and 10.1% median (17.8%, 19.4%, and 22.5% mean). These shells sit 5.5 to 16 mm from the implant, in the steepest part of the ^{192}Ir dose gradient, and are not patient anatomy, so the percentages are not clinical $D_{2\text{cc}}$ values. On the same shells and the same 40 dual-traced scans, Human Team 2's traces give 19.9% (bladder) and 25.2% (rectum) median against NeedleFinder 2026's 14.3% and 11.4%. Under Protocol A (fixed plan transferred), the bladder and rectum proxy differences are 3.8% and 5.4% median (Table 4; Appendix D, Table 9).

5.6.2 *Can calibrated uncertainty intervals guide a 1-minute clinician tip check? (Table 4, Part B; Appendix D, Table 9; Figure 8b).* Because 98.8% of tip error lies longitudinally along the already-traced shaft, correcting an uncertain tip in 3D Slicer requires

only sliding the endpoint along the track (a single click, which we assume takes 8 seconds). Our out-of-fold tip uncertainty score $\hat{\sigma}_{\text{tip}}$ identifies > 10 mm tip outliers with an **AUROC of 0.805**, and our nested **split-conformal 3D prediction intervals** achieve close-to-nominal empirical coverage across all 54 unseen test patients despite within-patient clustering (**79.3%** at 80% nominal, **84.4%** at 85%, **90.2% coverage at the 90% nominal level**, and **95.4% at the 95% level**, with 31.5%–35.1% tighter median error bars than unscaled intervals; Table 4, Part B; Appendix D, Table 9, Part B2).

In our simulated clinician-in-the-loop workflow where a physicist inspects the top k highest-uncertainty tips per patient and clicks the true distal termination (Table 4, Part B; Appendix D, Table 9, Part B1; Figure 8b):

- Reviewing $k = 2$ **tips per patient** (18 s modeled, including 2 s to open the review list) lowers mean tip error from 6.23 mm to 5.07 **mm** (1.95 mm median).
- Reviewing $k = 5$ **tips per patient** (42 s modeled) cuts > 10 mm tip outliers in half (16.7% → 8.0%) and lowers mean tip error to 3.72 **mm** (1.67 mm median).
- Reviewing $k = 8$ **tips per patient** (66 s modeled) brings the **95th-percentile tip error across the entire 972-catheter cohort below 10 mm (9.90 mm vs. 27.27 mm unreviewed)** and reduces mean tip error to 2.77 **mm** (1.51 mm median).

5.6.3 *Does the method generalize zero-shot to other organs and MRI sequences? (Table 4, Part D; Appendix D, Table 8; Figure 8c).* To test whether NeedleFinder 2026 overfits to gynecologic T2W MRI, we ran the gynecologic-trained models **zero-shot (with zero retraining)** across three new settings:

- **Cross-organ transfer to 3T prostate brachytherapy MRI (`Case 100`, 14 plastic needles):** This prostate scan is one of the 51 external volumes used, without labels, for Stage 1 pre-training, so we first evaluate a variant pre-trained without it (the zero-exposure row of Table 4, Part D, and Appendix D, Table 8). With automatic template cropping, that variant found **13 of 14 prostate catheters with no false alarms (96.3% F1)**, with 0.33 **mm median shaft error** and 0.99 **mm median 3D tip error**; with the clinician's manual crop it found all 14. The full pipeline, whose pre-training had seen this scan without labels, found all 14 under automatic cropping (errors in Appendix D, Table 8). **Why does it transfer on this scan?** When we zeroed out the Stage 0 physical symmetry channels ($X_{1:3} = 0$), prostate detection collapsed to **0% (0/14)**, whereas the 1-channel `ResUNet + clDice` baseline hallucinated 11 false vascular tracks (56.0% precision, 71.8% F1). Thus, on this single scan, the Stage 0 physical symmetry channels carried the transfer from gynecologic to prostate anatomy.
- **Secondary MRI pulse sequences (63 of the 243 secondary scans evaluated):** On 3D balanced steady-state free precession (`3D wbSSFP / True-FISP`), where tissue contrast differs markedly from TSE T2W, zero-shot NeedleFinder 2026 achieved **97.9% recall (46/47), 88.5% precision, and 92.9% F1** (0.72 mm median shaft error, 3.40 mm median tip error vs. 10.62 mm for PMB 2019). On 49 thick-slice 2D axial T2W scans ($\Delta z = 2.4$–6.0 mm), Z-upsampling to 1.6 mm

recovered **88.4% (780/882) of physically visible catheters** (76.9% F1 vs. 57.6% for clDice). On the third secondary sequence we evaluated, 12 post-contrast axial T1 VIBE scans from 2 patients (217 catheters), zero-shot F1 fell to 56.1% (58.5% recall, 53.8% precision, 8.08 mm median tip error), below Stage 0 alone (61.5%) and the 2019 U-Net + DBSCAN re-implementation (59.4%): post-contrast T1 catheter contrast lies outside what the pipeline was trained on.

– **Zero-shot tracking on post-implant Pelvic CT (40 CT volumes, 38 paired patients):** Extending Stage 0 to detect both bright metallic stylets and dark air lumens in Hounsfield units enabled zero-shot CT reconstruction across 801 MRI-paired reference catheters, matching **68.9% at ≤ 3.5 mm and 79.2% at ≤ 5.0 mm** across non-rigid MRI-to-CT anatomical deformation (2.01 mm median shaft registration residual).

5.6.4 *Can we track needles across serial scans during insertion and measure tissue biomechanics in 4D? (Table 4, Part D; Appendix D, Table 8; Figure 8c, d, e).* In 4 patients (`Cases 68–71`), 23 sequential T2W scans were acquired *during* the implant procedure as needles were progressively inserted (5 to 7 timepoints per patient, 1.31 s mean monitoring runtime per scan). Evaluating each patient's 4D series strictly with the outer-fold model where that patient was held out in the locked test vault (`Case 068` in Fold 4, `Case 069` in Fold 0, `Case 070` in Fold 1, `Case 071` in Fold 2), coupling NeedleFinder 2026 with a **causal 4D Kalman filter** that propagates catheter states forward in time boosted **pooled precision across the 108 catheters by +12.4 percentage points** (76.0% → 88.4%), reduced mean 3D tip error from 11.23 to 9.75 mm, and improved pooled F1 from **81.6% to 82.8%**, a difference we did not test given only 4 patients (reaching **93.3% F1** on `Case 070`, up from 85.7% with independent 3D tracking, while trading single-frame recall on newly inserted final-frame needles in `Case 071` for 100.0% precision).

Tracking 212 persisted catheter transitions across these sequential insertions, measured on the automatically tracked catheters rather than on expert traces, also suggested two biomechanical patterns in vivo (Figure 8d, e):

– **Inter-catheter "tissue push":** Inserting a new needle appears to push surrounding soft tissue aside. Already-implanted catheters located within ≤ 10 mm of a newly inserted needle shifted sideways by a **median of 1.44 mm** (mean 1.61 mm, 90th percentile 2.81 mm), compared with 1.10 **mm** at 10–20 mm distance and 1.00 **mm** beyond 20 mm (Mann–Whitney test of catheters within 12 mm versus beyond 12 mm, $p = 9.97 \times 10^{-4}$; the 212 transitions come from 4 patients and are treated as independent, so this p-value is optimistic).

– **Depth-dependent needle bending:** Deviation of a catheter from its initial entry angle grows non-linearly with insertion depth, rising from below measurement resolution at 0–30 mm depth and 0.15 mm at 30–60 mm to 1.22 **mm at** 60–90 **mm** and 3.29 **mm median (**4.56 **mm mean,** 10.49 **mm P90) beyond** 90 **mm depth** (Spearman $\rho = +0.555$ across depth samples pooled from 4 patients, nominal $p = 5.6 \times 10^{-9}$)—one reason to expect linear extrapolation of the entry angle to be unreliable on deep pelvic implants.

6 DISCUSSION

6.1 *Why Combining Physics, Peak Regression, and Graph Reasoning Works*

Appendix C (Table 7) places NeedleFinder 2026 in the context of published catheter digitization methods across CT, ultrasound, and MRI. On CT and ultrasound, metallic stylets, air lumens, or specular reflections create strong contrast against soft tissue, allowing voxel segmentation networks to reach 0.1 to 0.7 mm shaft errors (Andersén *et al.*, 2020; Zhang *et al.*, 2020d; Weishaupt *et al.*, 2022; Quetin *et al.*, 2025; Xiao *et al.*, 2026a). On clinical T2W MRI without exogenous contrast markers, a plastic catheter is a 2 mm dark proton-density and susceptibility void imaged across 1.2 to 6.0 mm slices. Prior MRI studies reported 0.74 to 0.97 mm shaft errors on gynecologic implants using positive-contrast markers or CT-assisted ground truth (Shaaer *et al.*, 2022; Kim *et al.*, 2026) and 0.75 to 0.93 mm on prostate implants (Dai *et al.*, 2020; Kostoulas *et al.*, 2025). NeedleFinder 2026 reaches a median shaft error below half a millimeter directly on standard T2W MRI on the 972 matched catheters of a 1,063-catheter cohort (Table 2), although these are different cohorts and reference standards, so the comparison is indicative only.

Our controlled ablations and peer-reviewed baseline reproductions explain why neither voxel segmentation (Zaffino *et al.*, 2019; Isensee *et al.*, 2021; Shit *et al.*, 2021), nor 1-channel MRI post-processing (Dai *et al.*, 2020; Aleong *et al.*, 2024; Kostoulas *et al.*, 2025; Kim *et al.*, 2026), nor mask-based tracking (Xiao *et al.*, 2026a) suffices on its own on T2W MRI, and why each part of our three-stage design is needed:

1. **Local physical symmetry before deep learning:** Stage 0 suppresses one-sided anatomical borders analytically and provides a representation that, on one prostate implant scan and 48 needle-free scans, let a gynecologic network transfer zero-shot without hallucinating blood vessels.
2. **Centerline peaks and hard-negative pre-training instead of binary masks:** By pre-training Stage 1 on 51 external pelvic MRI scans where real blood vessels and fascia are penalized as hard negatives alongside synthetic catheter voids, and by predicting sharp sub-pixel Gaussian peaks (P_{peak}) supervised by two expert teams, touching catheters stay separate instead of merging into a single blob.
3. **Graph message passing, fragment stitching, and bundle-level context:** Rather than trusting local voxel connectivity, Stage 2A enforces smooth triplet collinearity and stitches fragments across 1-to-3-slice signal dropouts, while the Stage 2B graph transformer judges every candidate trajectory against the geometry of the entire 15-to-30-catheter bundle.
4. **Dedicated 1D axial tip profiling and literature fusion:** Because 98.8% of tip error is longitudinal along the shaft, sampling a fine 0.5 mm multi-channel profile along the cranial trajectory tangent allows a 1D ResNet + self-attention localizer to pinpoint where the catheter void terminates and to quantify its own uncertainty—while incorporating sub-voxel inverse-variance-weighted polynomial fitting (Aleong *et al.*, 2024; Kim *et al.*, 2026), directional axial TV continuity (Dai *et al.*, 2020), and iterative multi-segment gap-bridging (Kostoulas *et al.*,

2025) refines shaft and tip accuracy and suppresses residual split fragments (4d, **89.4% F1**).

6.2 *How NeedleFinder 2026 Changes the Clinical Workflow in 3D Slicer*

Packaged as a one-click module in **3D Slicer** (Fedorov *et al.*, 2012), NeedleFinder 2026 runs in a few seconds on a 24-core workstation CPU or a GPU when its production Rust and ONNX Runtime code path is run headless, outside the Slicer interface (Figure 7c). Instead of spending 45 to 50 minutes manually clicking 15 to 30 catheters slice by slice while the patient waits with the implant in place, the medical physicist follows a structured verification workflow that we have modeled, not timed, at about half a minute for the two-tip check:

1. **Click *Detect Needles* (4 seconds for the production code path run headless; interface overhead not yet timed):** The software crops the template plane automatically and reconstructs the 3D catheter bundle. Because our non-inferiority analysis shows that the reconstructed shafts (Table 3) are statistically non-inferior to human inter-observer agreement with Team 1 as reference, this modeled workflow assumes no manual shaft editing; we did not measure how often shafts are edited.
2. **Check the few tips with the widest uncertainty bars (18 to 66 seconds simulated):** The module is designed to list the catheter tips with the widest split-conformal uncertainty intervals ($\hat{\sigma}_{\text{tip}}$) first. Because the shaft is already traced, correcting a flagged tip means sliding its endpoint along the track, which we model at 8 seconds per tip. Correcting only the two most uncertain tips per patient (18 seconds) is the configuration we carried through the TG-43U1 dose simulation: it lowers median target dose error $|\Delta D_{90}|$ from 4.2% to **2.8%** (mean from 8.2% to 5.9%). Extending the check to the eight most uncertain tips (66 seconds) was evaluated on tip geometry only; it brings the 95th-percentile tip error below 10 mm (9.90 mm, mean 2.77 mm). In the dose simulation the flagged tip is placed exactly at its reference depth; in the tip-geometry curve a residual of up to 0.85 mm along the shaft is retained. Either way the simulated physicist never errs, so these numbers are upper bounds on the benefit of review.
3. **Reconcile the catheter count (assumed 5 seconds):** The clinician knows how many needles went through the template (N_{implant}). If the software reports more, the physicist deletes the extra tracks (2.7 unmatched predictions per patient at the default threshold, of which 1.2 are genuine false alarms on vessels or fascia); we did not simulate this deletion, and the Protocol B dose simulation loads every predicted track, so its unreviewed numbers include these tracks. If the software reports fewer, one click near the missing catheter promotes the highest-scoring candidate track that Stage 2B had rejected there, rather than tracing from scratch. Only misses that still exist in the Stage 2A candidate pool can be recovered this way, so in simulation 0.65 such promotions per patient raise 1-to-1 recall from 91.4% to **93.0%** (94.2% closest-match recall) at the cost of 18 additional false tracks (precision 86.9% → 85.8%). Moreover, during progres-

sive needle insertion in intraoperative MRI suites, the 1.31 s causal 4D Kalman mode could in principle alert the implanting physician after each interim scan if a newly inserted needle pushes a neighboring catheter out of position or deflects toward an organ at risk; we evaluated it only retrospectively on 23 scans from 4 patients.

6.3 *Limitations and Future Directions*

Four limitations of the present study point directly to future work:

1. **Longitudinal tip ambiguity on thick slices.** While our 1D axial tip localizer cuts mean tip error to 6.23 mm and median tip error to 2.26–2.27 mm (58.7%–59.0% within ≤ 3.0 mm), 16.7% of unreviewed catheters still have a longitudinal tip error exceeding 10 mm when a catheter passes through a multi-slice signal dropout or terminates directly beneath dark bowel gas on 3 to 6 mm thick slices. Although simulated uncertainty-guided review of 5 to 8 tips contains this tail in our oracle model, combining T2W planning scans with thin-slice isotropic 3D sequences (≤ 1 mm) or zero-echo-time sequences (Kaza *et al.*, 2022; Lee *et al.*, 2023) is the natural path to reducing unreviewed tip outliers further.
2. **Single-institution clinical implant cohort.** While we tested zero-shot transfer on 48 external public prostate scans, one 3T prostate brachytherapy MRI, three other MRI pulse sequences, 4D insertion series, and paired CT scans, all implanted brachytherapy patients were treated at Brigham and Women's Hospital. Prospective multi-center validation across 1.5T, 3T, and emerging low-field MRI systems (Major *et al.*, 2026) from multiple scanner vendors is the essential next step.
3. **Standardized TG-43U1 anatomical envelopes and simulated review timings.** Because clinical DICOM RTSTRUCT and RTPLAN files were not archived across the historical 54-patient repository, our TG-43U1 dosimetric study used standardized target and organ-at-risk envelopes derived from the active implant geometry; the organ-at-risk shells lie in the steepest part of the dose gradient, which is why their proxy D_{2cc} differences are several times larger than the target D_{90} differences and must not be read as clinical organ doses. Our clinician tip-review curves used a quantitative 1-click oracle rather than prospective stopwatch trials in the operating room.
4. **Role of synthetic bundle pre-training (`SynthBrachy3T`).** Once Stage 1 is pre-trained on 51 external pelvic MRI scans and paired with collinear fragment stitching and the 1D tip localizer, `SynthBrachy3T` provides its largest detection boost when clinical bundle labels are scarce ($k = 0$ to 5 patients, lifting zero-shot long-shaft $F1_{adj}$ by +6.3 points to 83.4%), while offering a smaller refinement at the full 43-patient training budget (+0.4 F1 points, while reducing dwell-weighted tip error from 6.65 to 5.72 mm and counterfactual $|\Delta D_{90}|$ from 12.8% to 8.2%).

7 CONCLUSION

NeedleFinder 2026 reconstructs interstitial brachytherapy catheters directly on clinical 3T T2-weighted MRI. It combines an analytical dark-core symmetry filter (Stage 0), a 3D U-Net pre-trained on needle-free pelvic scans and fine-tuned on two expert teams' traces (Stage 1), and graph tracking with bundle-level scoring and a 1D axial tip localizer (Stage 2). Under strictly nested cross-validation on 54 patients and 1,063 catheters, it finds about nine catheters in ten with a median shaft error below half a millimeter and a median tip error of about 2 mm, in a few seconds on a workstation CPU (Table 2; Figure 7), and it outperforms our re-implementations of five peer-reviewed methods and one tracking preprint on the same splits. On 40 doubly traced scans its disagreement with one expert team is no larger than the disagreement between the two teams, and paired post-implant CT confirms that half of its apparent false alarms are real catheters (Table 3). Its calibrated tip uncertainty supports a brief simulated check in 3D Slicer: moving the two most uncertain tips per patient to their reference depth, modeled at 18 seconds, lowers simulated target dose error from 4.2% to 2.8% median (Table 4; Appendix D, Table 9). Without retraining, the pipeline also runs on one prostate implant, two other T2-type MRI sequences, serial intra-procedural scans and, with lower recall, post-contrast T1 MRI and pelvic CT.

DATA AVAILABILITY

The 54 clinical MRI scans and their expert annotations were acquired at Brigham and Women's Hospital under Institutional Review Board approval and contain protected health information; they cannot be shared publicly. De-identified trajectory annotations and per-patient evaluation results may be made available by the corresponding author upon reasonable request, subject to institutional ethics approval. The 48 needle-free pelvic MRI scans are publicly available from the Medical Segmentation Decathlon (`Task05_Prostate`, http://medicaldecathlon.com (Antonelli *et al.*, 2022)). The `SynthBrachy3T` synthetic bundle generator is released with our open-source repository.

CODE AVAILABILITY

The 3D Slicer extension, the native Rust Stage 0 kernel (`needlefinder_rust`), and the exported ONNX models (`NeedleUNet3D`, `NeedleSliceMPNN`, `TipAware BundleGraphTransformer`, and `Dedicated1DTipLocalizer`) are available at https://github.com/needlefinder/NeedleFinder under an open-source license. All scripts for reference-free preprocessing (`prepare_clean_dataset.py`), self-supervised pre-training (`pretrain_ssl_unet3d.py`), strictly nested 5 × 5 training and evaluation (`train_nested_stage1.py`, `run_stage2_and_tip_clean.py`, `export_clean_unified_and_table3.py`), peer-reviewed baseline reproduction and literature fusion (`run_part1_peer_baselines_and_unc_maps.py`, `run_part2_literature_fusion_inside_needlefinder.py`), specificity

benchmarking (`benchmark_public_pelvic_mri_clean.py`), runtime profiling (`benchmark_54cases_timing_clean.py`), calibration and failure auditing (`calibration_and_failure_clean.py`), and clinical translation Experiments 1–4 (`run_exp1_tg43_dosimetry.py` through `run_exp4_4d_insertion_and_ct.py`) are archived in the `benchmarks_clean/` directory of the repository.

ETHICS STATEMENT

This retrospective study analyzed imaging data acquired during routine clinical care under a protocol approved by the Institutional Review Board of Brigham and Women's Hospital / Mass General Brigham.

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AUTHOR CONTRIBUTIONS

[To be completed following the CRediT taxonomy.]

CONFLICTS OF INTEREST

The authors declare no conflicts of interest.

DECLARATION OF GENERATIVE AI AND AI-ASSISTED TECHNOLOGIES

During the preparation of this work the authors used large language models (Anthropic Claude, OpenAI ChatGPT and Codex, and Google Gemini) for four tasks: verifying bibliographic details and reported figures of cited literature against the original publications; organizing and restructuring the manuscript; editing language for clarity and readability; and adversarially reviewing draft sections, tables, and evaluation scripts for errors, inconsistencies, and unsupported claims. These tools did not design the method, acquire or annotate the data, run the experiments, or generate any result reported here. After using these tools, the authors reviewed and edited every sentence, re-derived every number from the archived benchmark outputs, and take full responsibility for the content of the publication. No AI tool is listed as an author, in accordance with Elsevier, IOP Publishing, and ICMJE authorship policies.

REFERENCES

Aleong, A.M., Berlin, A., Borg, J., Helou, J., Beiki-Ardakani, A., Rink, A., Raman, S., Chung, P., and Weersink, R.A. 2024. Rapid multi-catheter segmentation for magnetic resonance image-guided catheter-based interventions. *Medical Physics*, **51**, 5361–73. https://doi.org/10.1002/mp.17117.

Andersén, C., Rydén, T., Thunberg, P., and Lagerlöf, J.H. 2020. Deep learning-based digitization of prostate brachytherapy needles in ultrasound images. *Medical Physics*, **47**, 6414–20. https://doi.org/10.1002/mp.14508.

Androulakis, I., Godart, J., Luthart, L., MEMC, C., Westerveld, H., Nout, R.A., Hoogeman, M., and Kolkman-Deurloo, I.K. 2025. Reconstruction errors in clinical intraoperative TRUS-based prostate HDR-BT detected using electromagnetic tracking. *Brachytherapy*, **24**, 177–85. https://doi.org/10.1016/j.brachy.2024.11.004.

Antonelli, M., Reinke, A., Bakas, S., Farahani, K., Kopp-Schneider, A., Landman, B.A., Litjens, G., Menze, B., Ronneberger, O., Summers, R.M., et al. 2022. The medical segmentation decathlon. *Nature Communications*, **13**, 4128. https://doi.org/10.1038/s41467-022-30695-9.

Arapi, V., Hardt-Stremayr, A., Weiss, S., and Steinbrener, J. 2023. Bridging the simulation-to-real gap for AI-based needle and target detection in robot-assisted ultrasound-guided interventions. *European Radiology Experimental*, **7**, 30. https://doi.org/10.1186/s41747-023-00344-x.

Arcos, J. de, Schmidt, E.J., Wang, W., Tokuda, J., Vij, K., Seethamraju, R.T., Damato, A.L., Dumoulin, C.L., Cormack, R.A., and Viswanathan, A.N. 2017. Prospective clinical implementation of a novel magnetic resonance tracking device for real-time brachytherapy catheter positioning. *International journal of radiation oncology, biology, physics*, **99**, 618–26. https://doi.org/10.1016/j.ijrobp.2017.05.054.

Beaulieu, L., Bert, C., Borot de Battisti, M., Cormack, R.A., Cunha, J.A.M., Damato, A.L., Deufel, C.L., Hautvast, G.L.T.F., Hsu, I.-C., Kolkman-Deurloo, I.-K., Moerland, M.A., Niatsetski, Y., Weersink, R.A., and Tanderup, K. 2025. AAPM task group 317 report: A joint AAPM and ESTRO report on brachytherapy catheter, needle, and applicator tracking technology. *Medical Physics*, **52**, e70037. https://doi.org/10.1002/mp.70037.

Ben-Haim, T., and Riklin Raviv, T. 2022. Graph neural network for cell tracking in microscopy videos. *Computer vision – ECCV 2022*, Lecture notes in computer science, vol 13681, (Springer), pp 610–26. https://doi.org/10.1007/978-3-031-19803-8_36.

Brasó, G., and Leal-Taixé, L. 2020. Learning a neural solver for multiple object tracking. *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition (CVPR)*, pp 6246–56. https://doi.org/10.1109/CVPR42600.2020.00628.

C, K., MJ, R., D, B., F, B., M, D.B., Laarse R, van der, Y, N., P, P., TP, H., J, P.-C., K, T., JL, V., and FA, S. 2014. Review of clinical brachytherapy uncertainties: Analysis guidelines of GEC-ESTRO and the AAPM. *Radiotherapy and oncology : journal of the*

European Society for Therapeutic Radiology and Oncology, **110**, 199–212. https://doi.org/10.1016/j.radonc.2013.11.002.

Cantin, A., Lavallée, M.-C., Poulin, E., Hautvast, G., Foster, W., and Beaulieu, L. 2024. Proof-of-concept of real-time electromagnetic guidance for gynecologic interstitial catheters in high dose rate brachytherapy. *Physics and Imaging in Radiation Oncology*, **32**, 100661. https://doi.org/10.1016/j.phro.2024.100661.

Chen, X., and Egger, J. 2014. Development of an open source software module for enhanced visualization during MR-guided interstitial gynecologic brachytherapy. *SpringerPlus*, **3**, 167. https://doi.org/10.1186/2193-1801-3-167.

Chino, J., Erickson, B., Gaffney, D., Hsu, I.-C., Kamrava, M., Kim, Y., Niedermayr, T.R., Roumeliotis, M., Rownd, J., Todor, D., Viswanathan, A., and Kidd, E.A. 2025. The American Brachytherapy Society (ABS) consensus guidance for hybrid intracavitary interstitial brachytherapy for locally advanced cervical cancer. *Brachytherapy*, **24**, 463–78. https://doi.org/10.1016/j.brachy.2025.05.001.

Dai, X., Lei, Y., Zhang, Y., Qiu, R.L.J., Wang, T., Dresser, S.A., Curran, W.J., Patel, P., Liu, T., and Yang, X. 2020. Automatic multi-catheter detection using deeply supervised convolutional neural network in MRI-guided high-dose-rate prostate brachytherapy. *Medical Physics*, **47**, 4115–24. https://doi.org/10.1002/mp.14307.

Deufel, C.L., Brost, E.E., Dupere, J.M., Petersen, I.A., Haddock, M.G., and Garda, A.E. 2024. EMVision: An electromagnetic tracking guidance system for accurate placement of interstitial brachytherapy applicators. *Brachytherapy*, **23**, 676–86. https://doi.org/10.1016/j.brachy.2024.06.006.

Deufel, C.L., Tian, S., Yan, B.B., Vaishnav, B.D., Haddock, M.G., and Petersen, I.A. 2020. Automated applicator digitization for high-dose-rate cervix brachytherapy using image thresholding and density-based clustering. *Brachytherapy*, **19**, 111–8. https://doi.org/10.1016/j.brachy.2019.09.002.

Dise, J., Liang, X., Scheuermann, J., Anamalayil, S., Mesina, C., Lin, L.L., and Teo, B.K. 2015. Development and evaluation of an automatic interstitial catheter digitization tool for adaptive high-dose-rate brachytherapy. *Brachytherapy*, **14**, 619–25. https://doi.org/10.1016/j.brachy.2015.05.004.

Dürrbeck, C., Sauer, B.C., Lotter, M., Kreppner, S., Strnad, V., Fietkau, R., and Bert, C. 2022. Automating implant reconstruction in interstitial brachytherapy of the breast: A hybrid approach combining electromagnetic tracking and image segmentation. *Radiotherapy and Oncology*, **176**, 172–8. https://doi.org/10.1016/j.radonc.2022.09.019.

Fedorov, A., Beichel, R., Kalpathy-Cramer, J., Finet, J., Fillion-Robin, J.-C., Pujol, S., Bauer, C., Jennings, D., Fennessy, F., Sonka, M., et al. 2012. 3D Slicer as an image computing platform for the quantitative imaging network. *Magnetic Resonance Imaging*, **30**, 1323–41. https://doi.org/10.1016/j.mri.2012.05.001.

Goulet, M., Duguay-Drouin, P., Mascolo-Fortin, J., Mégrourèche, J., Octave, N., and Tsui, J.M.G. 2025. Clinical application of deep learning-assisted needles reconstruction in

prostate ultrasound brachytherapy. *International Journal of Radiation Oncology, Biology, Physics*, **122**, 199–207. https://doi.org/10.1016/j.ijrobp.2024.12.026.

Hellebust, T.P., Kirisits, C., Berger, D., Pérez-Calatayud, J., De Brabandere, M., De Leeuw, A., Dumas, I., Hudej, R., Lowe, G., Wills, R., and Tanderup, K. 2010. Recommendations from gynaecological (GYN) GEC-ESTRO working group: Considerations and pitfalls in commissioning and applicator reconstruction in 3D image-based treatment planning of cervix cancer brachytherapy. *Radiotherapy and Oncology*, **96**, 153–60. https://doi.org/10.1016/j.radonc.2010.06.004.

Hrinivich, W.T., Morcos, M., Viswanathan, A., and Lee, J. 2019. Automatic tandem and ring reconstruction using MRI for cervical cancer brachytherapy. *Medical Physics*, **46**, 4324–32. https://doi.org/10.1002/mp.13730.

Hu, H., Yang, Q., Li, J., Wang, P., Tang, B., Wang, X., and Lang, J. 2021. Deep learning applications in automatic segmentation and reconstruction in CT-based cervix brachytherapy. *Journal of Contemporary Brachytherapy*, **13**, 325–30. https://doi.org/10.5114/jcb.2021.106118.

Hu, X., Li, F., Samaras, D., and Chen, C. 2019. Topology-preserving deep image segmentation. *Advances in neural information processing systems 32 (NeurIPS 2019)*, vol 32. URL: https://papers.nips.cc/paper/8803-topology-preserving-deep-image-segmentation.

Isensee, F., Jaeger, P.F., Kohl, S.A.A., Petersen, J., and Maier-Hein, K.H. 2021. nnU-Net: A self-configuring method for deep learning-based biomedical image segmentation. *Nature Methods*, **18**, 203–11. https://doi.org/10.1038/s41592-020-01008-z.

J, R., A, O., S, R., AI, N., M, D., M, S., J, V., F, B., and J, P.-C. 2015. A practical MRI-based reconstruction method for a new endocavitary and interstitial gynaecological template. *Journal of contemporary brachytherapy*, **7**, 407–14. https://doi.org/10.5114/jcb.2015.55340.

Jacobsen, M.C., Beriwal, S., Dyer, B.A., Klopp, A.H., Lee, S.I., McGinnis, G.J., Robbins, J.B., Rauch, G.M., Sadowski, E.A., Simiele, S.J., Stafford, R.J., Taunk, N.K., Yashar, C.M., and Venkatesan, A.M. 2022. Contemporary image-guided cervical cancer brachytherapy: Consensus imaging recommendations from the society of abdominal radiology and the american brachytherapy society. *Brachytherapy*, **21**, 369–88. https://doi.org/10.1016/j.brachy.2022.04.005.

JC, D., P, P., K, T., P, P., D, B., C, K., EM, P., Limbergen E, van, C, H.-M., and R, P. 2012. Recommendations from gynaecological (GYN) GEC-ESTRO working group (IV): Basic principles and parameters for MR imaging within the frame of image based adaptive cervix cancer brachytherapy. *Radiotherapy and oncology : journal of the European Society for Therapeutic Radiology and Oncology*, **103**, 113–22. https://doi.org/10.1016/j.radonc.2011.12.024.

Jung, H., Gonzalez, Y., Shen, C., Klages, P., Albuquerque, K., and Jia, X. 2019a. Deep-learning-assisted automatic digitization of applicators in 3D CT image-based high-dose-rate brachytherapy of gynecological cancer. *Brachytherapy*, **18**, 841–51. https://doi.org/10.1016/j.brachy.2019.06.003.

Jung, H., Shen, C., Gonzalez, Y., Albuquerque, K., and Jia, X. 2019b. Deep-learning assisted automatic digitization of interstitial needles in 3D CT image based high dose-rate brachytherapy of gynecological cancer. *Physics in Medicine & Biology*, **64**, 215003. https://doi.org/10.1088/1361-6560/ab3fcb.

Kallis, K., Kaltsas, T., Kreppner, S., Lotter, M., Strnad, V., Fietkau, R., and Bert, C. 2019. Impact of inter- and intra-observer variabilities of catheter reconstruction on multi-catheter interstitial brachytherapy of breast cancer patients. *Radiotherapy and Oncology*, **135**, 25–32. https://doi.org/10.1016/j.radonc.2019.02.017.

Kapur, T., Egger, J., Damato, A., Schmidt, E.J., and Viswanathan, A.N. 2012. 3-t MR-guided brachytherapy for gynecologic malignancies. *Magnetic resonance imaging*, **30**, 1279–90. https://doi.org/10.1016/j.mri.2012.06.003.

Karius, A., Kreppner, S., Strnad, V., Schweizer, C., Lotter, M., Fietkau, R., and Bert, C. 2024. Inter-observer effects in needle reconstruction for temporary prostate brachytherapy: Dosimetric implications and adaptive CBCT-TRUS registration solutions. *Brachytherapy*, **23**, 421–32. https://doi.org/10.1016/j.brachy.2024.05.002.

Kaza, E., Lee, C.Y., King, M.T., Dyer, M.A., Cormack, R.A., and Buzurovic, I. 2022. First pointwise encoding time reduction with radial acquisition (PETRA) implementation for catheter detection in interstitial high-dose-rate (HDR) brachytherapy. *Brachytherapy*, **21**, 501–10. https://doi.org/10.1016/j.brachy.2022.01.003.

Kim, G., Antaki, M., Schmidt, E.J., Roumeliotis, M., Viswanathan, A.N., and Lee, J. 2026. Automatic digitization of applicator and catheters for MRI-guided cervical cancer brachytherapy. *Brachytherapy*, **25**, 85–94. https://doi.org/10.1016/j.brachy.2025.09.017.

Kirchhoff, Y., Rokuss, M.R., Roy, S., Kovacs, B., Ulrich, C., Wald, T., Zenk, M., Vollmuth, P., Kleesiek, J., Isensee, F., and Maier-Hein, K. 2024. Skeleton recall loss for connectivity conserving and resource efficient segmentation of thin tubular structures. *Computer vision – ECCV 2024, lecture notes in computer science*, (Springer), pp 218–34. https://doi.org/10.1007/978-3-031-72980-5_13.

Kostoulas, V., Guijt, A., Kerkhof, E.M., Pieters, B.R., Bosman, P.A.N., and Alderliesten, T. 2025. Dealing with segmentation errors in needle reconstruction for MRI-guided brachytherapy. *Medical imaging 2025: Image-guided procedures, robotic interventions, and modeling, proc. SPIE 13408*, p 1340826. https://doi.org/10.1117/12.3043274.

Ladbury, C., Harkenrider, M., Taunk, N., Fisher, C., Mayadev, J., Venkat, P., Yashar, C., Gaffney, D., Beriwal, S., and Glaser, S. 2023. A practical guide to hybrid interstitial/intracavitary brachytherapy for locally-advanced cervical cancer. *Brachytherapy*, **22**, 640–8. https://doi.org/10.1016/j.brachy.2023.07.001.

Lee, C.Y., Kaza, E., Harris, T.C., O'Farrell, D.A., King, M.T., Dyer, M.A., Cormack, R.A., and Buzurovic, I. 2023. Catheter reconstruction and dosimetric verification of MRI-only treatment planning (MRTP) for interstitial HDR brachytherapy using PETRA sequence. *Physics in Medicine & Biology*, **68**, 035002. https://doi.org/10.1088/1361-6560/acaf48.

Li, X., Young, A.S., Raman, S.S., Lu, D.S., Lee, Y.H., Tsao, T.C., and Wu, H.H. 2020. Automatic needle tracking using mask r-CNN for MRI-guided percutaneous interventions. *International journal of computer assisted radiology and surgery*, **15**, 1673–84. https://doi.org/10.1007/s11548-020-02226-8.

Liu, D., Tupor, S., Singh, J., Chernoff, T., Leong, N., Sadikov, E., Amjad, A., and Zilles, S. 2022. The challenges facing deep learning-based catheter localization for ultrasound guided high-dose-rate prostate brachytherapy. *Medical Physics*, **49**, 2442–51. https://doi.org/10.1002/mp.15522.

Major, T., Vízkeleti, J., and Takácsi-Nagy, Z. 2026. Geometric accuracy of applicator and needle reconstruction in gynecological brachytherapy using low-field magnetic resonance imaging. *Journal of Contemporary Brachytherapy*, **18**, 288–95. https://doi.org/10.5114/jcb.2026.164049.

Malajovich, I., Anamalayil, S., Dolney, O.V., Kevin Teo, B.K., Arscott, W.T., and Taunk, N.K. 2020. Techniques for and uncertainties of MRI-based reconstruction of titanium tandem and ring brachytherapy applicators. *Brachytherapy*, **19**, 651–8. https://doi.org/10.1016/j.brachy.2020.05.005.

Mastmeyer, A., Pernelle, G., Barber, L., Pieper, S., Fortmeier, D., Wells, S., Handels, H., and Kapur, T. 2017a. Model-based catheter segmentation in MRI-images. URL: https://arxiv.org/abs/1705.06712.

Mastmeyer, A., Pernelle, G., Ma, R., Barber, L., and Kapur, T. 2017b. Accurate model-based segmentation of gynecologic brachytherapy catheter collections in MRI-images. *Medical Image Analysis*, **42**, 173–88. https://doi.org/10.1016/j.media.2017.06.011.

Mehrtash, A., Damato, A., Pernelle, G., Barber, L., Farhat, N., Viswanathan, A., Cormack, R., and Kapur, T. 2014. EM-navigated catheter placement for gynecologic brachytherapy: An accuracy study. *Medical imaging 2014: Image-guided procedures, robotic interventions, and modeling, proc. SPIE 9036*, vol 9036, p 90361F. https://doi.org/10.1117/12.2044381.

Moradi, M.M., Siavashpour, Z., Takhtardeshir, S., Showkatian, E., Jaberi, R., Ghaderi, R., Mofid, B., and Taghizadeh-Hesary, F. 2025. Fully automatic reconstruction of prostate high-dose-rate brachytherapy interstitial needles using two-phase deep learning-based segmentation and object tracking algorithms. *Clinical and Translational Radiation Oncology*, **51**, 100925. https://doi.org/10.1016/j.ctro.2025.100925.

Ning, M.S., Vedam, S., Ma, J., Stafford, R.J., Bruno, T.L., Cunningham, M., Starks, C., Lawyer, A., Venkatesan, A.M., Wang, L., Wang, J., Olivieri, N.D., Guzman, A.B., Incalcaterra, J.R., Thaker, N.G., Joyner, M.M., Lin, L.L., Jhingran, A., Eifel, P.J., and Klopp, A.H. 2020. Clinical utility and value contribution of an MRI-positive line marker for image-guided brachytherapy in gynecologic malignancies. *Brachytherapy*, **19**, 305–15. https://doi.org/10.1016/j.brachy.2019.12.005.

Nunez, D.A., Trager, M., Beaudry, J., Cohen, G.N., Dauer, L.T., Gorovets, D., Hassan Rezaeian, N., Kollmeier, M.A., Leong, B., McCann, P., Williamson, M., Zelefsky, M.J., and Damato, A.L. 2021. Quantifying clinical severity of physics errors in

high-dose rate prostate brachytherapy using simulations. *Brachytherapy*, **20**, 1062–9. https://doi.org/10.1016/j.brachy.2021.05.007.

Pernelle, G., Mehrtash, A., Barber, L., Damato, A., Wang, W., Seethamraju, R.T., Schmidt, E.J., Cormack, R.A., Wells, W.M., Viswanathan, A.N., and Kapur, T. 2013. Validation of catheter segmentation for MR-guided gynecologic cancer brachytherapy. *Medical image computing and computer-assisted intervention – MICCAI 2013*, Lecture notes in computer science, vol 8151, (Springer), pp 380–7. https://doi.org/10.1007/978-3-642-40760-4_48.

Pötter, R., Tanderup, K., Kirisits, C., Leeuw, A. de, Kirchheiner, K., Nout, R., Tan, L.T., Haie-Meder, C., Mahantshetty, U., Segedin, B., et al. 2018. The EMBRACE II study: The outcome and prospect of two decades of evolution within the GEC-ESTRO gynaecological working group and the EMBRACE studies. *Clinical and Translational Radiation Oncology*, **9**, 48–60. https://doi.org/10.1016/j.ctro.2018.01.001.

Pötter, R., Tanderup, K., Schmid, M.P., Jürgenliemk-Schulz, I., Haie-Meder, C., Fokdal, L.U., Sturdza, A.E., Hoskin, P., Mahantshetty, U., Segedin, B., et al. 2021. MRI-guided adaptive brachytherapy in locally advanced cervical cancer (EMBRACE-I): A multicentre prospective cohort study. *The Lancet Oncology*, **22**, 538–47. https://doi.org/10.1016/S1470-2045(20)30753-1.

Prisciandaro, J., Zoberi, J.E., Cohen, G., Kim, Y., Johnson, P., Paulson, E., Song, W., Hwang, K.-P., Erickson, B., Beriwal, S., Kirisits, C., and Mourtada, F. 2022. AAPM task group report 303 endorsed by the ABS: MRI implementation in HDR brachytherapy—considerations from simulation to treatment. *Medical Physics*, **49**, e983–1023. https://doi.org/10.1002/mp.15713.

Quetin, S., Jafarzadeh, H., Kalinowski, J., Bekerat, H., Bahoric, B., Maleki, F., and Enger, S.A. 2025. Automatic catheter digitization in breast brachytherapy. *Medical Physics*, **52**, e18107. https://doi.org/10.1002/mp.18107.

Richardson, S.L., Baghwala, A., Cohen, G.N., Dempsey, C., Libby, B., Melhus, C.S., Miller, R.A., Scanderbeg, D.J., and Simiele, S.J. 2025. AAPM medical physics practice guideline 13.a: HDR brachytherapy, part B. *Journal of Applied Clinical Medical Physics*, **26**, e70118. https://doi.org/10.1002/acm2.70118.

Richart, J., Carmona-Meseguer, V., García-Martínez, T., Herreros, A., Otal, A., Pellejero, S., Tornero-López, A., and Pérez-Calatayud, J. 2018. Review of strategies for MRI based reconstruction of endocavitary and interstitial applicators in brachytherapy of cervical cancer. *Reports of practical oncology and radiotherapy : journal of Greatpoland Cancer Center in Poznan and Polish Society of Radiation Oncology*, **23**, 547–61. https://doi.org/10.1016/j.rpor.2018.06.005.

Roumeliotis, M., Antaki, M., Kim, G., Gardner, U.G., Sheikh, K., Lee, J., Seethamraju, R., Tokuda, J., Hu, C., Schmidt, E., and Viswanathan, A.N. 2026. Prospective operational feasibility, safety, and workflow of magnetic resonance-guided tracking technology for interstitial gynecologic brachytherapy. *Brachytherapy*, **25**, 741–8. https://doi.org/10.1016/j.brachy.2026.03.005.

Rylander, S., Buus, S., Pedersen, E.M., Bentzen, L., and Tanderup, K. 2017. Dosimetric impact of contouring and needle reconstruction uncertainties in US-, CT- and MRI-based high-dose-rate prostate brachytherapy treatment planning. *Radiotherapy and oncology : journal of the European Society for Therapeutic Radiology and Oncology*, **123**, 125–32. https://doi.org/10.1016/j.radonc.2017.01.007.

S, H., SK, N., JC, L., J, G., and K, T. 2009. Applicator reconstruction in MRI 3D image-based dose planning of brachytherapy for cervical cancer. *Radiotherapy and oncology : journal of the European Society for Therapeutic Radiology and Oncology*, **91**, 187–93. https://doi.org/10.1016/j.radonc.2008.09.002.

Sargazi, V., Naseri, S., Gholamhosseinian, H., and Momennezhad, M. 2026. Applicator reconstruction in cervical cancer brachytherapy: A systematic review of current methods, challenges, and AI-driven future directions. *Brachytherapy*, **25**, 262–74. https://doi.org/10.1016/j.brachy.2025.10.004.

Schindel, J., Zhang, W., Bhatia, S.K., Sun, W., and Kim, Y. 2013. Dosimetric impacts of applicator displacements and applicator reconstruction-uncertainties on 3D image-guided brachytherapy for cervical cancer. *Journal of Contemporary Brachytherapy*, **5**, 250–7. https://doi.org/10.5114/jcb.2013.39453.

Shaaer, A., Paudel, M., Smith, M., Tonolete, F., Nicolae, A., Leung, E., and Ravi, A. 2020. Evaluation of an MR-only interstitial gynecologic brachytherapy workflow using MR-line marker for catheter reconstruction. *Brachytherapy*, **19**, 642–50. https://doi.org/10.1016/j.brachy.2020.06.007.

Shaaer, A., Paudel, M., Smith, M., Tonolete, F., and Ravi, A. 2022. Deep-learning-assisted algorithm for catheter reconstruction during MR-only gynecological interstitial brachytherapy. *Journal of Applied Clinical Medical Physics*, **23**, e13494. https://doi.org/10.1002/acm2.13494.

Shit, S., Koner, R., Wittmann, B., Paetzold, J., Ezhov, I., Li, H., Pan, J., Sharifzadeh, S., Kaissis, G., Tresp, V., and Menze, B. 2022. Relationformer: A unified framework for image-to-graph generation. *Computer vision – ECCV 2022, lecture notes in computer science*, (Springer), pp 422–39. https://doi.org/10.1007/978-3-031-19836-6_24.

Shit, S., Paetzold, J.C., Sekuboyina, A., Ezhov, I., Unger, A., Zhylka, A., Pluim, J.P.W., Bauer, U., and Menze, B.H. 2021. clDice – a novel topology-preserving loss function for tubular structure segmentation. *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition (CVPR)*, pp 16555–64. https://doi.org/10.1109/CVPR46437.2021.01629.

Singh, A., Karshafian, R., Ravi, A., and Paudel, M.R. 2026. Dosimetric evaluation of systematic source position uncertainty along the source path and robust planning during MRI-only cervical HDR brachytherapy. *Journal of Applied Clinical Medical Physics*, **27**, e70810. https://doi.org/10.1002/acm2.70810.

Tanderup, K., Hellebust, T.P., Lang, S., Granfeldt, J., Pötter, R., Lindegaard, J.C., and Kirisits, C. 2008. Consequences of random and systematic reconstruction uncertainties in 3D image based brachytherapy in cervical cancer. *Radiotherapy and Oncology*, **89**, 156–63. https://doi.org/10.1016/j.radonc.2008.06.010.

Vijayan, S., Theeler, J.J., Fu, Z., and Kim, Y. 2026. Systematic review of artificial intelligence in brachytherapy. *Journal of Contemporary Brachytherapy*, **18**, 85–114. https://doi.org/10.5114/jcb.2026.160595.

W, Q., M, Y., M, D., D, T., and A, F. 2013. Needle segmentation using 3D hough transform in 3D TRUS guided prostate transperineal therapy. *Medical physics*, **40**, 042902. https://doi.org/10.1118/1.4795337.

Wang, F., Xing, L., Bagshaw, H., Buyyounouski, M., and Han, B. 2020. Deep learning applications in automatic needle segmentation in ultrasound-guided prostate brachytherapy. *Medical Physics*, **47**, 3797–805. https://doi.org/10.1002/mp.14328.

Wang, W., Viswanathan, A.N., Damato, A.L., Chen, Y., Tse, Z., Pan, L., Tokuda, J., Seethamraju, R.T., Dumoulin, C.L., Schmidt, E.J., and Cormack, R.A. 2015. Evaluation of an active magnetic resonance tracking system for interstitial brachytherapy. *Medical physics*, **42**, 7114–21. https://doi.org/10.1118/1.4935535.

Wang, Y., Jian, W., Zhu, L., Cai, C., Zhang, B., and Wang, X. 2024. Attention-gated deep-learning-based automatic digitization of interstitial needles in high-dose-rate brachytherapy for cervical cancer. *Advances in Radiation Oncology*, **9**, 101340. https://doi.org/10.1016/j.adro.2023.101340.

Weishaupt, L.L., Sayed, H.K., Mao, X., Choo, R., Stish, B.J., Enger, S.A., and Deufel, C.L. 2022. Approaching automated applicator digitization from a new angle: Using sagittal images to improve deep learning accuracy and robustness in high-dose-rate prostate brachytherapy. *Brachytherapy*, **21**, 520–31. https://doi.org/10.1016/j.brachy.2022.02.005.

Xiao, Z., Liu, B., Wang, J., Yao, Q., Sun, H., Zhou, F., Wang, J., Wu, Q., and Jiang, P. 2026a. Multi-catheter digitization in brachytherapy via few-shot synthetic-to-real learning and structure-aware tracking. *arXiv preprint arXiv:2607.11290*. URL: https://arxiv.org/abs/2607.11290.

Xiao, Z., Zhou, F., Wang, J., He, C., Liu, B., Sun, H., Ji, Z., Jiang, Y., Wang, J., and Wu, Q. 2026b. Multi-needle localization for pelvic seed implant brachytherapy based on tip-handle detection and matching. *IEEE Journal of Biomedical and Health Informatics*, 1–14. https://doi.org/10.1109/JBHI.2026.3694701.

Xie, H., Wang, J., Chen, Y., Tu, Y., Chen, Y., Zhao, Y., Zhou, P., Wang, S., Bai, Z., and Tang, Q. 2023. Automatic reconstruction of interstitial needles using CT images in post-operative cervical cancer brachytherapy based on deep learning. *Journal of Contemporary Brachytherapy*, **15**, 134–40. https://doi.org/10.5114/jcb.2023.126514.

Zaffino, P., Pernelle, G., Mastmeyer, A., Mehrtash, A., Zhang, H., Kikinis, R., Kapur, T., and Spadea, M.F. 2019. Fully automatic catheter segmentation in MRI with 3D convolutional neural networks: Application to MRI-guided gynecologic brachytherapy. *Physics in Medicine & Biology*, **64**, 165008. https://doi.org/10.1088/1361-6560/ab2f47.

Zhang, D., Yang, Z., Jiang, S., Zhou, Z., Meng, M., and Wang, W. 2020a. Automatic segmentation and applicator reconstruction for CT-based brachytherapy of cervical

cancer using 3D convolutional neural networks. *Journal of Applied Clinical Medical Physics*, **21**, 158–69. https://doi.org/10.1002/acm2.13024.

Zhang, H., Donnelly, E.D., Strauss, J.B., Kang, Z., Gopalakrishnan, M., Lee, P.C., Khelashvili, G., Nair, C.K., Lee, B.H., and Sathiaseelan, V. 2019. Clinical implementation, logistics and workflow guide for MRI image based interstitial HDR brachytherapy for gynecological cancers. *Journal of applied clinical medical physics*, **20**, 37–49. https://doi.org/10.1002/acm2.12736.

Zhang, Y., He, X., Tian, Z., Jeong, J.J., Lei, Y., Wang, T., Zeng, Q., Jani, A.B., Curran, W.J., Patel, P., Liu, T., and Yang, X. 2020b. Multi-needle detection in 3D ultrasound images using unsupervised order-graph regularized sparse dictionary learning. *IEEE transactions on medical imaging*, **39**, 2302–15. https://doi.org/10.1109/TMI.2020.2968770.

Zhang, Y., Lei, Y., Qiu, R.L.J., Wang, T., Wang, H., Jani, A.B., Curran, W.J., Patel, P., Liu, T., and Yang, X. 2020c. Multi-needle localization with attention u-net in US-guided HDR prostate brachytherapy. *Medical physics*, **47**, 2735–45. https://doi.org/10.1002/mp.14128.

Zhang, Y., Tian, Z., Lei, Y., Wang, T., Patel, P., Jani, A.B., Curran, W.J., Liu, T., and Yang, X. 2020d. Automatic multi-needle localization in ultrasound images using large margin mask RCNN for ultrasound-guided prostate brachytherapy. *Physics in Medicine & Biology*, **65**, 205003. https://doi.org/10.1088/1361-6560/aba410.

Zheng, Y., Jiang, S., Yang, Z., and Wei, L. 2021. Automatic needle detection using improved random sample consensus in CT image-guided lung interstitial brachytherapy. *Journal of Applied Clinical Medical Physics*, **22**, 121–31. https://doi.org/10.1002/acm2.13231.

Zhou, W., Dai, Q., Curiel, O., Tsao, T.C., Chiang, J., Lu, D.S., and Wu, H.H. 2026. Keypoint detection network for needle localization on intra-procedural MRI in MRI-guided liver interventions. *International journal of computer assisted radiology and surgery*, **21**, 1775–88. https://doi.org/10.1007/s11548-026-03592-5.

Zhou, W., Li, X., Zabihollahy, F., Lu, D.S., and Wu, H.H. 2024. Deep learning-based automatic pipeline for 3D needle localization on intra-procedural 3D MRI. *International Journal of Computer Assisted Radiology and Surgery*, **19**, 2227–37. https://doi.org/10.1007/s11548-024-03077-3.

A APPENDIX A: TILT INVARIANCE AND NETWORK SPECIFICATIONS

1. Oblique catheter tilt invariance of the Stage 0 circular dark-core filter. Consider a cylindrical 6F plastic catheter of physical radius $r_c = 1.0$ mm inclined at an angle α relative to the axial slice normal $\hat{\mathbf{z}}$. Its intersection with an axial plane $z = z_0$ is an ellipse with semi-minor axis $a = r_c = 1.0$ mm perpendicular to the tilt projection and semi-major axis $b = r_c \sec\alpha$ along the tilt projection. For clinical perineal template and freehand insertions ($\alpha \leq 25^\circ$–30°), $\sec\alpha \leq 1.155$, so the in-plane dark core extends at most $b \leq 1.16$ mm from the centerline. Because Stage 0 evaluates six diameters across four physical radii $\mathcal{R} = \{1.35, 1.85, 2.50, 3.30\}$ mm and takes the outer radial maximum over the rank-2 and rank-3 directional order statistics,

$$S_{\text{ring}}(\mathbf{p}) = \max_{r\in\mathcal{R}} \left[0.45\, m_{(2),r} + 0.55\, m_{(3),r}\right],$$

any sampling radius $r \geq 1.85$ mm clears the obliquely elongated dark core along all six diameters for $\alpha \leq 30^\circ$, while tolerating one adjacent touching catheter ($m_{(1),r}$). When slice thickness is coarse ($\Delta z \in [4.0, 6.0]$ mm) and through-plane partial-volume averaging shifts the apparent dark core across $\Delta z \tan\alpha$ within the slice, initial cubic z-upsampling to 1.6 mm spacing restores axial continuity prior to Stage 0 evaluation.

2. `SynthBrachy3T` physical K-space and bundle simulation parameters. To pre-train the Stage 2B bundle graph transformer without clinical bundle annotations, `SynthBrachy3T` generates 300 realistic candidate bundles per fold. Each synthetic implant samples $N_{\text{true}} \sim \mathcal{U}\{12, 32\}$ physical catheters entering through a 60×50 mm^2 perineal template grid around a central vaginal obturator ($R_{\text{obt}} \sim \mathcal{U}[10, 15]$ mm), with correlated 3D quadratic deflections, 15%–25% touching twin pairs (≤ 3.0 mm clearance), 20% short obturator guide channels, random 1-to-3-slice mid-shaft signal dropouts, and $N_{\text{FP}} \sim \mathcal{U}\{6, 22\}$ vascular and fascial false-positive trajectories sampled at higher peripheral curvature and peripheral radial distances.

3. Summary of neural network hyperparameters across Stages 1 and 2.

- **Stage 1 (`NeedleUNet3D`):** 6 input channels, 4 resolution scales with `[24, 48, 96, 144]` channels, squeeze-and-excitation residual blocks, in-plane-only (1, 2, 2) pooling at the first two scales, 4 output heads ($P_{\text{mask}}, P_{\text{peak}}, V_x, V_y, P_{\text{obt}}$). Pre-trained for 45 epochs on 51 external pelvic MRI volumes (AdamW, learning rate 10^{-3}, weight decay 10^{-4}) and fine-tuned for 60 epochs per inner fold on dual-rater consensus traces.
- **Stage 2A (`NeedleSliceMPNN` + Collinear Stitcher):** 16-D node features, 10-D directed edge features across $\Delta k \in \{1, 2, 3, 4\}$ slices (≤ 24 mm), 4 triplet message-passing layers of hidden width 64 with cosine bending-angle attention; collinear stitcher bridges axial gaps ≤ 20 mm with midpoint error ≤ 2.2 mm and Δslope ≤ 0.14.
- **Stage 2B (`TipAwareBundleGraphTransformer`):** 26-D trajectory node tokens, 8-D pairwise geometric interaction edge features, 2 edge-biased local graph attention layers + 2 global bundle self-attention layers (width 96, 4 heads), temperature-calibrated on the 9 inner-validation patients per fold.

– **Stage 2C (`Dedicated1DTipLocalizer`):** 91 axial steps at 0.5 mm spacing over [−25.0, +20.0] mm along the cranial tangent, 22 channels (24 in 4d), 3 dilated 1D residual blocks ($d \in \{1, 2, 4\}$, kernel size 5, width 64) + 4-head 1D self-attention, trained with KL-divergence stopping loss and smooth-L_1 sub-slice offset regression, calibrated via split-conformal quantiles $\hat{q}_{1-\alpha}$ on the inner validation split.

B APPENDIX B: STAGE-BY-STAGE ABLATION AND ERROR AUDIT

Table 5 reports the complete 12-row stage-by-stage architectural ablation ladder across all 54 patients (1,063 catheters), on the 759 full-depth active-dwell shafts, and on the 23-patient severe catheter-adhesion subgroup, together with the ground-truth mask oracle upper bound for Xiao *et al.* (2026a).

TABLE 5 **Controlled architectural ablation and stage-by-stage progression on all 54 BWH patients (1 063 catheters, strictly nested 5 × 5 out-of-fold).** Traces the sequential addition of each mechanism from Stage 0 analytical dark-core symmetry through Stage 2C 1D tip profiling and literature fusion, alongside the oracle ground-truth mask upper bound.

	Detection, 1-to-1 (%)			E_{shaft} (mm)			E_{tip} (mm)		Time (s)
Method / Stage Configuration	Rec.	Prec.	F1	Mean	Median	P95	Mean	Median	CPU
Sequential stage-by-stage progression (whole cohort, 1 063 catheters)									
Stage 0: Dark-core prior (`k_lim_auto`)	82.8	69.1	75.3	0.60	0.47	1.41	9.91	3.08	**0.27**
Stage 1B: SSL + Multi-Rater U-Net + tubes	**94.2**	61.5	74.4	**0.57**	**0.46**	1.31	9.29	2.95	2.84
Stage 2A: Triplet MPNN only	93.9	73.5	82.5	0.58	0.47	1.31	8.95	2.83	4.09
+ Stage 2B: Bundle transformer + calibration	90.9	83.6	87.1	**0.57**	**0.46**	**1.30**	7.39	2.70	4.12
+ Stage 2B: Multi-rater consensus supervision	91.7	81.7	86.4	0.58	**0.46**	1.32	7.38	2.57	4.12
+ Stage 2A/B: Collinear split-fragment stitch	91.4	86.9	89.1	0.59	0.47	1.36	6.92	2.56	4.13
+ Stage 2C: 1D Tip Localizer (τ = 35 %)	93.6	83.4	88.2	0.59	0.47	1.37	6.71	2.32	4.15
+ Stage 2C: 1D Tip Localizer (τ = 50 %, 4b)	91.4	86.9	89.1	0.59	0.47	1.37	**6.23**	2.27	4.15
+ Stage 2C: 1D Tip Localizer (τ = 55 %)	91.3	**87.6**	**89.4**	0.59	0.47	1.37	6.33	2.27	4.15
+ Lit. Fusion (Dai TV + Kostoulas + Kim, 4d)[a]	91.4	87.4	**89.4**	0.58	0.47	1.36	**6.23**	**2.26**	4.18
Xiao 2026 on GT masks (oracle upper bound)[b]	97.7[r]	83.5[r]	90.0[r]	0.20[r]	0.11[r]	0.83[r]	8.94[r]	1.68[r]	0.12[r]

[a] Combines Kim *et al.* (2026) inverse-variance sub-voxel Poly3 fitting, Dai *et al.* (2020) axial TV continuity + Aleong *et al.* (2024) 5-model epistemic uncertainty features, and Kostoulas *et al.* (2025) iterative multi-segment gap-bridging.
[b] Oracle upper-bound check following Xiao *et al.* (2026) Table 2 Part B (tracker run directly on ground-truth binary masks). Source: `2026-10-23-unified-clean-nested-benchmark.json` and `2026-10-24-peer-reviewed-baselines-and-fusion.json`.

1. Itemized stage-by-stage progression analysis:

– **Stage 0 alone (physical dark-core filter):** Using zero learned parameters, Stage 0 with automatic template cropping already finds 82.8% of all catheters (95.3% of long shafts) with 69.1% precision (**75.3% F1**, 0.47 mm median shaft error, 3.08 mm median / 9.91 mm mean tip error). However, it cannot distinguish round blood vessels from catheters.

- **Adding Stage 1B (self-supervised, multi-rater 3D U-Net):** Regressing sub-pixel centerline peaks and 3D directions boosts recall to **94.2%** (97.0% on long shafts) and cuts mean tip error to 9.29 mm (2.95 mm median), but because every candidate seed is still tracked greedily, each patient carries 11.6 false vascular tracks on average (61.5% precision, 74.4% F1).
- **Adding Stage 2A (triplet message-passing graph network):** Penalizing kinks across consecutive slice links eliminates 43% of false tracks while keeping **93.9% recall**, lifting precision by +12.0 percentage points to 73.5% (**82.5% F1**, 8.95 mm mean / 2.83 mm median tip error).
- **Adding Stage 2B (bundle graph transformer, multi-rater supervision, and collinear fragment stitching):** Evaluating the candidate bundle as a whole (87.1% F1 with temperature calibration; 86.4% F1 with multi-rater Stage 2 supervision) and stitching collinear fragments separated by 1-to-3-slice dropouts raises precision to **86.9%** and F1 to **89.1%** (91.4% recall) while reducing mean tip error to 6.92 mm (2.56 mm median).
- **Adding Stage 2C (dedicated 1D axial tip localizer, 4b):** Inspecting the 0.5 mm sub-slice axial profile at the cranial end of each trajectory cuts mean 3D tip error from 6.92 mm down to 6.23 **mm** and median tip error from 2.56 mm down to 2.27 **mm** (95th percentile reduced to 27.27 mm).
- **Adding Literature-Inspired Fusion (4d: sub-voxel inverse-variance cubic fitting + epistemic uncertainty & axial TV continuity + iterative multi-segment gap-bridging):** Inverse-variance-weighted sub-voxel cubic fitting (Aleong *et al.*, 2024; Kim *et al.*, 2026) reduces mean shaft error from 0.589 mm to 0.583 **mm** ($p = 1.84 \times 10^{-8}$; long-shaft median 0.45 → 0.44 mm); epistemic uncertainty & axial TV continuity features (Dai *et al.*, 2020; Aleong *et al.*, 2024) reduce median 3D tip error to 2.26 **mm** (axial tip median 1.99 → 1.94 mm, tips ≤ 3 mm 58.7% → 59.0%); and iterative multi-segment gap-bridging & polynomial split suppression (Kostoulas *et al.*, 2025) eliminate 6 residual split-fragment false alarms, raising precision to **87.4%** and F1 to **89.4%** ($p = 0.028$).

2. Complete catheter-level root-cause audit of residual errors across all 54 patients:

- **What causes the 146 unmatched predictions (2.7 per patient)?** Nearly a third (**31.5%**, 46/146) are residual split or near-duplicate segments of an already-detected catheter, **20.5% (**30/146**)** correspond to real catheters supported by Expert Team 2 within 5.0 mm (and, among the 100 unmatched predictions on the 38 CT-paired patients, 52.0% correspond to a catheter on CT), and 2.7% (4/146) lie within 3.5–6.0 mm of an unmatched reference track. Only **45.2% (**66/146**, or 1.22 per patient)** are genuine background false alarms (> 6.0 mm from any reference catheter), occurring along dark pelvic vessels or fascial intersections.
- **What causes the 91 missed catheters (1.7 per patient)?** Before Stage 2B thresholding, our 1,960-track candidate pool contains **97.4% (**1,035/1,063**)** of all reference catheters within 3.5 mm (and **98.8%** [750/759] of the long active-dwell shafts). Fewer than a third of missed catheters (**30.8%,** 28/91) are lost upstream due to severe multi-slice signal dropouts. The remaining **69.2%**

(63/91) are traced by Stages 1 and 2A but filtered out in Stage 2B—most often because their bundle confidence falls just below the $\tau = 50\%$ threshold (46 catheters) or because they lose one-to-one matching inside a tight cluster of touching needles (15 catheters). Lowering τ to 35% recovers more than a quarter of these misses, raising whole-cohort recall to **93.6%** and long-shaft recall to **94.7%** (precision 83.4%, F1 88.2%).

– **Why do some tips have errors > 10 mm?** Decomposing every 3D tip error vector into its longitudinal component ($\Delta s_{\parallel}$) and lateral component ($\Delta r_{\perp}$) shows that **98.8% of total squared 3D tip error across all 972 matched catheters (and 99.3% in the > 10 mm tail) is purely longitudinal along the catheter track**, whereas lateral tip error is 0.72 mm median. When a tip error exceeds 10 mm (162 catheters, 16.7%), it is driven by mid-shaft signal dropout (41.4%), dark bowel/air pockets sitting directly atop the catheter tip (25.3%), intrinsic human inter-rater tip ambiguity > 6 mm (16.7%), crowded tip arrays (12.3%), or top-of-scan truncation (4.3%).

C APPENDIX C: SPECIFICITY AND MULTI-MODALITY LITERATURE

Table 6 details the 5-row zero-needle false-positive specificity study across all 48 public needle-free 3T T2W pelvic MRI volumes (`MSD Task05_Prostate` (Antonelli *et al.*, 2022)), and Table 7 compares NeedleFinder 2026 against published catheter digitization methods across CT, ultrasound, electromagnetic tracking, and MRI.

TABLE 6 **Zero-needle out-of-distribution specificity on 48 public clinical 3T pelvic T2W MRI volumes (MSD `Task05_Prostate`).** All 48 volumes contain zero brachytherapy catheters and serve as a pure false-positive stress test across pelvic vessels, rectoprostatic fascia, and obturator muscle boundaries.

	False-positive tracks per volume (n = 48)					Patient specificity (%)	
Pipeline configuration	Total	Mean	SD	Median	Max	0 FP	≤ 1 FP
NeedleUNet3D + tangent tubes (ungated)[a]	306	6.38	4.77	6.0	19	14.6	18.8
Unpruned Stage 0 + Stage 1 pool[b]	294	6.12	4.70	5.0	23	4.2	10.4
Ungated Unified **NeedleFinder 2026** (τ = 50 %)[c]	197	4.10	4.45	3.0	23	16.7	27.1
Ungated Unified **NeedleFinder 2026** (τ = 60 %)[c]	171	3.56	3.48	3.0	17	16.7	29.2
Stage 1 U-Net + Stage 0 radial gate[d]	18	0.38	0.53	**0.0**	2	64.6	97.9
Frozen Unified **NeedleFinder 2026** + Dark-Core Gate (τ = 50 %)[e]	16	0.33	0.75	**0.0**	3	79.2	91.7
Clean SSL + Multi-Rater Stage 1 + Stage 0 gate[f]	**6**	**0.125**	**0.33**	**0.0**	**1**	**87.5**	**100.0**

[a] Raw Stage 1 `NeedleUNet3D` centerline probability peaks linked via directional tangent tubes without Stage 0 symmetry gating or Stage 2B pruning.

[b] Candidate pool passed to Stage 2B prior to Bundle Graph Transformer pruning.

[c] Unified **NeedleFinder 2026** pipeline evaluated without the Stage 0 dark-core gate at τ = 50 % (4.10 FP/scan) and τ = 60 % (3.56 FP/scan).

[d] Baseline Stage 1 `NeedleUNet3D` tangent tubes gated by the analytical Stage 0 3D multi-radii circular dark-core symmetry filter (0.38 FP/scan).

[e] Full Unified **NeedleFinder 2026** pipeline with Stage 0 dark-core symmetry gating at τ = 50 % (16 total FPs = 0.333/scan; 38/48 [79.2 %] 0-FP scans; 44/48 [91.7 %] ≤ 1-FP scans).

[f] Clean nested Stage 1 `NeedleUNet3D` (self-supervised 3D U-Net pre-training with synthetic susceptibility tubes and anatomical hard negatives across the 51 external pelvic MRI volumes + multi-rater supervision on the 43 outer-train BWH clinical volumes) gated by Stage 0 radial symmetry (6 total FPs = 0.125/scan; 42/48 [87.5 %] 0-FP scans; 48/48 [100.0 %] ≤ 1-FP scans). Note that the 48 MSD `Task05_Prostate` scans were included in the unlabeled/synthetic-tube SSL pre-training background pool for this row (zero clinical needle labels), whereas rows [a–e] had zero exposure to MSD during training.

TABLE 7 **Published automatic catheter and needle reconstruction studies in brachytherapy (2013–2026), grouped by imaging modality.** Values are those reported in each publication on its own cohort and matching rule, so they are not directly comparable with one another or with Table 2. Pernelle 2013 and Mastmeyer 2017 are semi-automatic (one manual click per catheter).

Study	Modality	Site	Pts	Cath.	Rec./Prec. (%)	E_{shaft}	E_{tip}	Time Stat.
CT								
Weishaupt *et al.* 2022	CT	Prostate	57[l]	353[a]	–	0.13	–	~60 s mean
Xie *et al.* 2023	CT	GYN post-op	70	210	–[b]	–	–	– Dice / HD95
Wang *et al.* 2024	CT	GYN IC/IS	17	56[c]	–	1.8	1.1	– mean
Quetin *et al.* 2025	CT	Breast	117	408[a]	–	0.70	1.37	118 s mean
Xiao *et al.* 2026[m]	CT	GYN recurrent	38	203[c]	91.1 / 90.7	0.31	0.88	11.6 s mean
Xiao *et al.* 2026 (JBHI)	CT	Pelvic seeds	100		– F1 90.3	–	1.05	9.4 s mean tip
Ultrasound								
Zhang *et al.* 2020	TRUS	Prostate	23	339	98 / –	0.09	0.33	<1 s mean
Andersén *et al.* 2020	TRUS	Prostate	1102[l]	7207[l]	100 / –	0.55[c]	–	– median RMS
Liu *et al.* 2022	TRUS	Prostate	242	3500	80[d] / –	–	–	– frac. ≤ 2 mm
Goulet *et al.* 2025	3D US	Prostate	102[o]		– 93.4 / –	0.44	–	– mean
MRI								
Pernelle *et al.* 2013	3T T2W	GYN IC/IS	10	101	92.1 / –	–[e]	–	0.3 s/cath. frac. < 2 mm
Mastmeyer *et al.* 2017	3T T2W	GYN IC/IS	54	760	93[f] / –	1.49[g]	–	3 s/cath. mean HD
Zaffino *et al.* 2019	3T T2W	GYN IC/IS	50	826	98.5 / 93.3[h]	2.0	–	2–9 min mean dist.
Dai *et al.* 2020	MRI	Prostate	20	299	–	0.93	0.37[n]	– mean / signed
Shaaer *et al.* 2022	1.5T + markers	GYN IS	20	360	100 / –	0.97	–	11 min[i] mean shift
Aleong *et al.* 2024	MRI	Prostate	35	214[o]	95 / –	–	–	0.9 s –
Kostoulas *et al.* 2025	MRI	Prostate	34[o]	261	100 / 100	0.75[j]	1.07[j]	– median
Kim *et al.* 2026	T2W	GYN hybrid	30	35[o]	98.5 / –	0.74	2.52	37 s mean
NeedleFinder 2026 (all 1 063)	3T T2W	GYN IC/IS	54	1063	91.4 / 86.9	0.47[j]	2.27[j]	**4.15 s** median
NeedleFinder 2026 (759 long)	3T T2W	GYN IC/IS	54	759	91.3 / 88.7[k]	0.45[j]	2.41[j]	**4.15 s** median

[a] Test-set catheters.
[b] Reported as Dice 0.88–0.90 and HD_{95} 0.71–0.77 mm.
[c] Root-mean-square distance to the clinical reconstruction.
[d] Fraction of catheters within 2 mm along 90 % of their length.
[e] Maximum Hausdorff distance below 2 mm for 92 % of catheters.
[f] Fraction within 3.4 mm.
[g] Mean Hausdorff distance.
[h] Cluster-level: 1.5 % false negatives and 6.7 % false positives among 826 catheters.
[i] Assisted reconstruction time, versus 46 min manual.
[j] Median (means: 0.59 ± 0.40 mm shaft and 6.23 ± 10.49 mm tip on all 1 063 catheters; 0.57 ± 0.39 mm shaft and 7.41 ± 12.23 mm tip on the 759 long shafts).
[k] Strict one-to-one recall at $\tau = 50$ % (92.5 % with inner-OOF cross-fit; 94.7 % at $\tau = 35$ %); precision counts predictions matched to short obturator tracks as correct (90.0 % $F1_{adj}$, 90.5 % with inner-OOF cross-fit). Times for **NeedleFinder 2026** are full-volume uncached per-patient averages across all 54 scans on CPU (4.15 s overall: 1.26 s on the 20 standard 320 × 256 scans, 5.85 s on the 34 high-resolution 640 × 416 / 768 × 768 scans) and one RTX 4090 GPU (2.67 s overall: 0.67 s standard, 3.85 s high-resolution).
[l] Treatments, not patients; 7207 of 24422 needles formed the evaluation set.
[m] Fully supervised model; their 10-shot synthetic-to-real variant reports 89.4 / 89.0, 0.33 mm and 0.90 mm.
[n] Signed along-axis tip displacement (mean ± SD 0.37 ± 1.68 mm), not an absolute 3D distance.
[o] Reported in plans (Wang 2024: 56 plans), fractions (Xiao 2026: 203 fx), scans (Goulet 2025: 102 scans; Kostoulas 2025: 34 scans), or volumes (Aleong 2024: 214 vol.; Kim 2026: 35 vol.) rather than individual catheters/patients.

D APPENDIX D: EXTENDED DOSIMETRY AND GENERALIZATION TABLES

Tables 8 and 9 provide all 11 cross-organ, multi-sequence, 4D intra-procedural insertion, and paired Pelvic CT generalization configurations, together with the complete protocol-by-protocol AAPM TG-43U1 ^{192}Ir dosimetric distributions (including bladder, rectum, and sigmoid proxy D_{2cc} differences), simulated clinician tip-review curves ($k \in \{0, 1, 2, 3, 5, 8, 10\}$), and split-conformal coverage at 80%–95% nominal levels.

TABLE 8 **Zero-shot transfer across organs, secondary MRI pulse sequences (including T1 VIBE failure boundary), 4D serial insertion, and paired post-implant Pelvic CT (Part C supporting Table 4).**

Part C: Cross-organ, seq., 4D & CT[a]	n_{vol}	n_{cath}	Rec. %	Prec. %	F1 %	Sh. mm	Tip mm	Key finding
Prostate 3T: Stage 0 (`k_lim_auto=40`)	1	14	**100.0**	87.5	93.3	0.34	2.09	14/14, 2 FPs
Prostate 3T: Zero-Exp. No-SSL (`auto=40`, τ=50)	1	14	92.9	**100.0**	96.3	0.33	**0.99**	13/14, 0 FPs
Prostate 3T: Zero-Exp. No-SSL (`ACSV=47`, τ=50)	1	14	**100.0**	**100.0**	**100.0**	0.34	1.57	14/14, 0 FPs
Prostate 3T: Stage 1 Full Nested (`auto=40`)	1	14	**100.0**	**100.0**	**100.0**	0.30	1.08	14/14, 0 FPs
Prostate 3T: Stage 0 zeroed ($X_{1:3}$ = 0, `auto=40`)	1	14	0.0	–	0.0	–	–	985× drop
Prostate 3T: 1-ch ResUNet + clDice (`auto=40`)	1	14	**100.0**	56.0	71.8	**0.24**	2.48	11 vascular FPs
3D wbSSFP / True-FISP ($\Delta z \leq 2.5$ mm)	2	47	**97.9**	88.5	**92.9**	0.72	3.40	vs. 10.62 PMB19
Thick-slice 2D AX T2 ($\Delta z \leq 6.0$ mm)	49	882	88.4	67.9	76.9	1.07	4.89	vs. 57.6 % 1-ch
Post-contrast AX T1 VIBE ($\Delta z \leq 4.0$ mm)	12	217	58.5	53.8	56.1	1.87	8.08	vs. 61.5 % Stage 0
4D Serial MRI: Indep. 3D (Cases 68–71)	23	108	88.0	76.0	81.6	0.45	2.71	4-pt pooled
4D Serial MRI: Causal Kalman (Cases 68–71)	23	108	77.8	**88.4**	82.8	0.75	3.57	+12.4 % prec.
Paired Pelvic CT (38 paired / 40 vols, ≤ 5 mm)	38	801	79.2	–	–	2.01	2.78	52.0 % extra conf.

[a] 'Sh. mm' and 'Tip mm' report median in-plane shaft and 3D tip errors (mm). 'Zero-Exp. No-SSL' denotes the 5-fold GYN ensemble trained with zero prostate exposure (excluding Case 100 from SSL pre-training; `auto=40` uses automatic `k_lim_auto` slice 40, whereas `ACSV=47` uses the clinical template crop slice 47). On 'Thick-slice 2D AX T2', 882 is the number of reference catheters exhibiting a physical dark-core signal in the thick-slice slab (out of 1 019 geometrically intersecting). In 4D monitoring (4 patients, Cases 68–71, 23 serial scans, 1.31 s mean monitoring runtime, each evaluated strictly with its held-out outer-fold model), per-case final-scan F1s (Independent 3D → Causal 4D Kalman) are Case 68: 77.2 % → 83.0 %, Case 69: 84.7 % → 82.7 % (precision 83.7 % → 93.9 %), Case 70: 85.7 % → 93.3 %, and Case 71: 76.2 % → 66.7 % (precision 72.7 % → 100.0 %); mean tip error improves from 11.23 → 9.75 mm. Persisted catheters within ≤ 10 mm of newly inserted needles exhibit 1.44 mm median (1.61 mm mean) lateral shift vs. 1.10 mm at 10–20 mm and 1.00 mm (1.19 mm mean) at > 20 mm (Mann–Whitney $p = 9.97 \times 10^{-4}$ for ≤ 12 mm vs. > 12 mm); depth deflection reaches 3.29 mm median (4.56 mm mean, 10.49 mm P90) at > 90 mm depth ($\rho = 0.555$, $p = 5.55 \times 10^{-9}$). In Pelvic CT, 38 of 40 scanned CT volumes pair with the 54-case MRI benchmark (801 MRI-paired reference catheters evaluated for cross-modality correspondence across non-rigid inter-scan deformation: 68.9 % matched at ≤ 3.5 mm, 79.2 % at ≤ 5.0 mm; 52/100 [52.0 %] extra MRI detections vs. `CY` confirmed on CT).

TABLE 9 **Dosimetric translation and simulated clinician tip triage (Parts A & B; complete breakdown supporting Table 4).** Part A evaluates AAPM TG-43U1 HDR ^{192}Ir dosimetric sensitivity (7.0 Gy prescription per fraction, 5.0 mm dwell step spacing) under Protocol A (reference dwell times transferred to predicted catheters) and Protocol B (dwell times re-optimized on predicted catheters, dose evaluated on reference geometry). Part B evaluates simulated uncertainty-guided 1-click tip relocation ($\hat{\sigma}_{tip}$) and unreviewed ($k = 0$) nested 5-fold split-conformal 3D tip prediction intervals across the $n = 972$ matched out-of-fold catheters.

	Dwell-wt. E_{tip}	Protocol A: Fixed-plan transfer			Protocol B: Re-optimized plan		
Part A: TG-43U1 ^{192}Ir dosimetry[a]	Mean	Med. $\lvert\Delta D_{90}\rvert$ (%)	$\gamma_{2/2}$ (%)	$\gamma_{3/3}$ (%)	$\lvert\Delta D_{90}\rvert$ (%)	Cran. D_{2cc}	$\gamma_{2/2}$ (%)
PMB 2019 3D U-Net + DBSCAN (n=54)	10.51	8.36 27.6 / 27.2	64.0	73.3	37.1 / 10.2	95.3 %	43.8
Stage 0: 3D radial symmetry (n=54)	9.04	7.51 23.0 / 21.3	68.2	76.5	17.8 / 8.9	54.0 %	48.6
NeedleFinder 2026 w/o Synth (4a, n=54)	6.65	5.49 18.5 / 13.2	73.4	81.5	12.8 / 5.8	35.2 %	59.5
NeedleFinder 2026 proposed (4b, n=54)[b]	5.72	4.66 15.2 / 11.5	76.6	84.3	8.2 / 4.2	29.5 %	66.0
NeedleFinder 2026 + Top-2 Tip Review (n=54)	**4.56**	**3.93 11.7 / 10.2**	**80.2**	**87.3**	**5.9 / 2.8**	**22.5 %**	**74.3**
Human Rater 2 (`LB/AMe`, n=40)	6.27	5.54 21.6 / 16.3	68.1	77.4	20.9 / 4.0	57.7 %	59.3
NeedleFinder 2026 proposed (4b, n=40)[b]	5.18	3.93 13.5 / 11.1	79.0	86.4	7.0 / 3.2	21.1 %	71.6
NeedleFinder 2026 + Top-2 Tip Review (n=40)	**4.12**	**3.33 10.5 / 10.1**	**81.9**	**88.7**	**5.1 / 2.6**	**17.3 %**	**78.2**

Part B1: Simulated $\hat{\sigma}_{tip}$ tip triage (n = 972)[c]	Reloc.	Mean	Med.	P90	P95	≤ 2 mm	≤ 3 mm	> 10 mm
Unreviewed **NeedleFinder 2026** (k=0 tips/pt)	0	6.23	2.27	16.20	27.27	45.6 %	58.7 %	16.7 %
Simulated tip review (k=1/pt, ~8 s)	35	5.54	2.06	14.60	23.54	48.6 %	61.9 %	14.4 %
Simulated tip review (k=2/pt, ~16 s)	69	5.07	1.95	12.58	21.48	51.3 %	64.8 %	12.8 %
Simulated tip review (k=3/pt, ~24 s)	105	4.57	1.81	10.91	19.16	54.0 %	67.2 %	11.1 %
Simulated tip review (k=5/pt, ~40 s)	169	3.72	1.67	7.79	14.26	59.5 %	72.6 %	8.0 %
Simulated tip review (k=8/pt, ~64 s)	251	2.77	1.51	5.98	9.90	65.5 %	78.5 %	5.0 %
Simulated tip review (k=10/pt, ~80 s)	301	**2.52**	**1.44**	**4.97**	**8.30**	**69.2 %**	**81.8 %**	**4.3 %**

Part B2: Split-conformal (k=0, n=972)[c]	Nom.	Emp.	Fixed	Mean r	Med. r	Fixed r	Low-$\hat{\sigma}$ r	Width red.
Conformal 80 % 3D tip interval (k=0)	80.0 %	79.3 %	80.2 %	7.31	6.46	7.73	4.46	−16.4 %
Conformal 85 % 3D tip interval (k=0)	85.0 %	84.4 %	85.5 %	9.80	8.71	11.43	5.96	−23.8 %
Conformal 90 % 3D tip interval (k=0)	90.0 %	**90.2 %**	89.9 %	13.71	12.11	17.68	8.41	−31.5 %
Conformal 95 % 3D tip interval (k=0)	95.0 %	**95.4 %**	95.2 %	21.05	18.42	28.39	12.89	−35.1 %

[a] TG-43U1 evaluation uses 5.0 mm dwell step spacing (3.5 mm distal tip offset, ≤ 45 mm active length) on a 1.5 mm isotropic 3D dose grid with implant-conformal HR-CTV$_{implant}$ (8.0 mm radial envelope, D_{90} = 7.0 Gy) and 2 cc OAR proxy shells. $\lvert\Delta D_{90}\rvert$ cells report mean / median percentage target dose shift; 'Cran. D_{2cc}' reports mean relative error $\lvert\Delta D_{2cc}\rvert$ (%) in the cranial (sigmoid/fundal) OAR proxy shell; $\gamma_{2/2}$ and $\gamma_{3/3}$ are global 3D gamma pass rates (2 %/2 mm and 3 %/3 mm). Dwell-weighted tip error $E_{tip}^{dwell\text{-}w}$ correlates with $\lvert\Delta D_{90}\rvert$ at Spearman $\rho = 0.8629$ ($p = 4.9 \times 10^{-17}$) and with $\gamma_{2\%/2mm}$ at $\rho = -0.8474$ ($p = 6.5 \times 10^{-16}$).

[b] Paired Wilcoxon vs. PMB 2019 ($p = 4.9 \times 10^{-6}$ on $\lvert\Delta D_{90}\rvert$, $p = 8.2 \times 10^{-7}$ on $\gamma_{2\%/2mm}$), vs. Stage 0 ($p = 7.9 \times 10^{-8}$ on $\lvert\Delta D_{90}\rvert$, $p = 1.4 \times 10^{-7}$ on $\gamma_{2\%/2mm}$), and vs. Human Rater 2 on $n = 40$ ($p = 0.0278$ on $\lvert\Delta D_{90}\rvert$, $p = 0.0109$ on $\gamma_{2\%/2mm}$).

[c] Part B1 simulates a retrospective 1-click distal tip relocation oracle ($E_{tip} \rightarrow \sqrt{\Delta r_{\perp}^2 + \min(\lvert\Delta s_{\parallel}\rvert, 0.85\,\text{mm})^2}$ when $E_{tip} > 2.0$ mm, E_{shaft} unchanged; 'Reloc.' = total tips relocated across 54 patients) ranked by out-of-fold $\hat{\sigma}_{tip}$. Part B2 reports unreviewed ($k = 0$) 5-fold split-conformal 3D tip prediction interval half-widths (radii r in mm) across the $n = 972$ matched OOF catheters ('Fixed': unscaled residual quantile; 'Low-$\hat{\sigma}$ r': median half-width on the lower-uncertainty half of catheters; 'Width red.': adaptive vs. fixed median half-width reduction). Simulated oracle count-guided recovery (0.65 clicks/patient) recovers +17 missed catheters (972 → 989/1 063, 93.0 % recall).